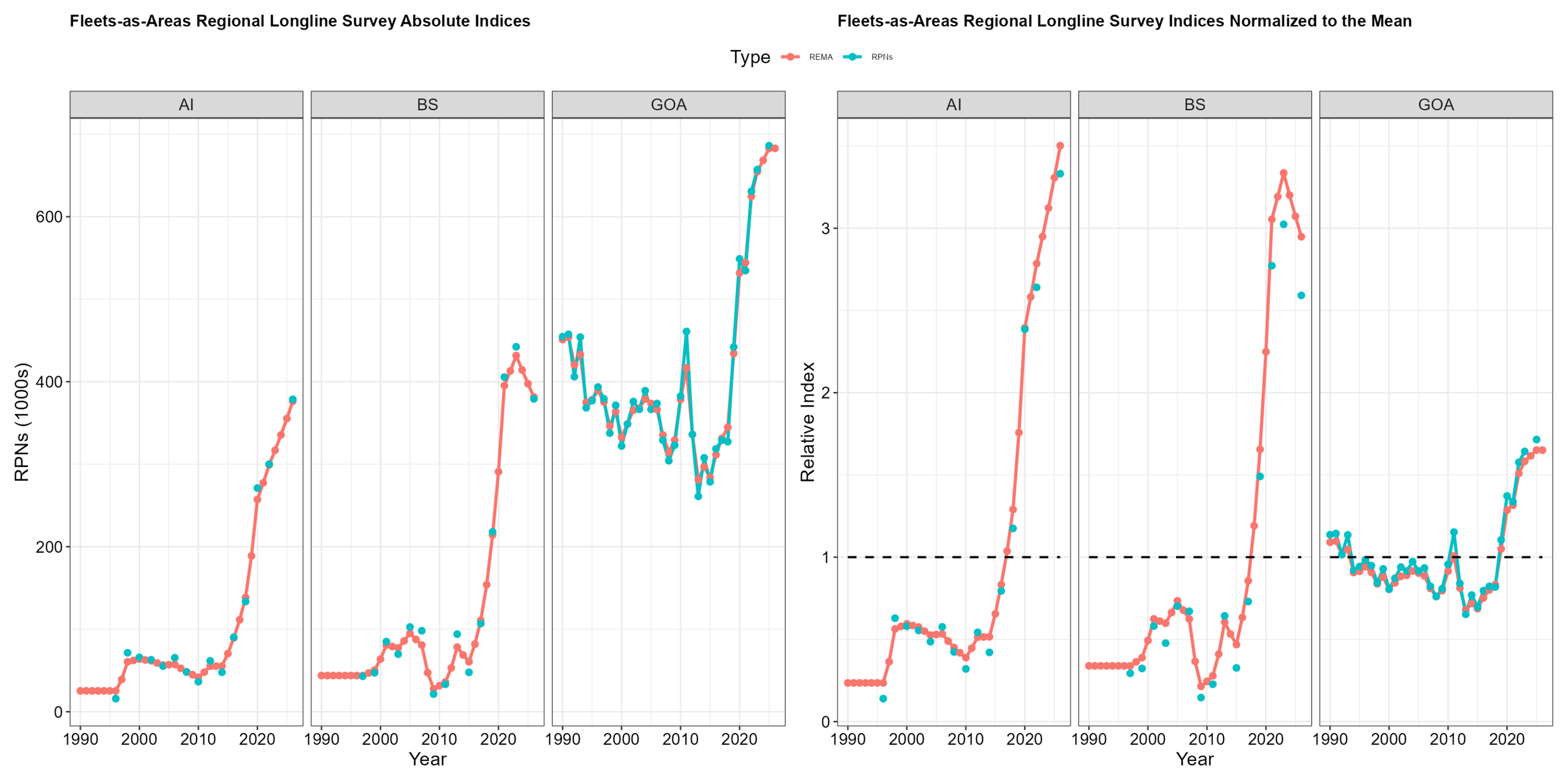
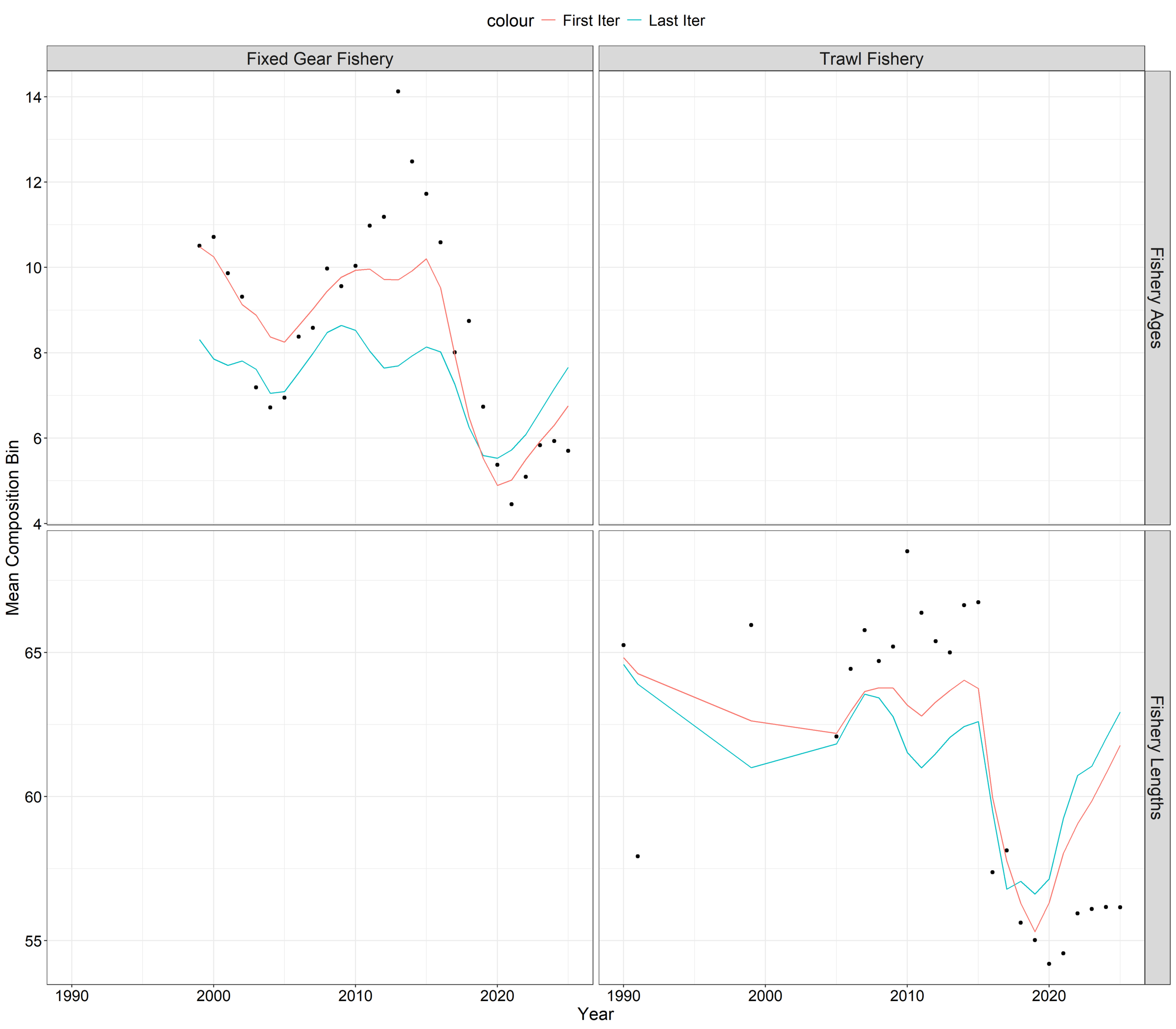
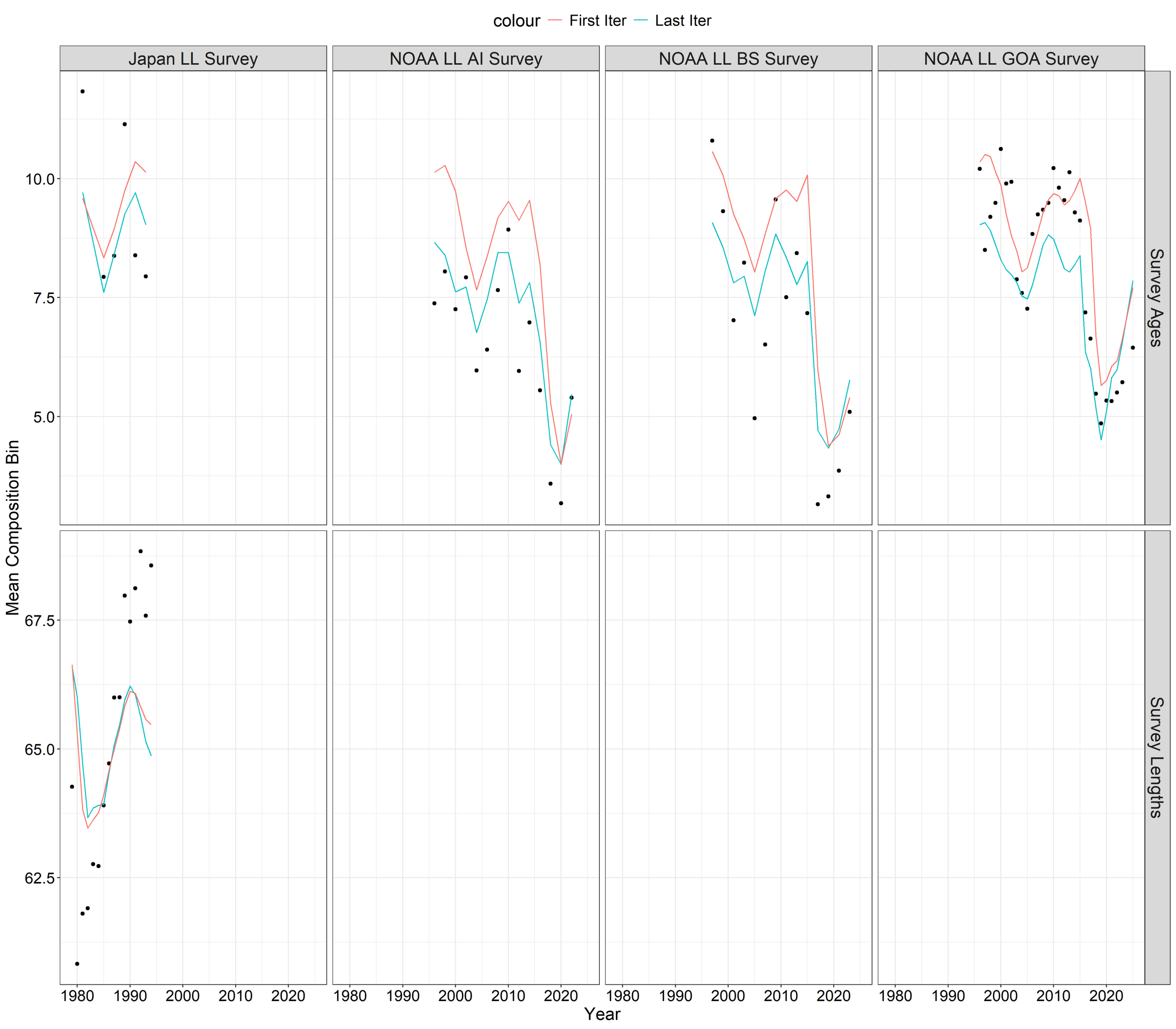
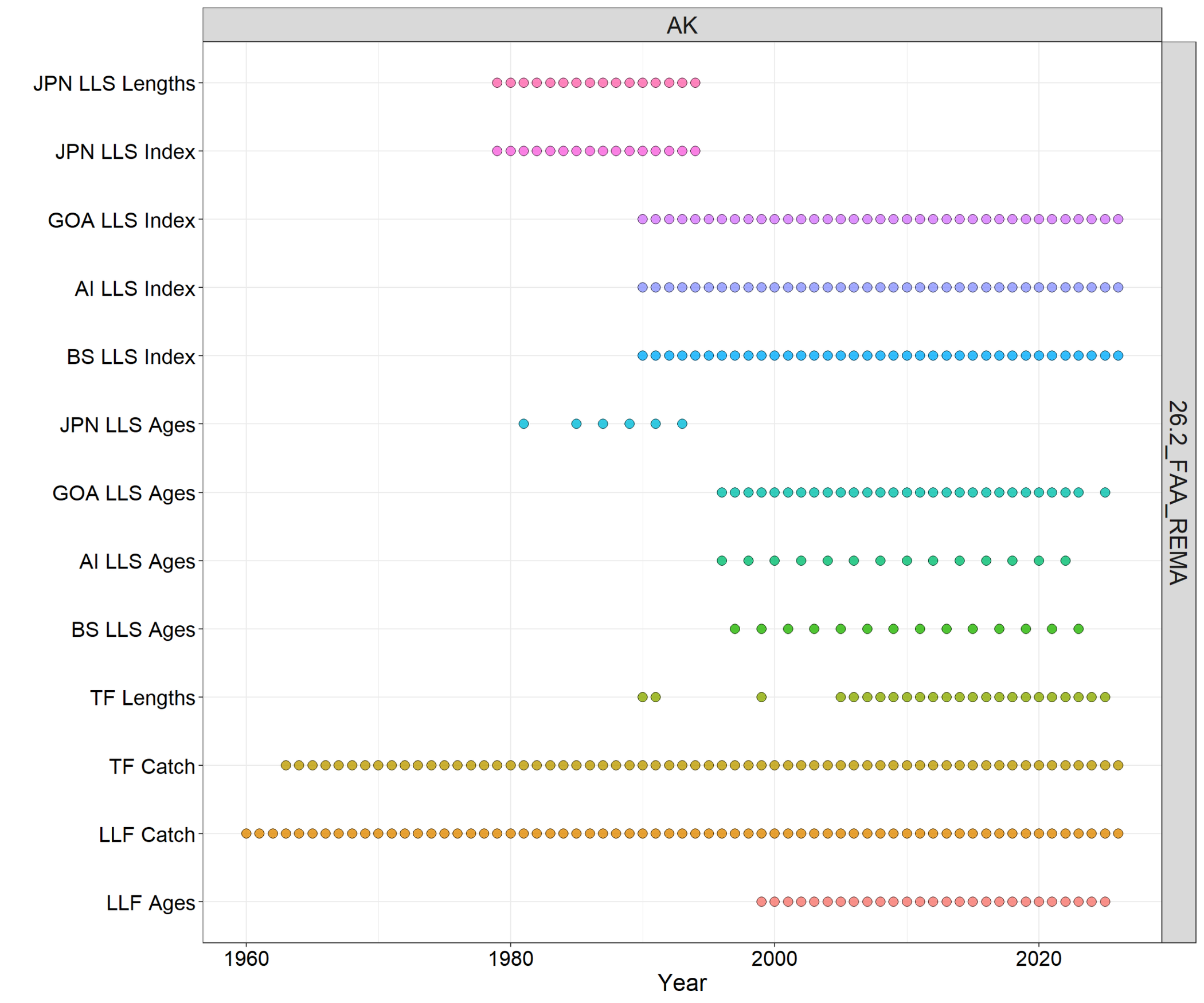
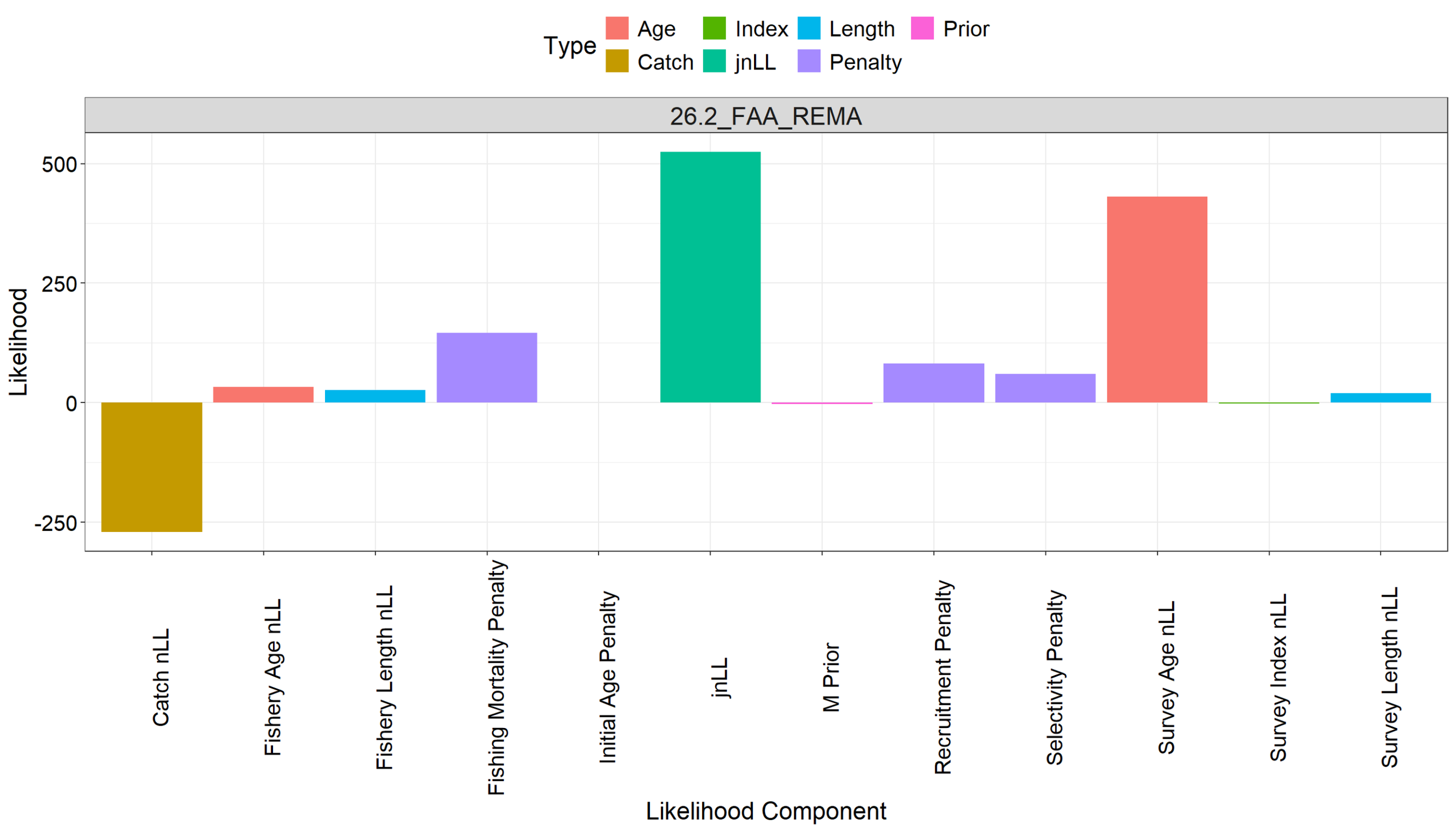

Fleets-as-Areas Regional Longline Survey Absolute Indices
Fleets-as-Areas Regional Longline Survey Indices Normalized to the Mean
Type
REMA
RPNs
AI
BS
GOA
RPNs (1000s)
Relative Index
Year
0
200
400
600
1
2
3
1990
2000
2010
2020


colour
First Iter
Last Iter
Fixed Gear Fishery
Trawl Fishery
Fishery Ages
Fishery Lengths
Mean Composition Bin
14
12
10
8
6
4
65
60
55
1990
2000
2010
2020
Year


colour
First Iter
Last Iter
Japan LL Survey
NOAA LL AI Survey
NOAA LL BS Survey
NOAA LL GOA Survey
Survey Ages
Survey Lengths
Mean Com
position Bin
10.0
7.5
5.0
67.5
65.0
62.5
1980
1990
2000
2010
2020
Year


AK
JPN LLS Lengths
JPN LLS Index
GOA LLS Index
AI LLS Index
BS LLS Index
JPN LLS Ages
GOA LLS Ages
AI LLS Ages
BS LLS Ages
TF Lengths
TF Catch
LLF Catch
LLF Ages
26.2_FAA_REMA
1960
1980
2000
2020
Year


Type
Age
Index
Length
Prior
Catch
jnLL
Penalty
26.2_FAA_REMA
500
250
0
-250
Likelihood
Catch nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL
Likelihood Component

Numbers-at-Age

Sex — Female ····· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

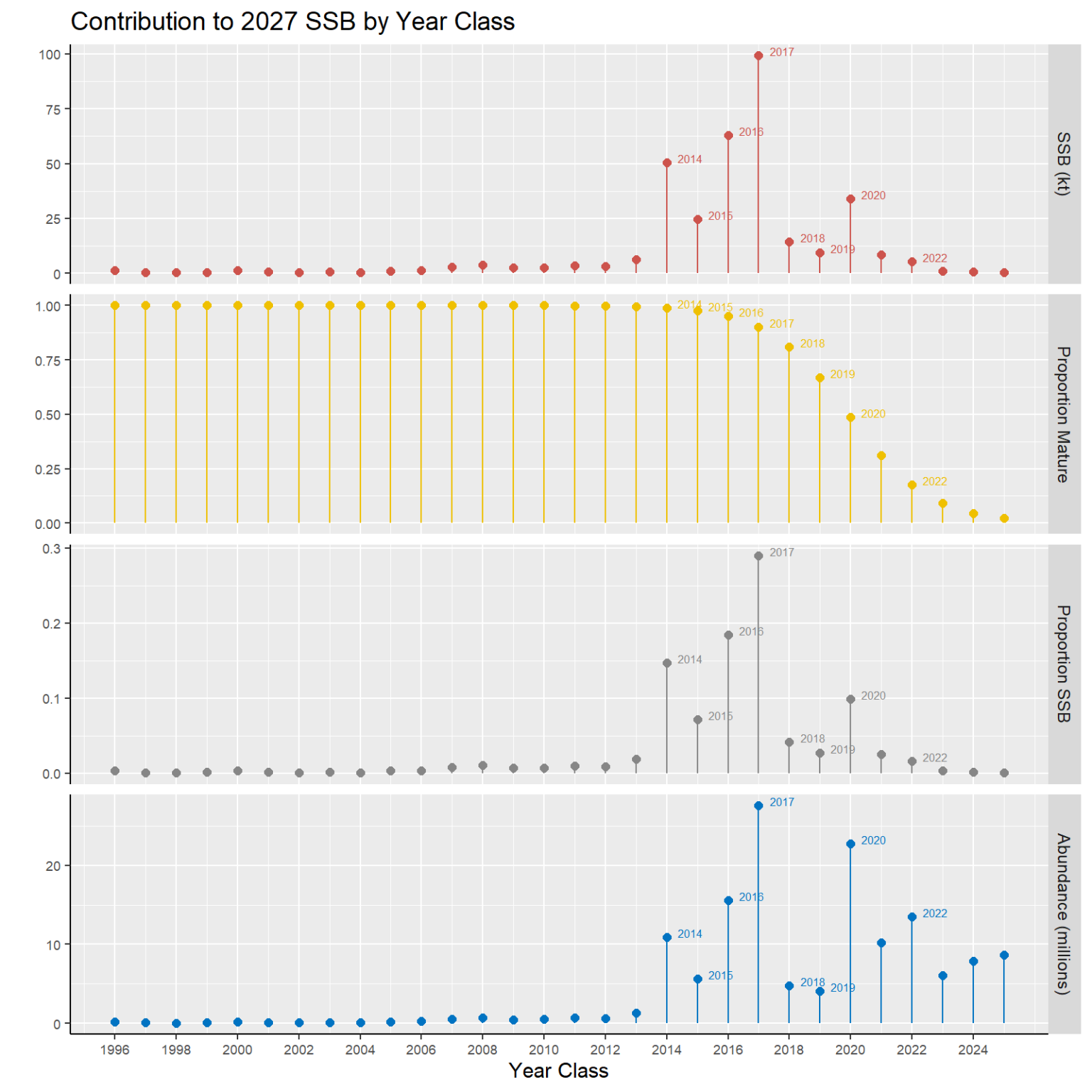
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

Numbers-at-Age (millions)

Year


Contribution to 2027 SSB by Year Class
SSB (kt)
Proportion Mature
Proportion SSB
Abundance (millions)
Year Class
2014
2015
2016
2017
2018
2019
2020
2022

Model 26.2_FAA_REMA

AK

Dynamic SSB0

Recruitment

SSB

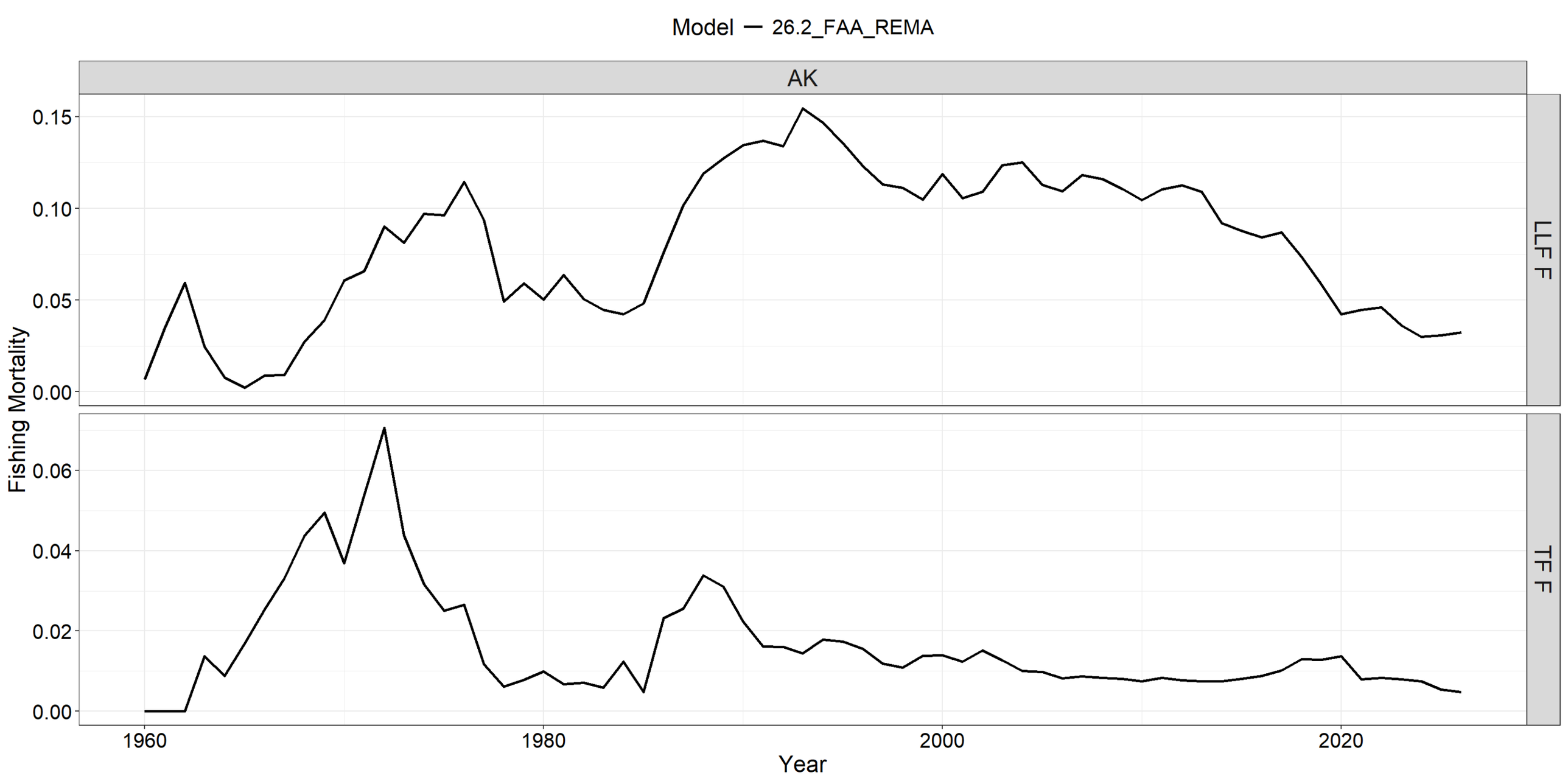
Total Biom

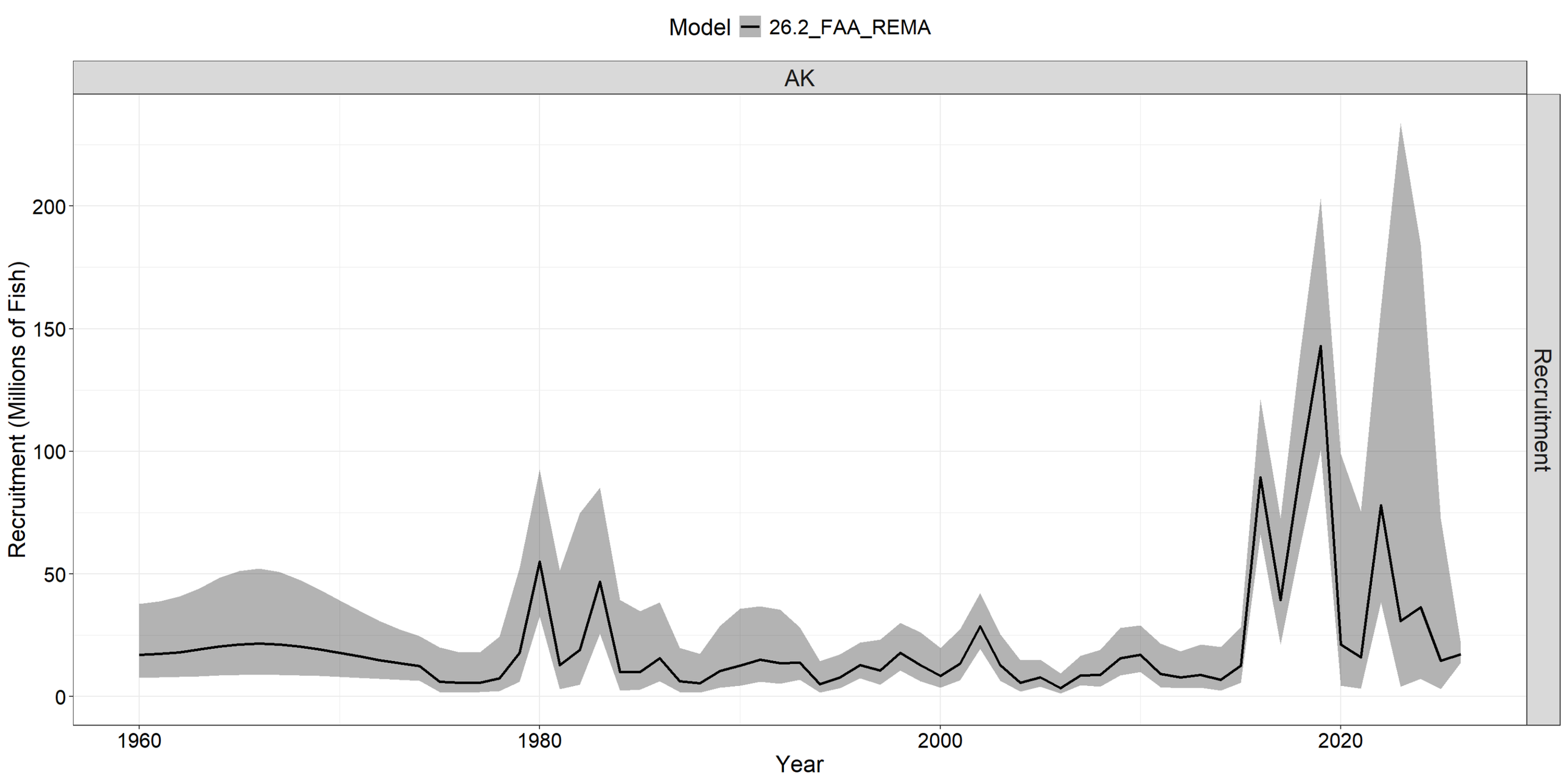
LLF F

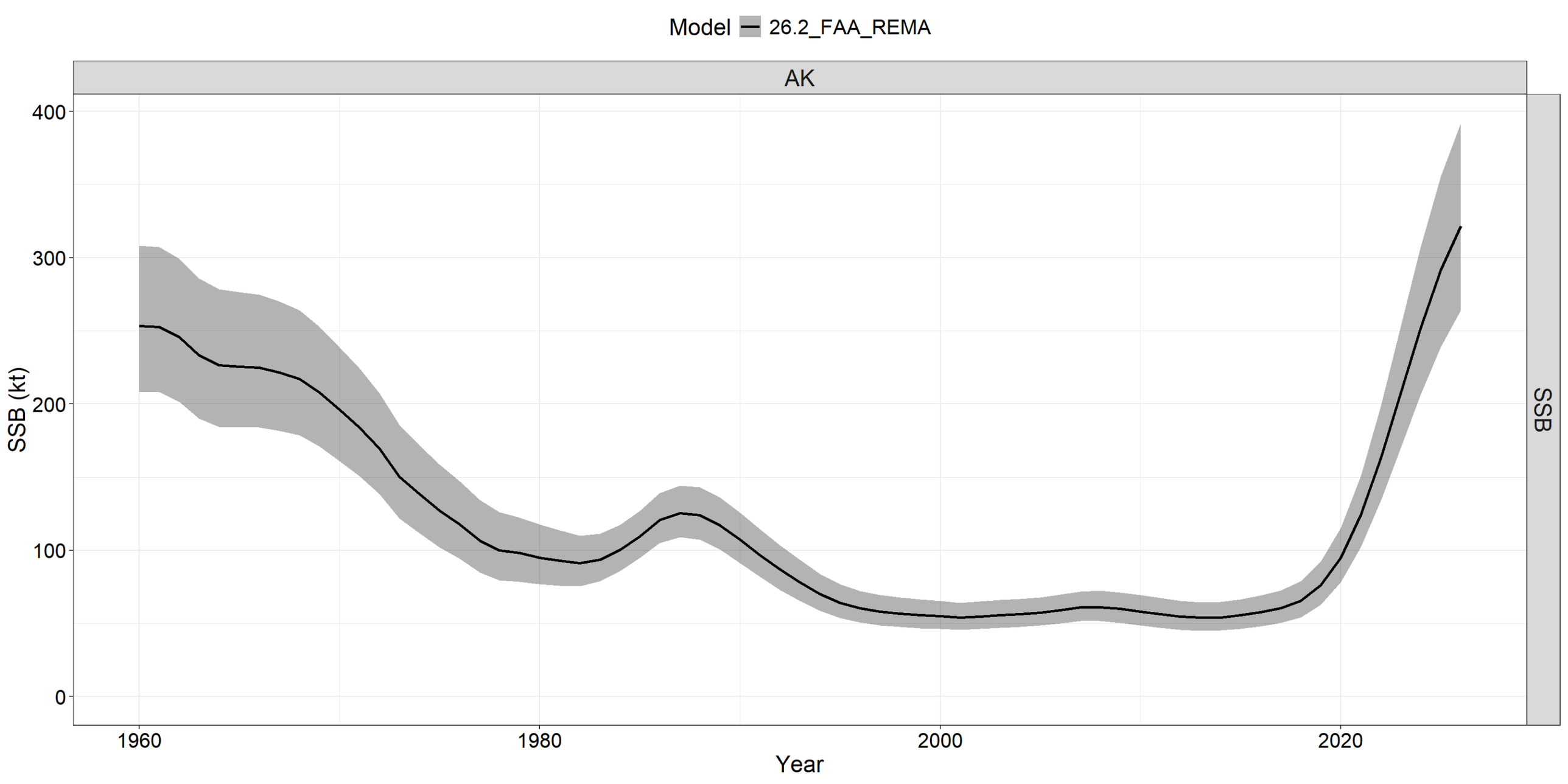
TF F

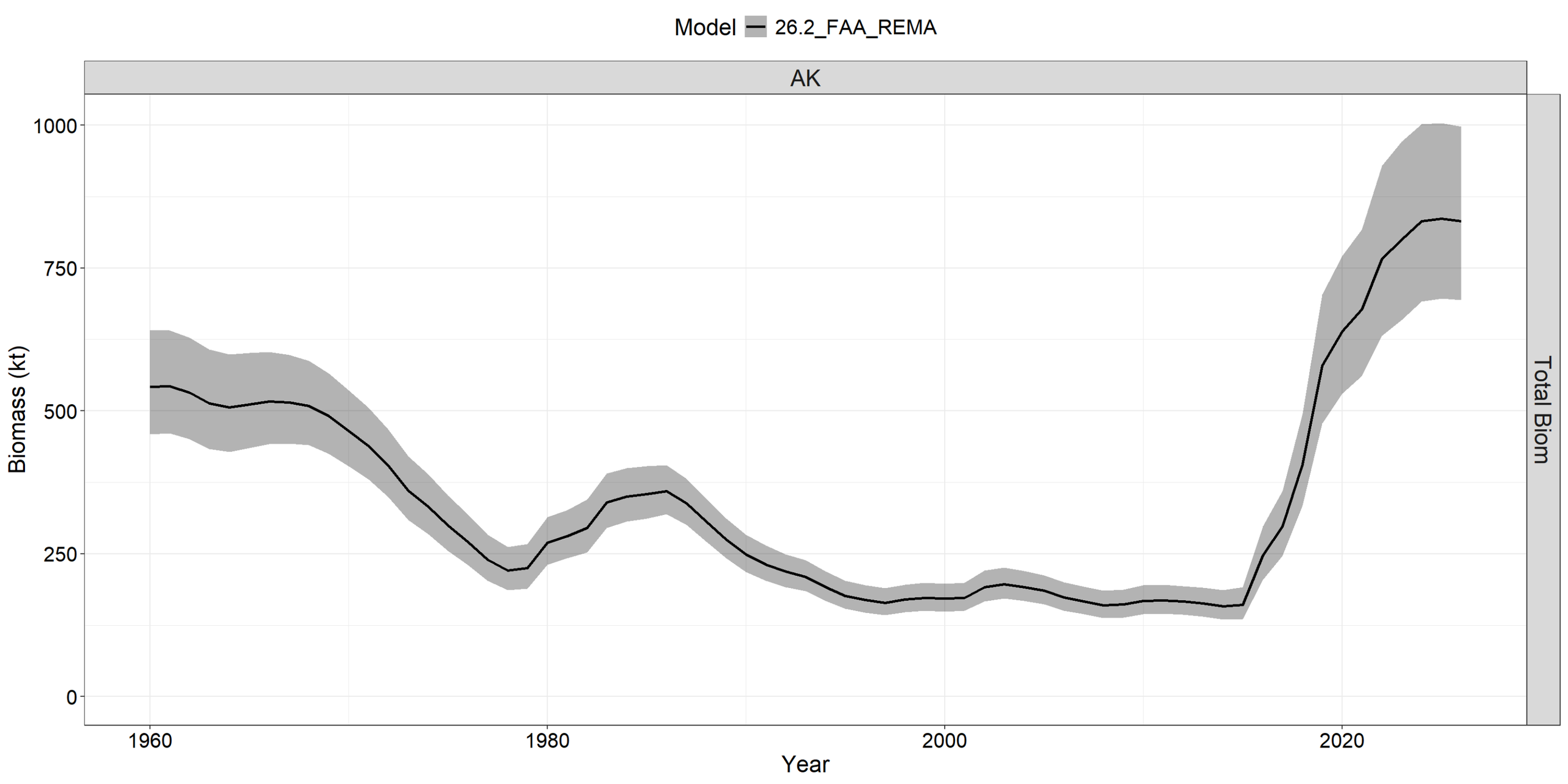
Value

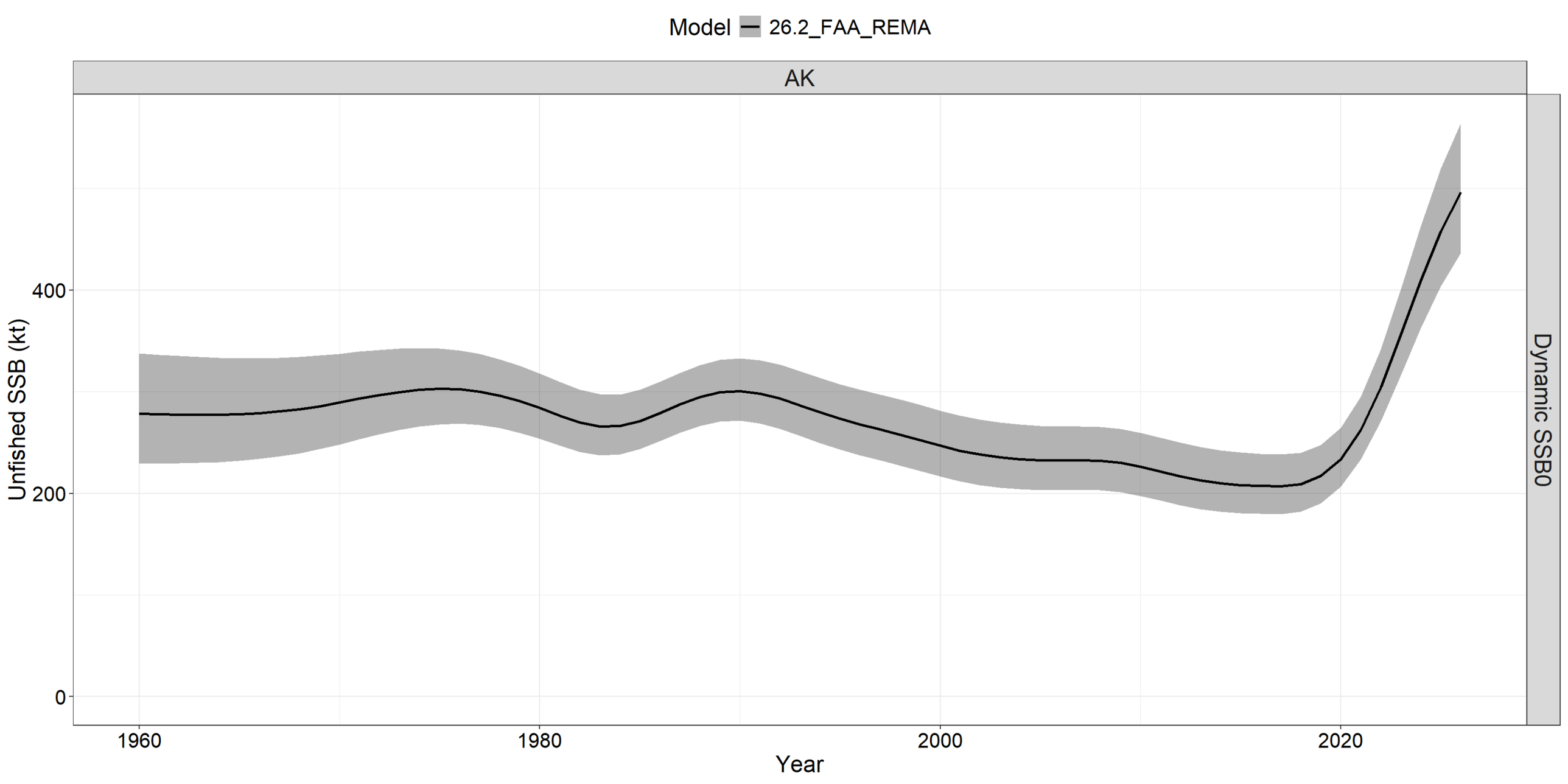
Year

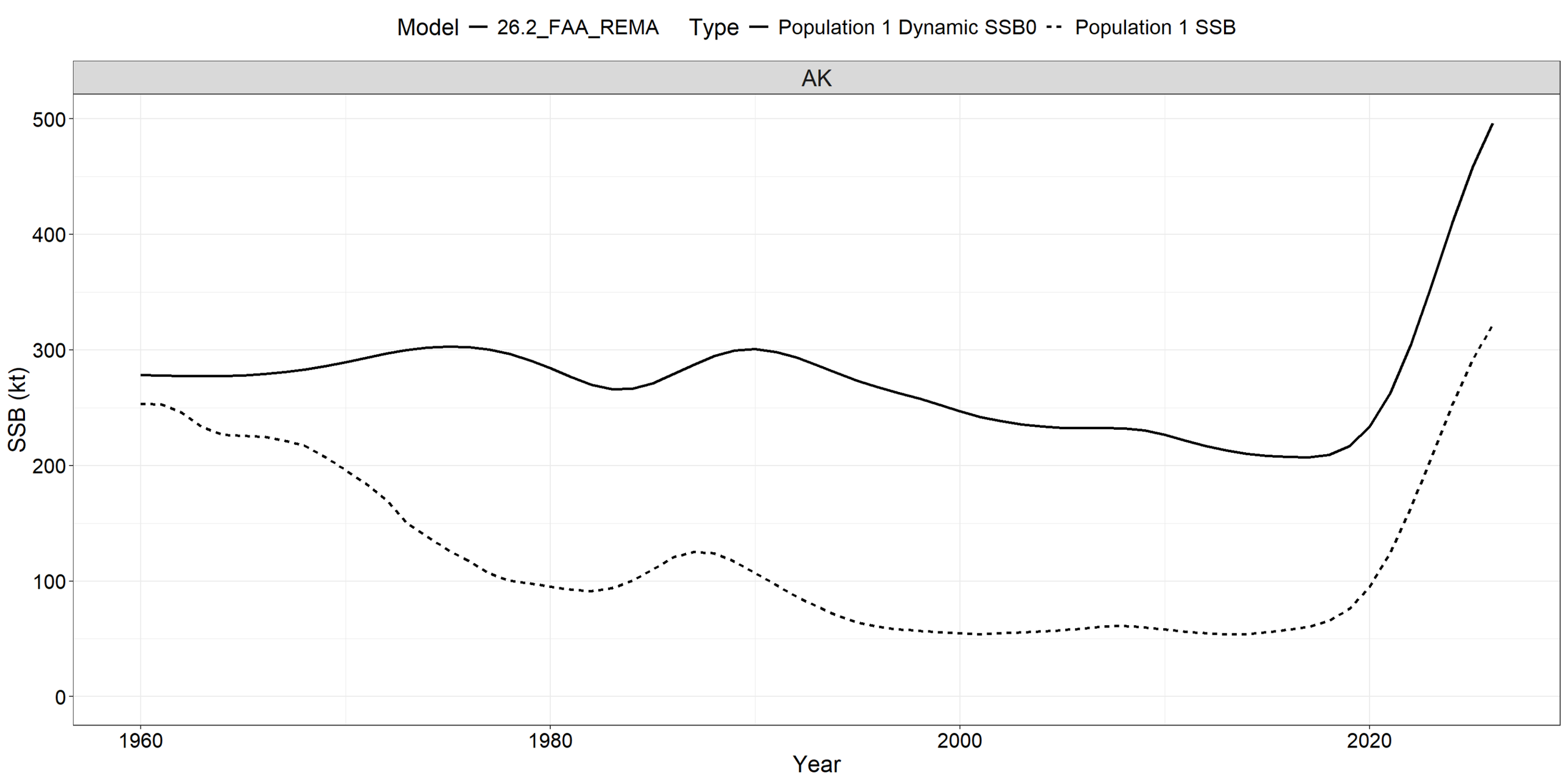

Model — 26.2_FAA_REMA
AK
LLF F
TF F
Fishing Mortality
0.15
0.10
0.05
0.00
0.06
0.04
0.02
0.00
1960
1980
2000
2020
Year

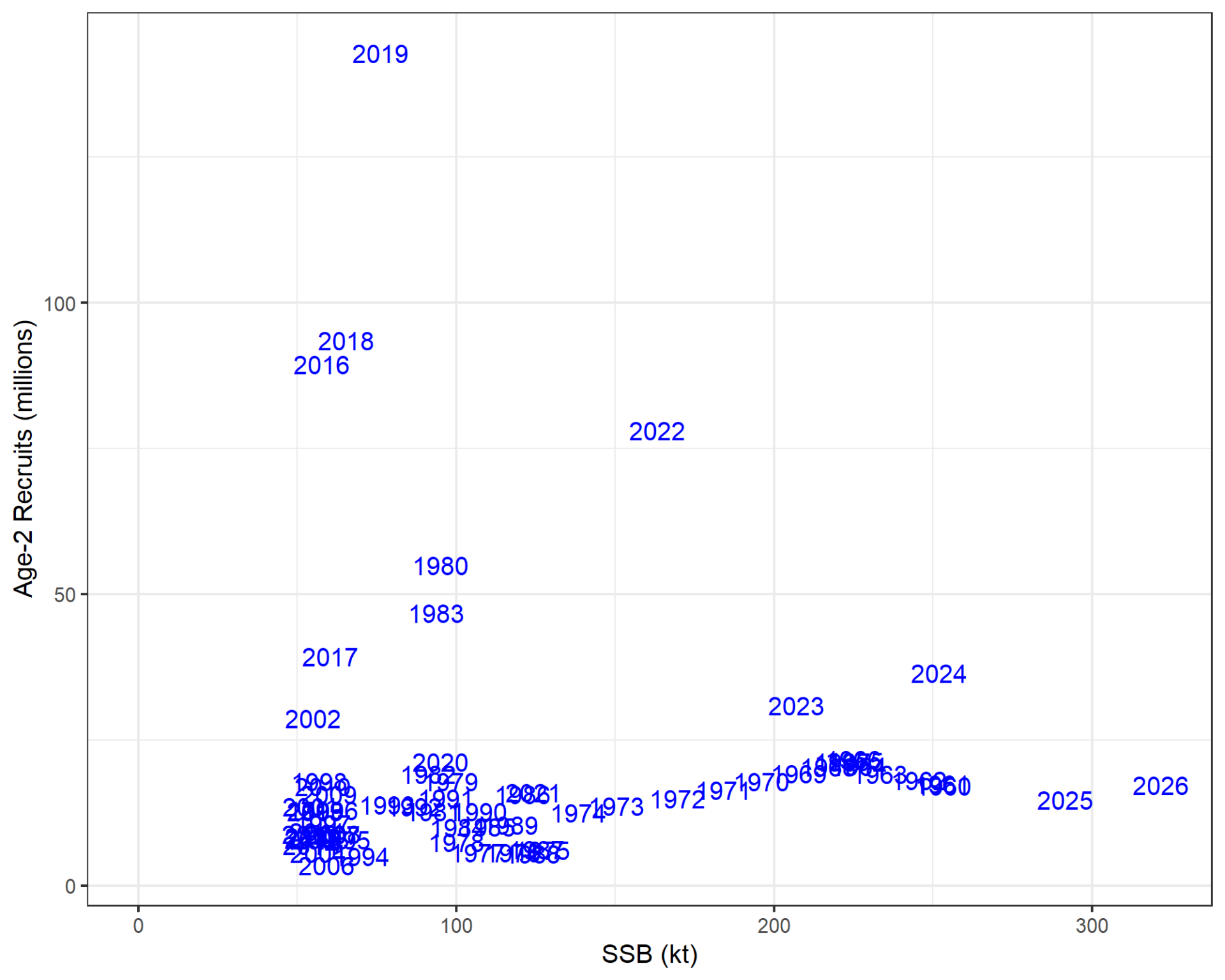

Model
26.2_FAA_REMA
AK
Recruitment (Millions of Fish)
0
50
100
150
200
1960
1980
2000
2020
Year
Recruitment

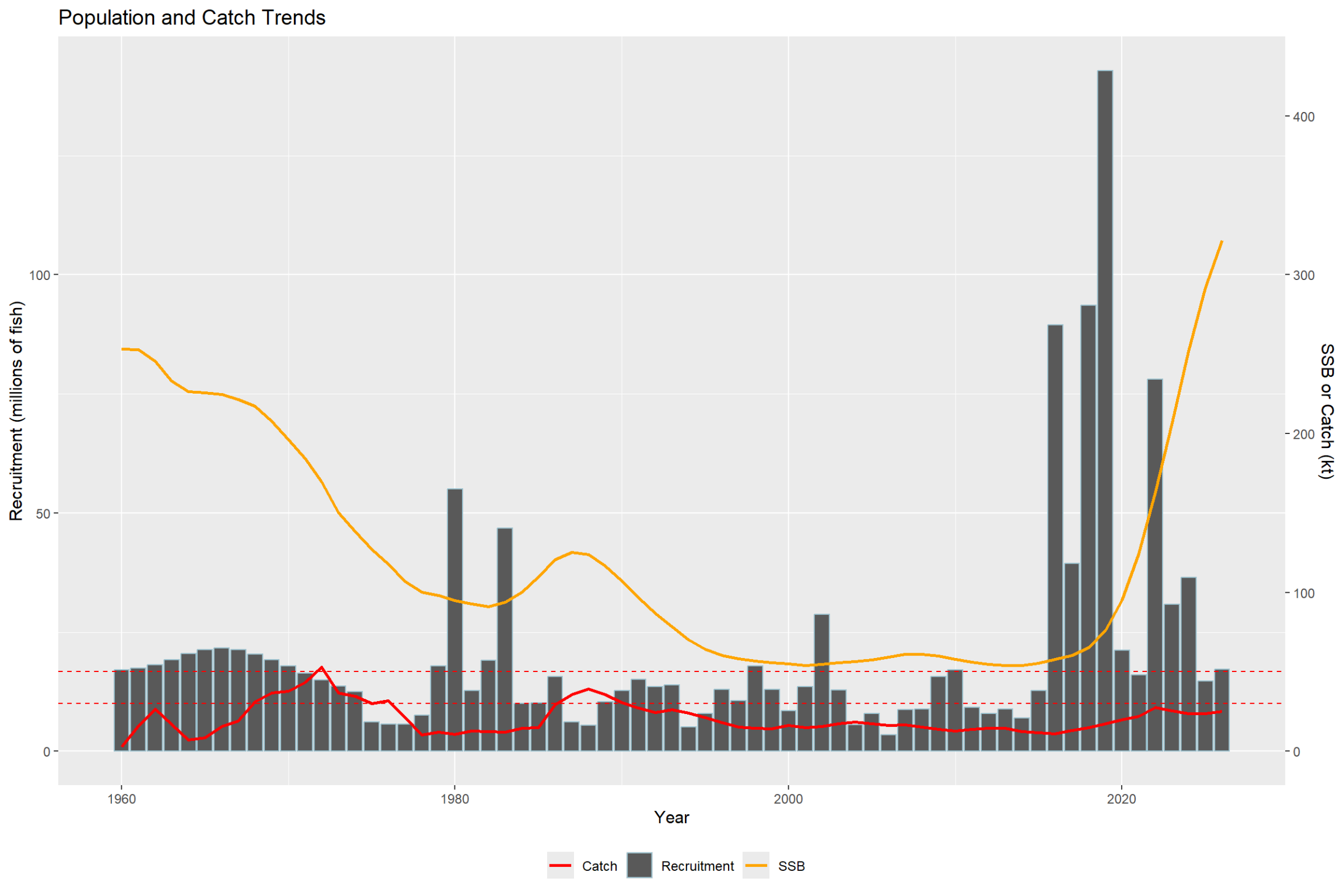

Model
26.2_FAA_REMA
AK
SSB
SSB (kt)
400
300
200
100
0
1960
1980
2000
2020
Year

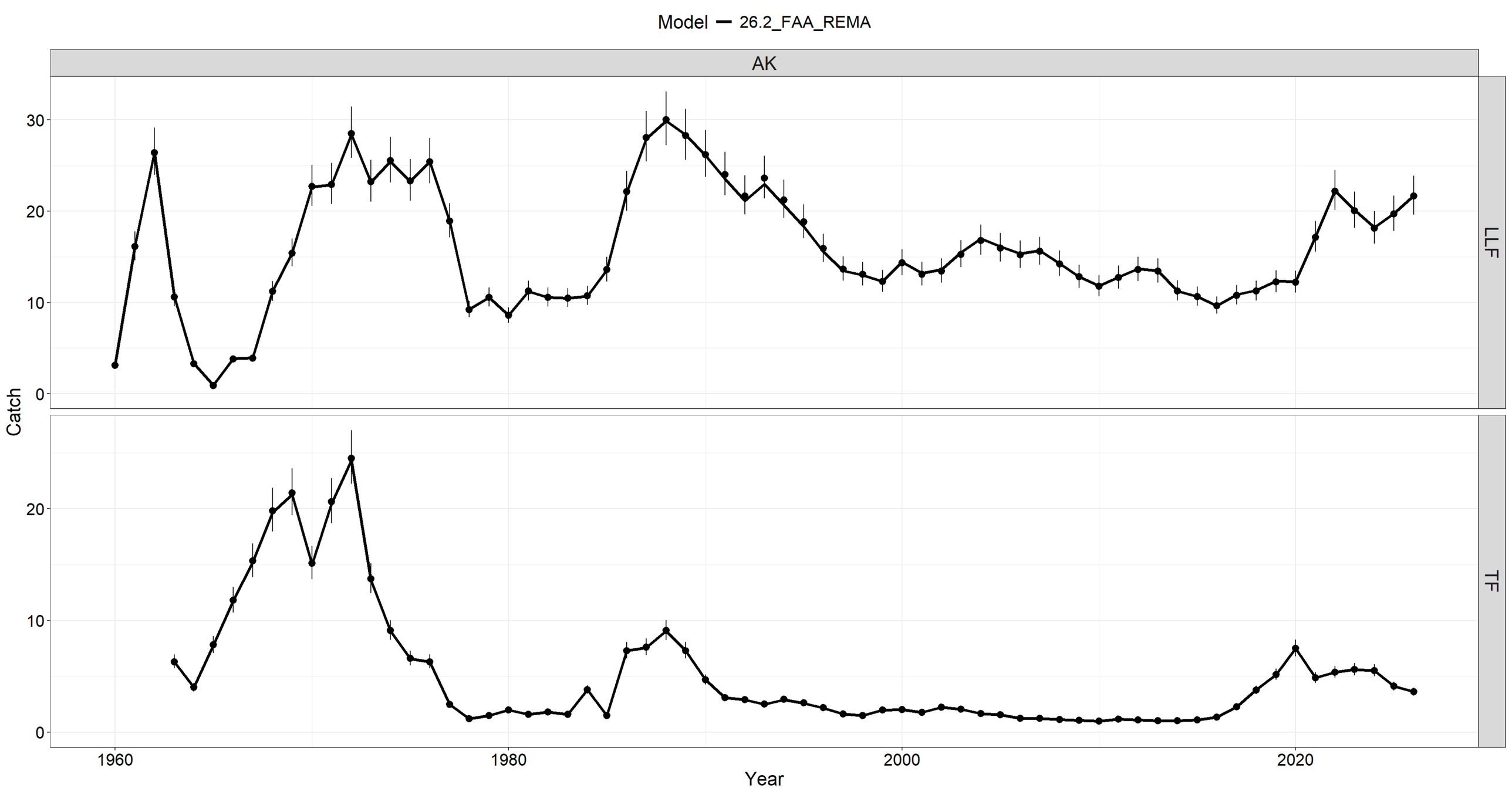

Model
26.2_FAA_REMA
AK
Total Biom
Biomass (kt)
1000
750
500
250
0
1960
1980
2000
2020
Year


Model
26.2_FAA_REMA
AK
Dynamic SSB0
Unfished SSB (kt)
400
200
0
1960
1980
2000
2020
Year


Model — 26.2_FAA_REMA Type — Population 1 Dynamic SSB0 -- Population 1 SSB
AK
500
400
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year


2019
2018
2016
2022
1980
1983
2017
2024
2023
2002
2020
2025
2026
1974
1973
1972
1971
1970
1969
100
50
0
0
100
200
300
Age-2 Recruits (millions)
SSB (kt)


Population and Catch Trends
Recruitment (millions of fish)
SSB or Catch (kt)
0
50
100
0
100
200
300
400
1960
1980
2000
2020
Year
Catch
Recruitment
SSB


Model — 26.2_FAA_REMA
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

Model — 26.2_FAA_REMA

BS LLS

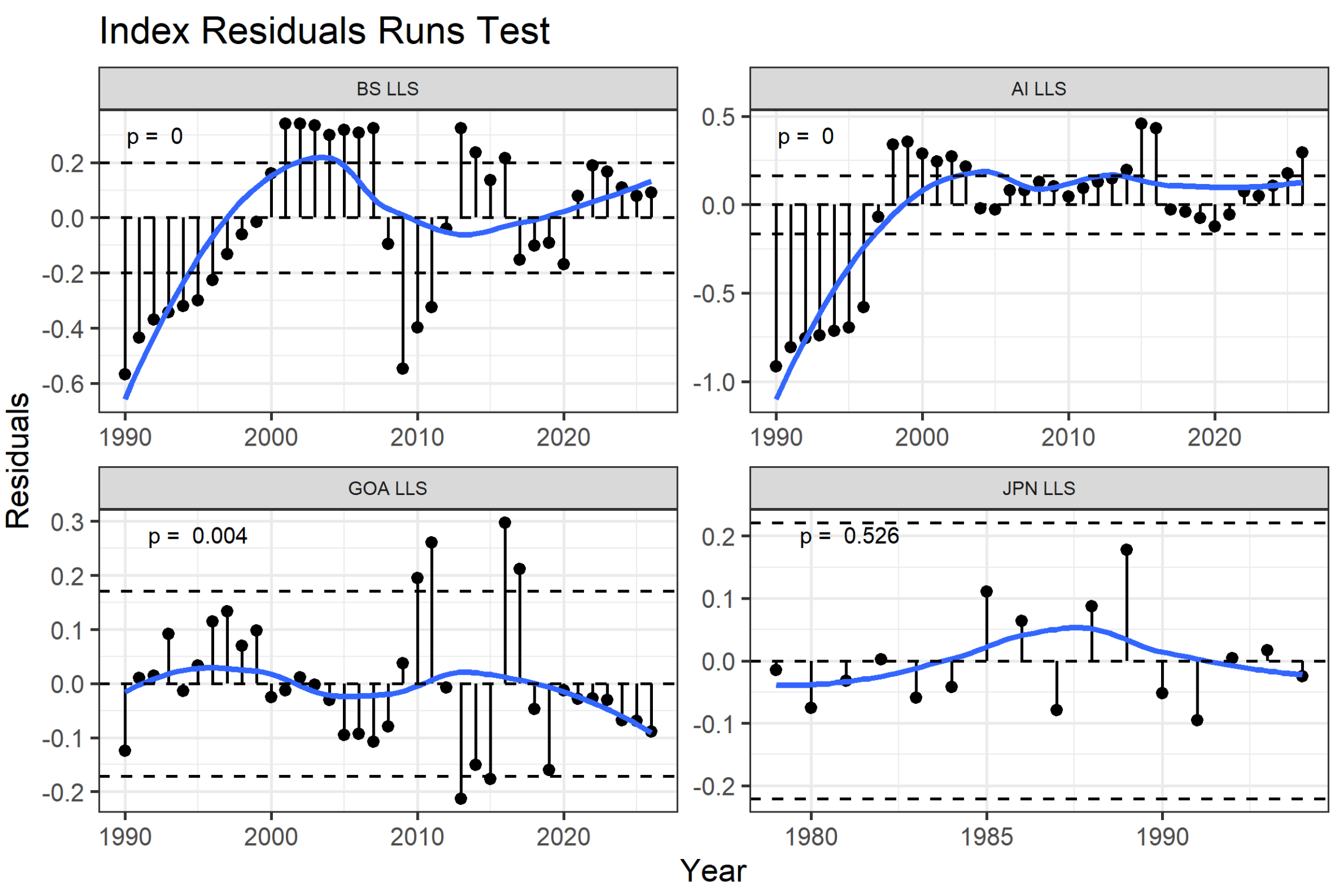
AI LLS

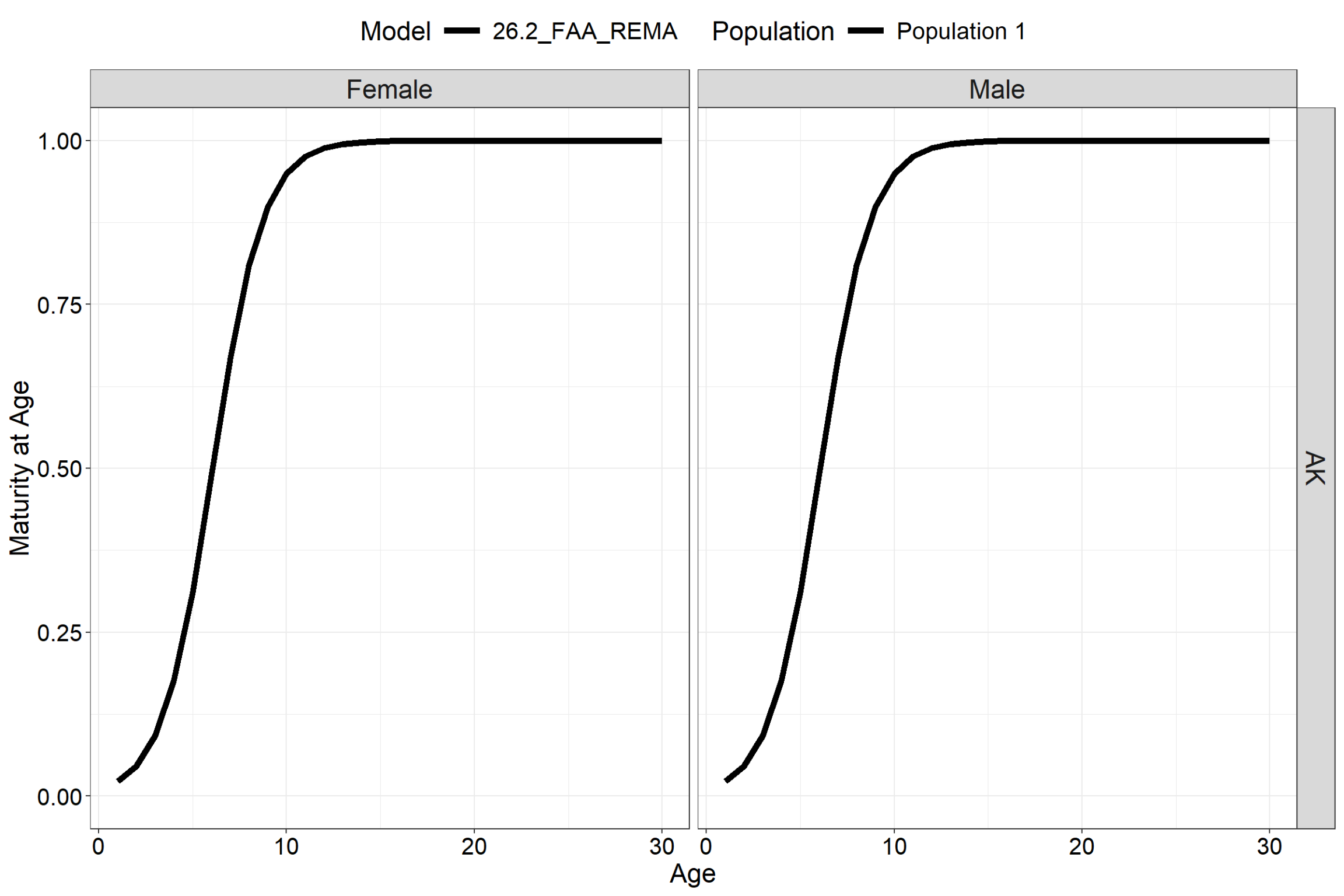
GOA LLS

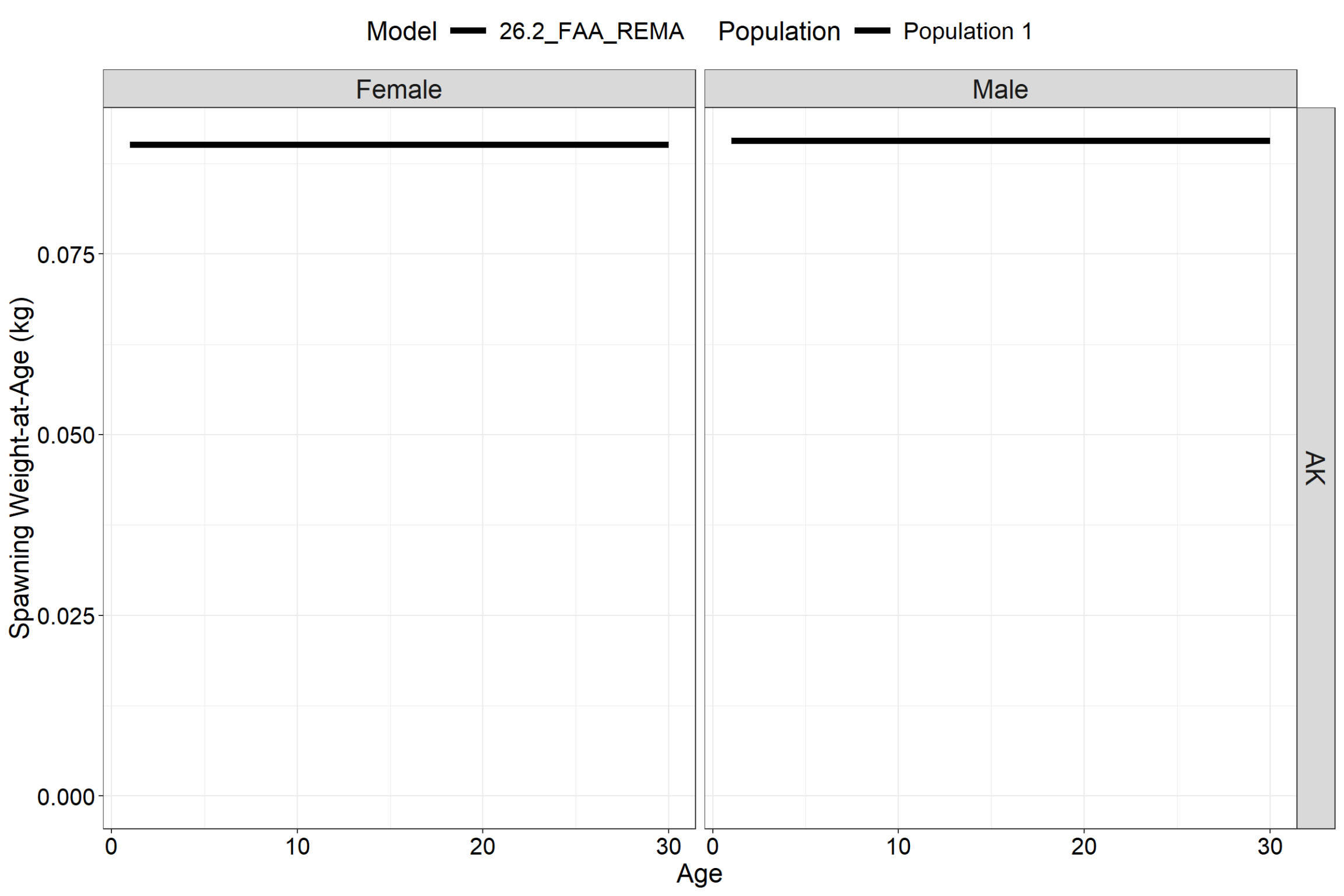
JPN LLS

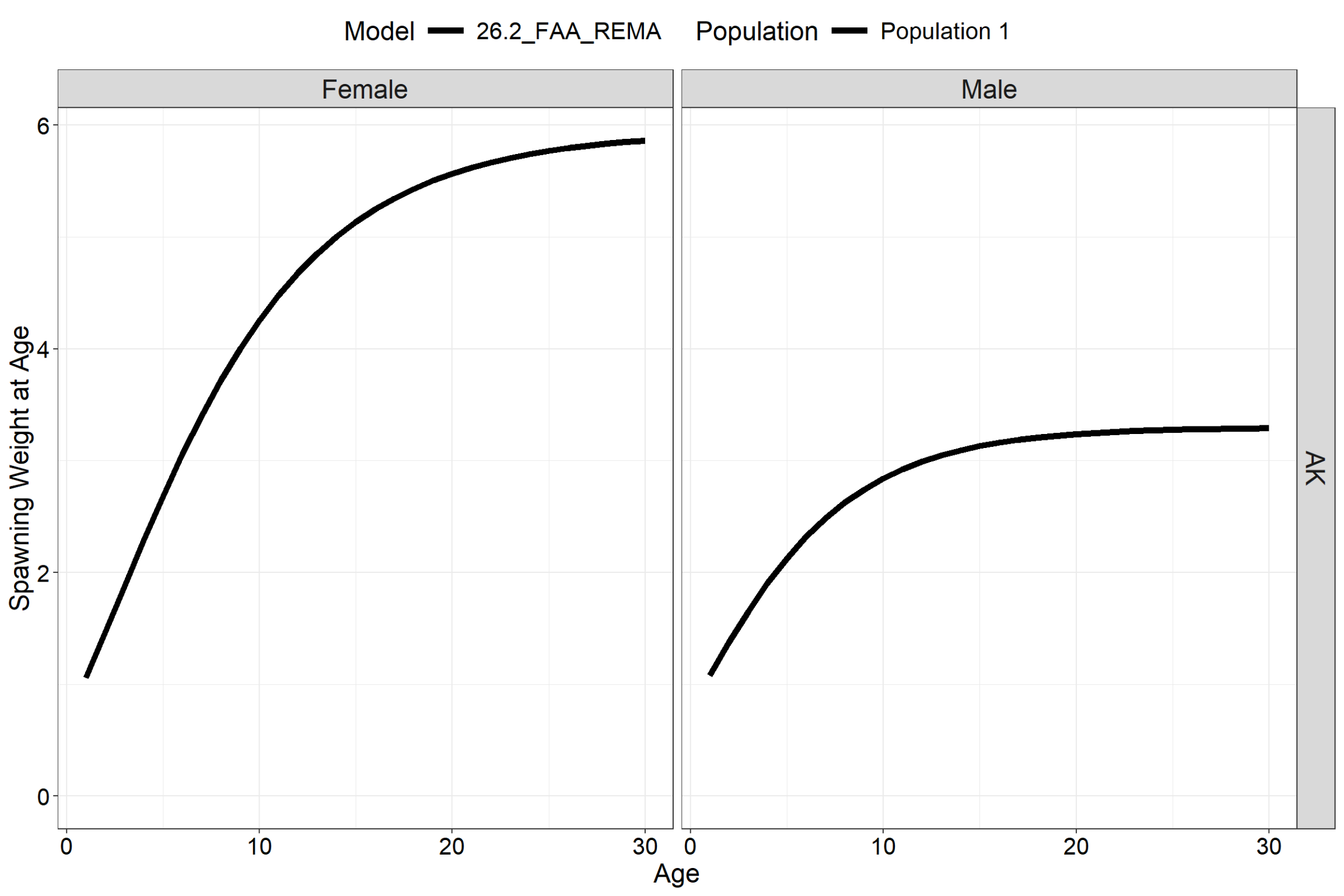
RPNs (1000s)

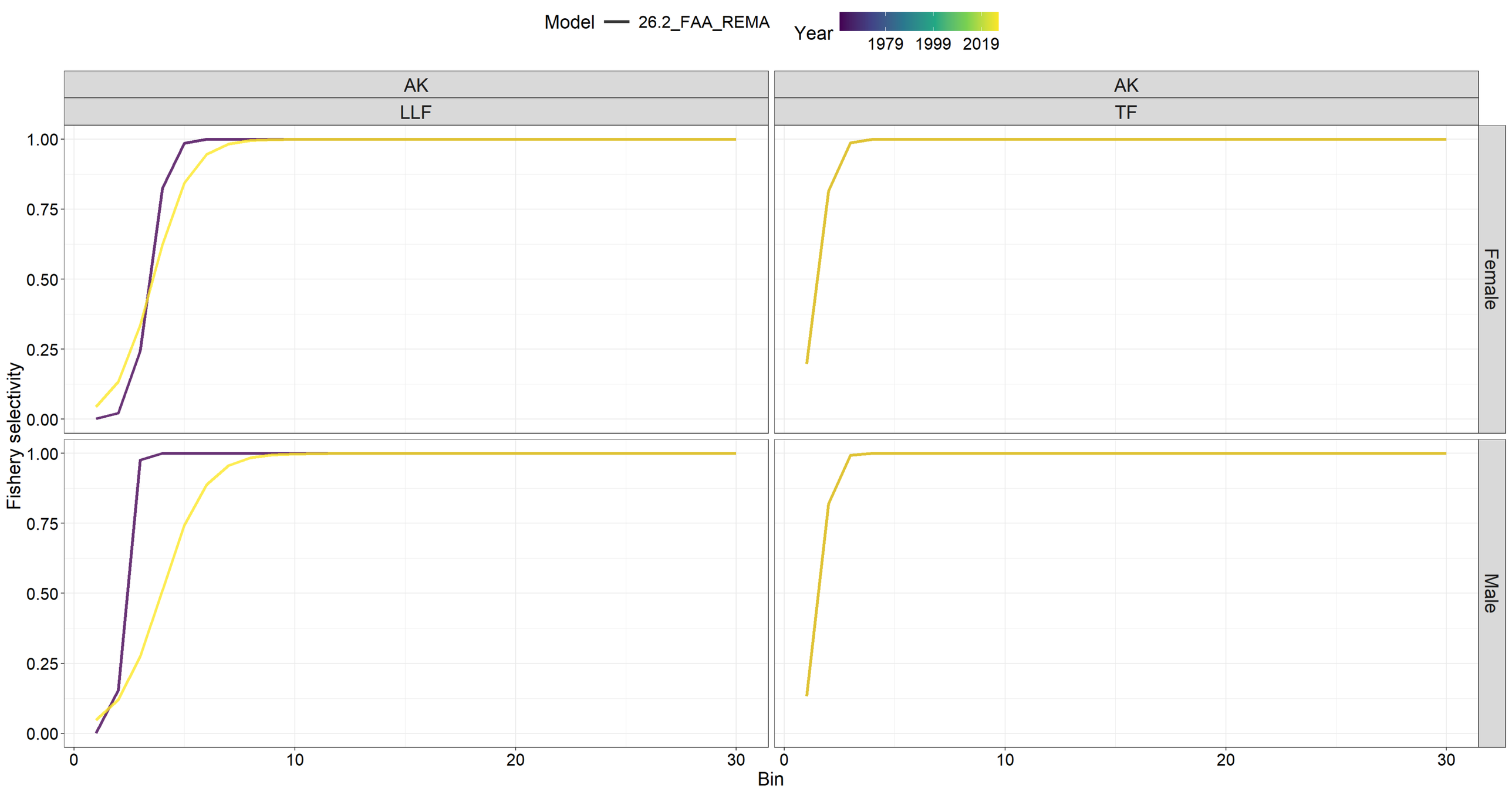
Year

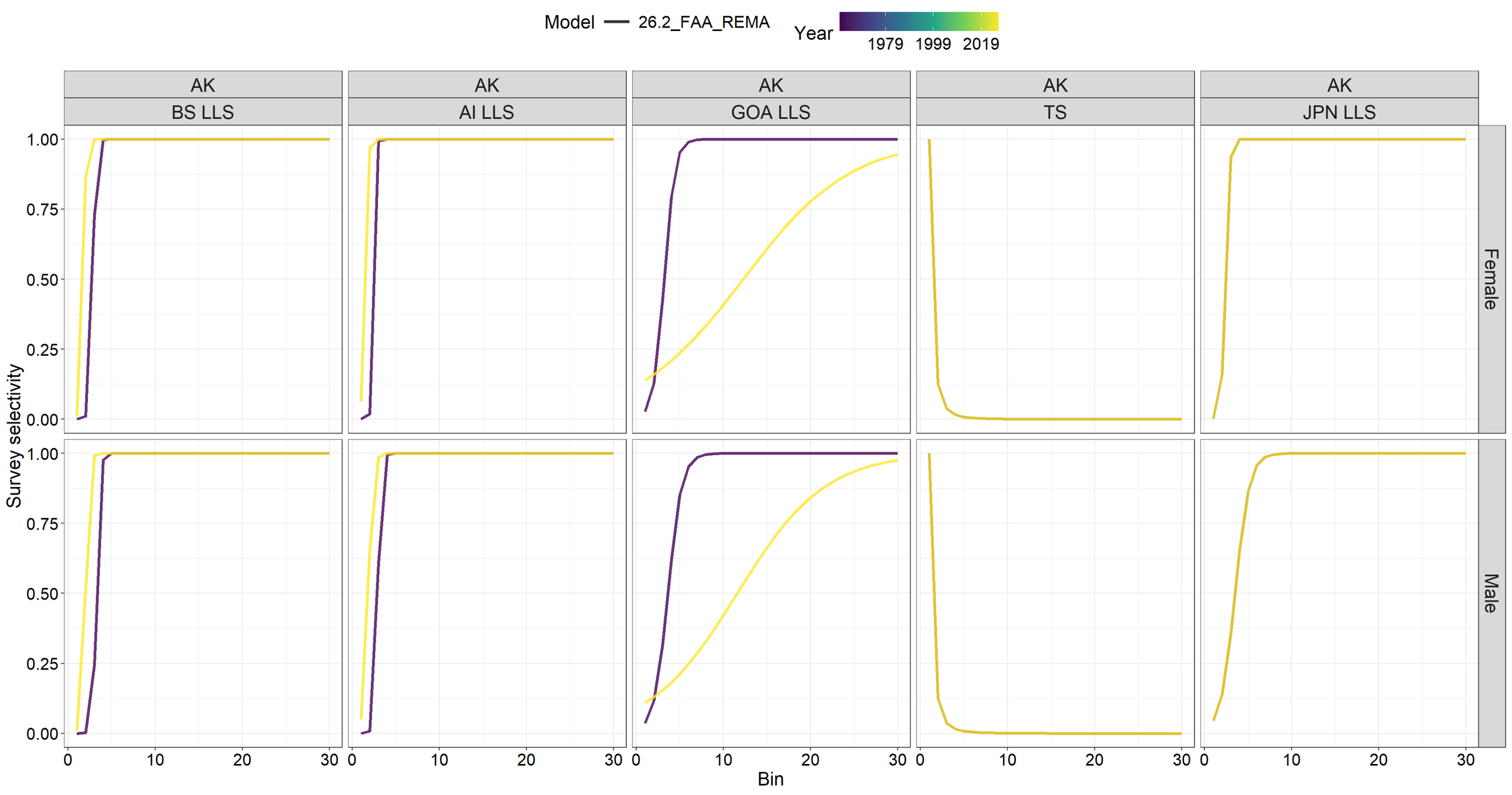

Index Residuals Runs Test
BS LLS
p = 0
AI LLS
p = 0
GOA LLS
p = 0.004
JPN LLS
p = 0.526
Residuals
Year
0.2
0.0
-0.2
-0.4
-0.6
0.5
-0.5
-1.0
0.3
0.1
-0.1
1990
2000
2010
2020
1980
1985


Model
26.2_FAA_REMA
Population
Population 1
Female
Male
AK
Maturity at Age
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Age


Model
26.2_FAA_REMA
Population
Population 1
Female
Male
AK
Spawning Weight-at-Age (kg)
0.075
0.050
0.025
0.000
0
10
20
30
Age


Model
26.2_FAA_REMA
Population
Population 1
Female
Male
AK
Spawning Weight at Age
Age
0
2
4
6
0
10
20
30


Model
26.2_FAA_REMA
Year
1979
1999
2019
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model
26.2_FAA_REMA
Year
1979
1999
2019
AK
BS LLS
AK
AI LLS
AK
GOA LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin

Phase Plane

$F/F_{35\%}$

$SSB/B_{35\%}$

1.5
1.0
0.5
0.0

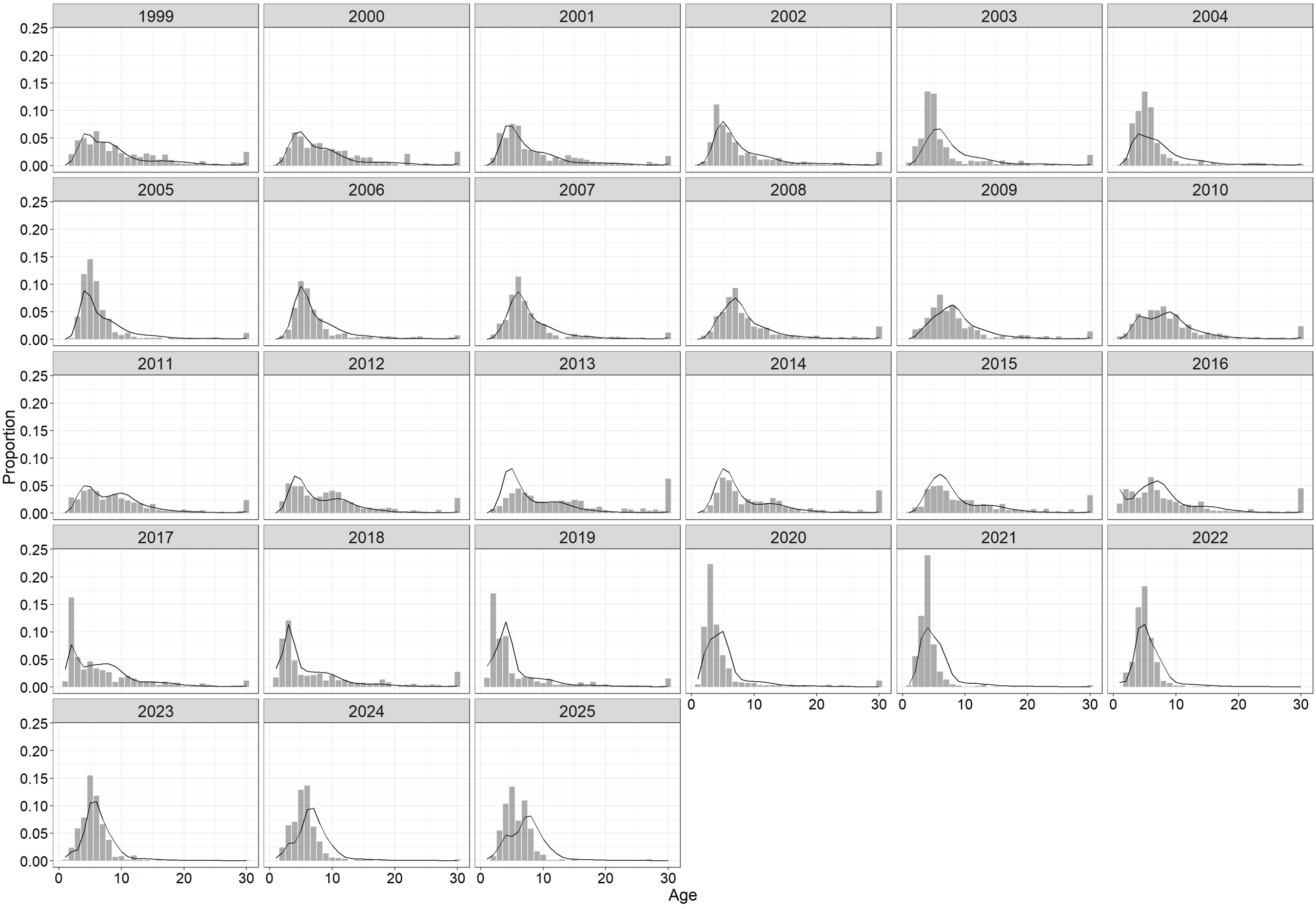
0.0
0.5
1.0
1.5
2.0
2.5

1960
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
2026

Year
2020
2000
1980
1960

Control Rule
OFL (F35%)
Max ABC (F40%)

Fixed Gear Fishery Age Comps Female AK



Fixed Gear Fishery Age Comps Male AK

1999 2000 2001 2002 2003 2004
2005 2006 2007 2008 2009 2010
2011 2012 2013 2014 2015 2016
2017 2018 2019 2020 2021 2022
2023 2024 2025

Proportion

0.15
0.10
0.05
0.00

0 10 20 30

Age

Fixed Gear Fishery Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=1.00
(0.97-1.03)

Sample quantiles

Theoretical quantiles

Female

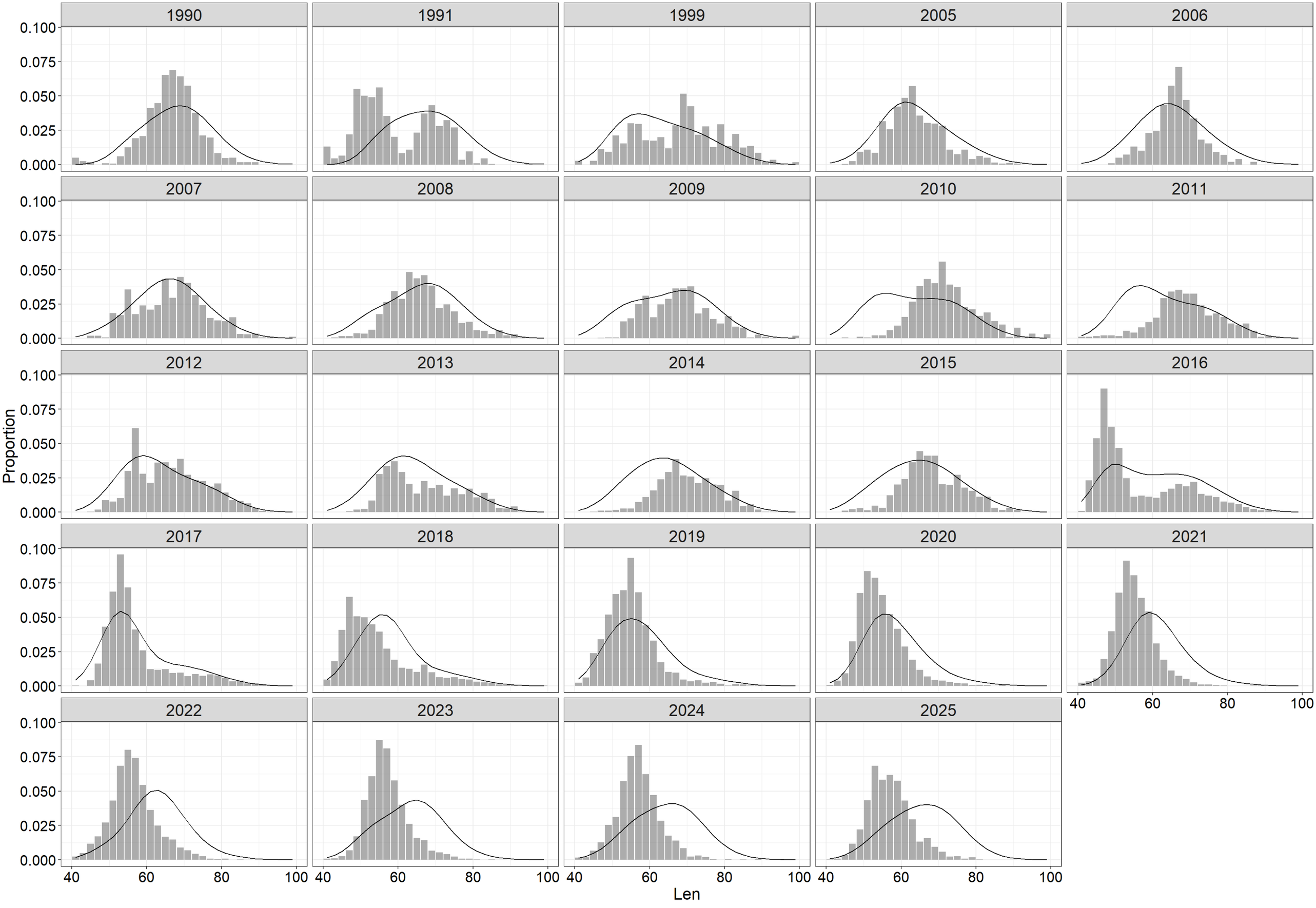
Male

AK

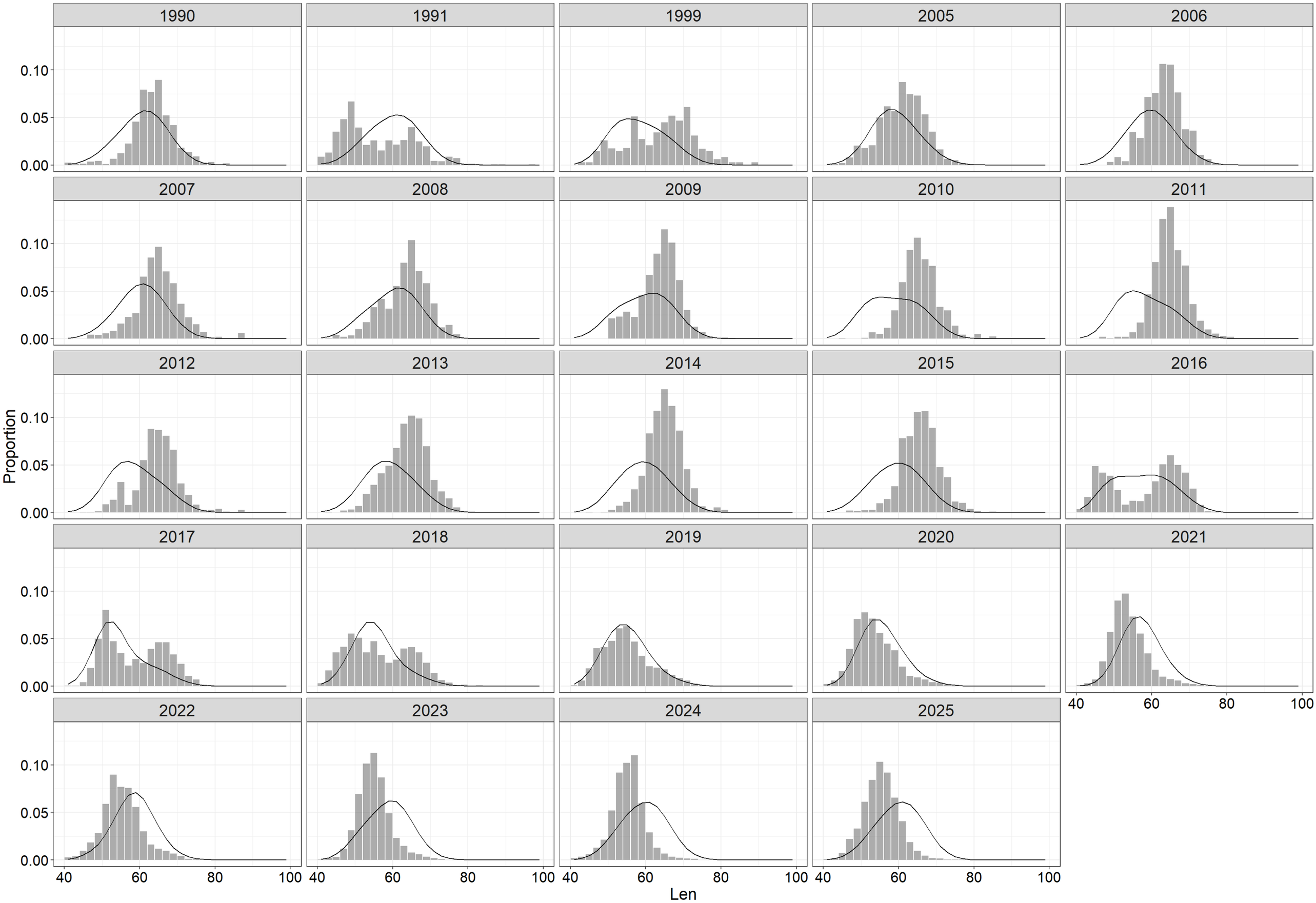
Proportions

Age

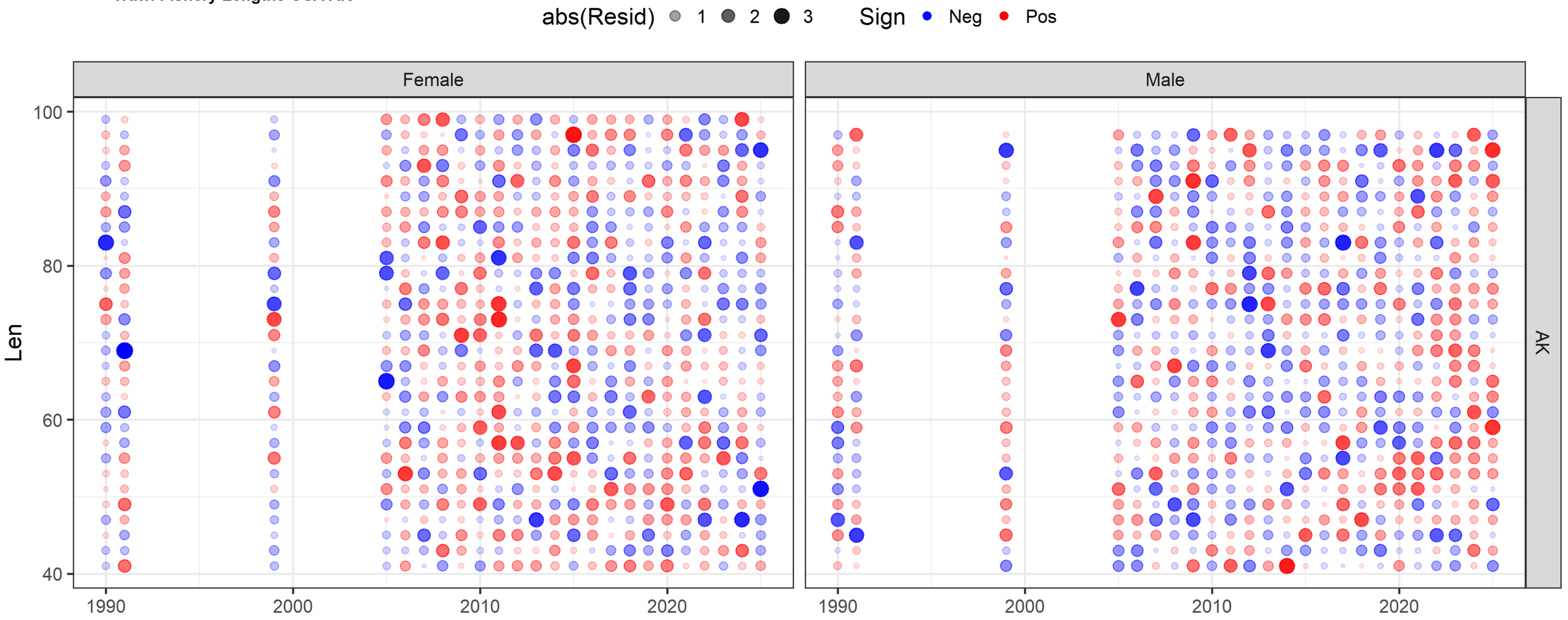
Trawl Fishery Lengths Female AK

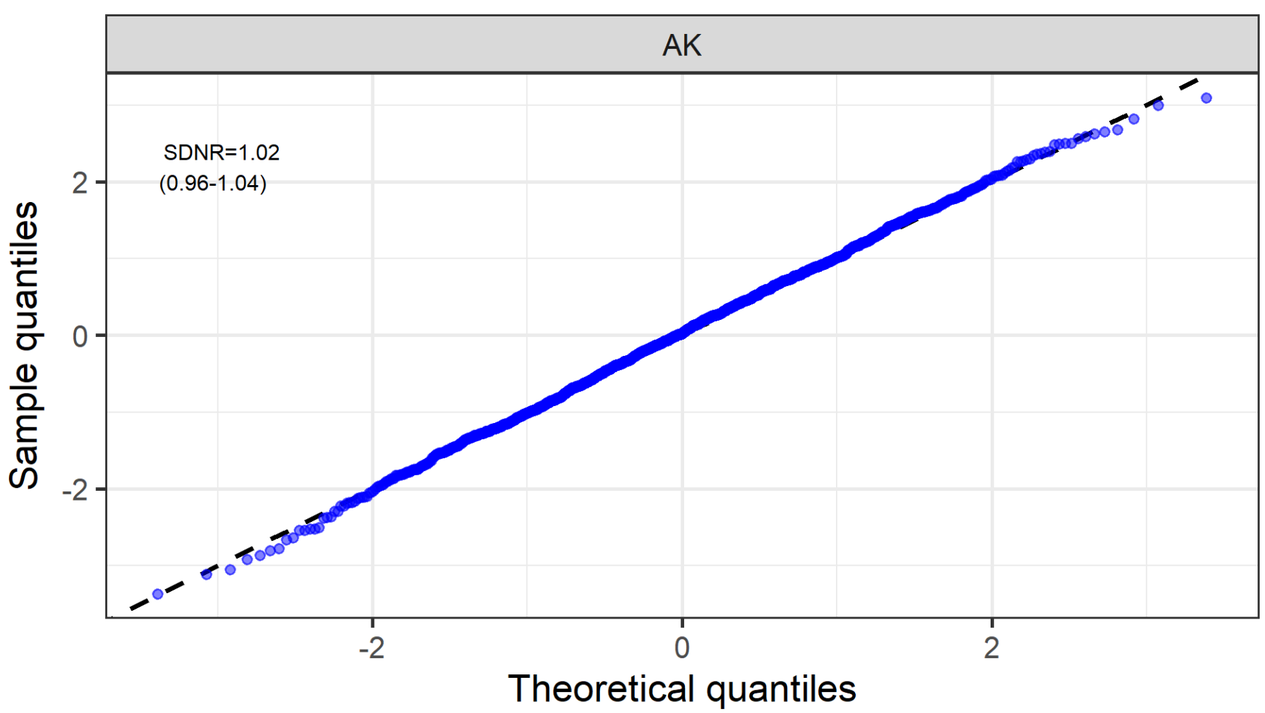


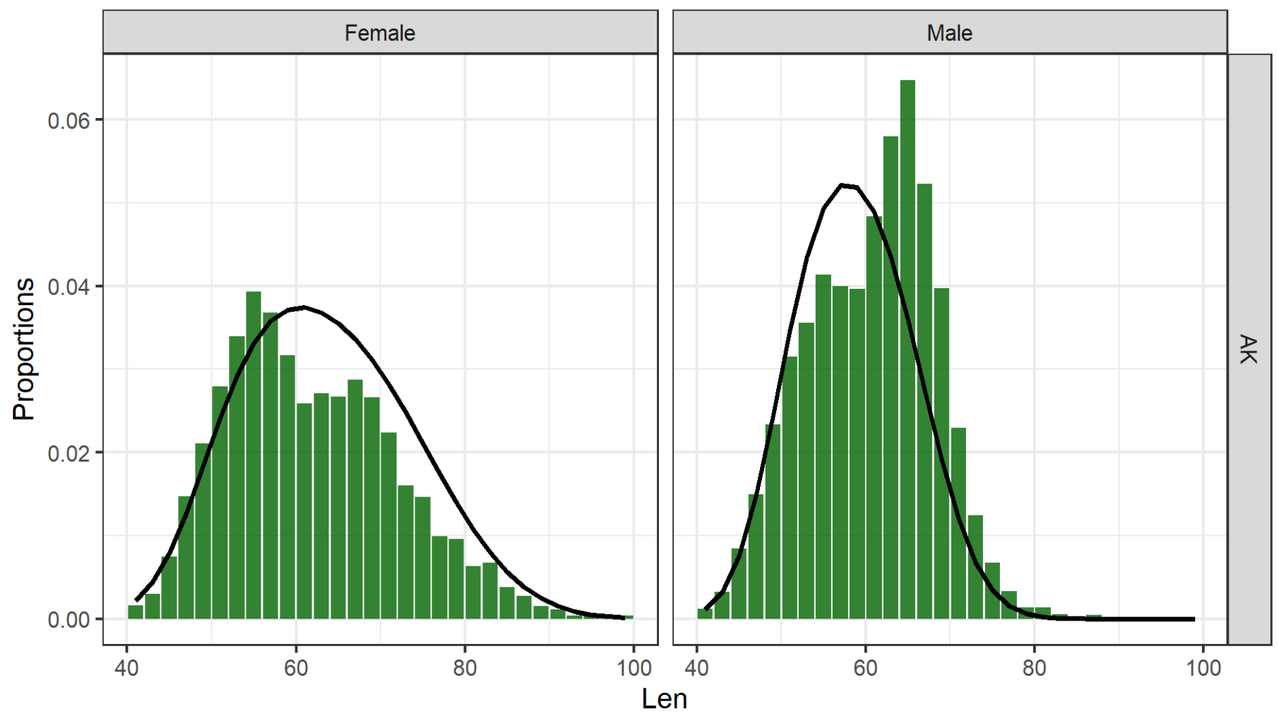
Trawl Fishery Lengths Male AK



Trawl Fishery Lengths OSA AK







BS LL Survey Age Comps Female AK

1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

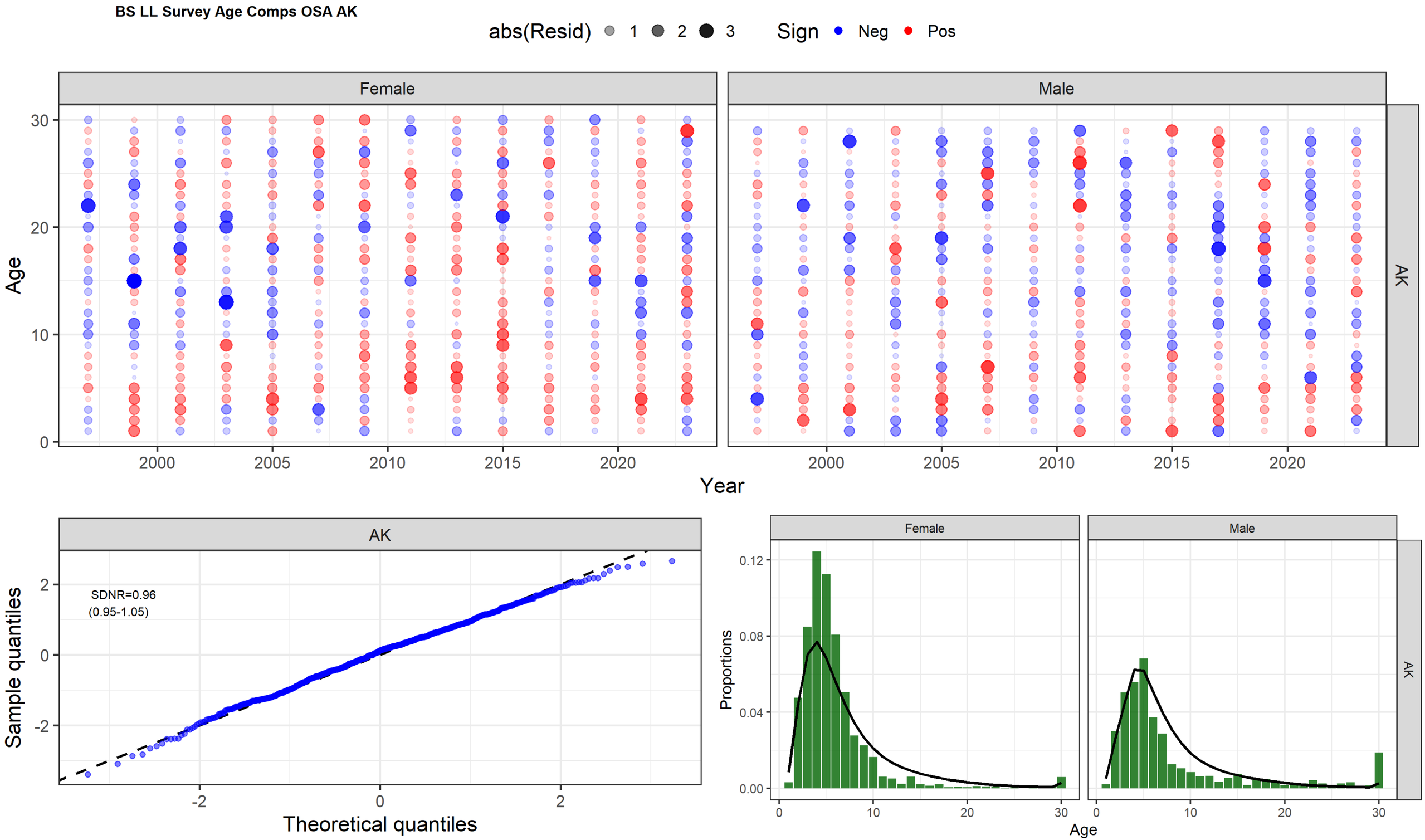
Proportion
Age

BS LL Survey Age Comps Male AK

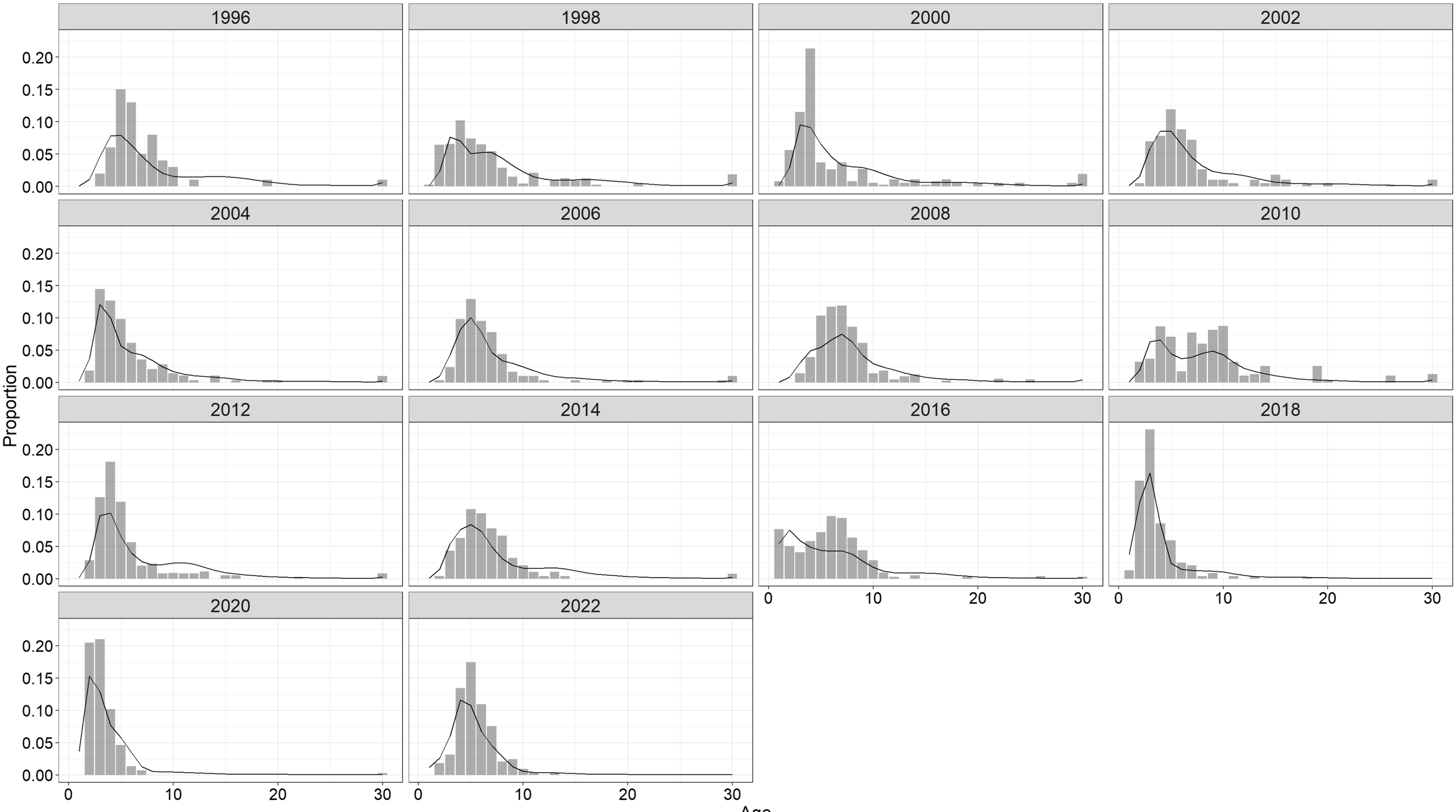
1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

Proportion
0.00 0.05 0.10 0.15
0 10 20 30
Age

BS LL Survey Age Comps OSA AK



AI LL Survey Age Comps Female AK



AI LL Survey Age Comps Male AK

1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion
0.15 0.10 0.05 0.00
0 10 20 30
Age

AI LL Survey Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=0.98
(0.95-1.05)

Sample quantiles

Theoretical quantiles

Female

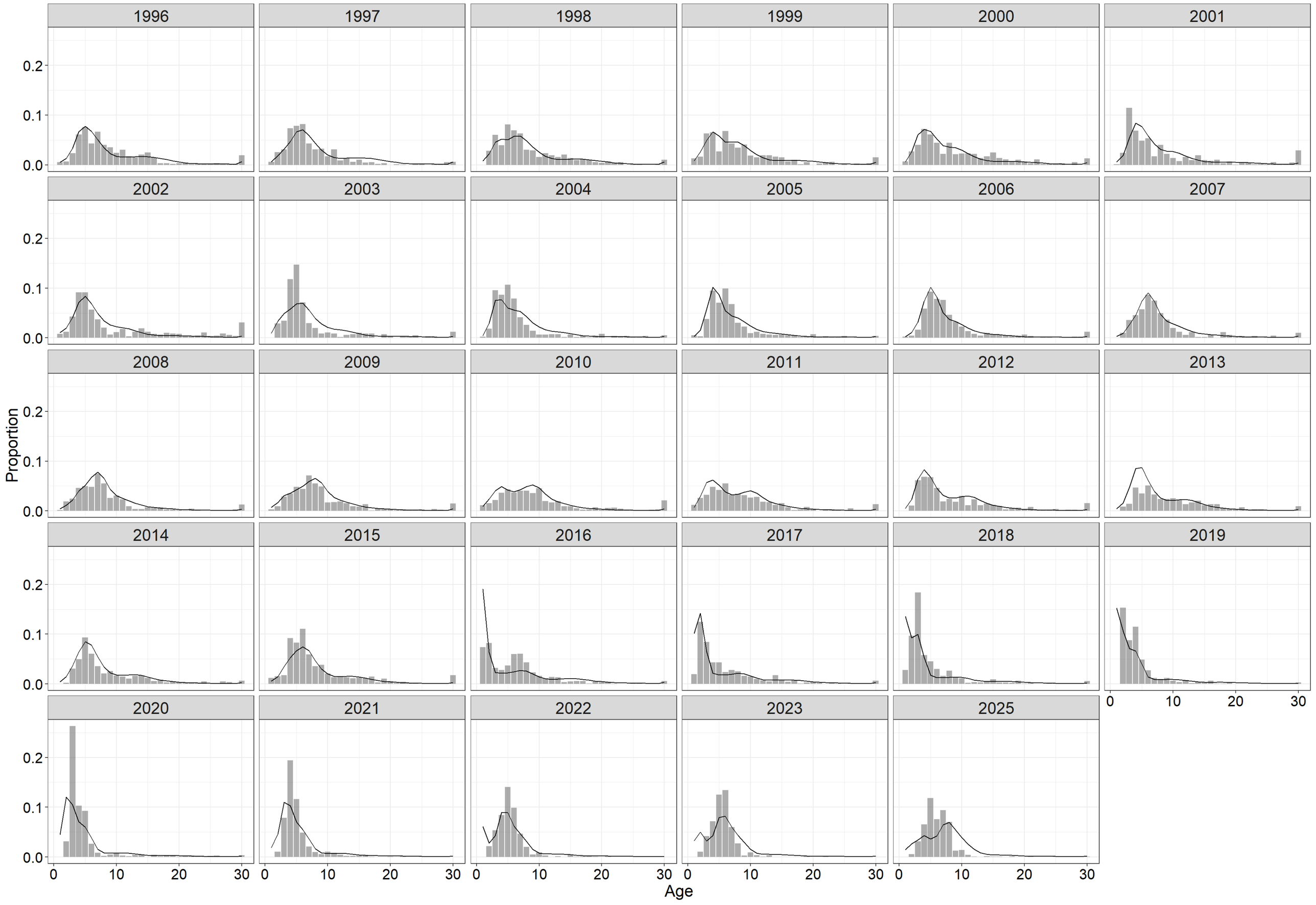
Male

AK

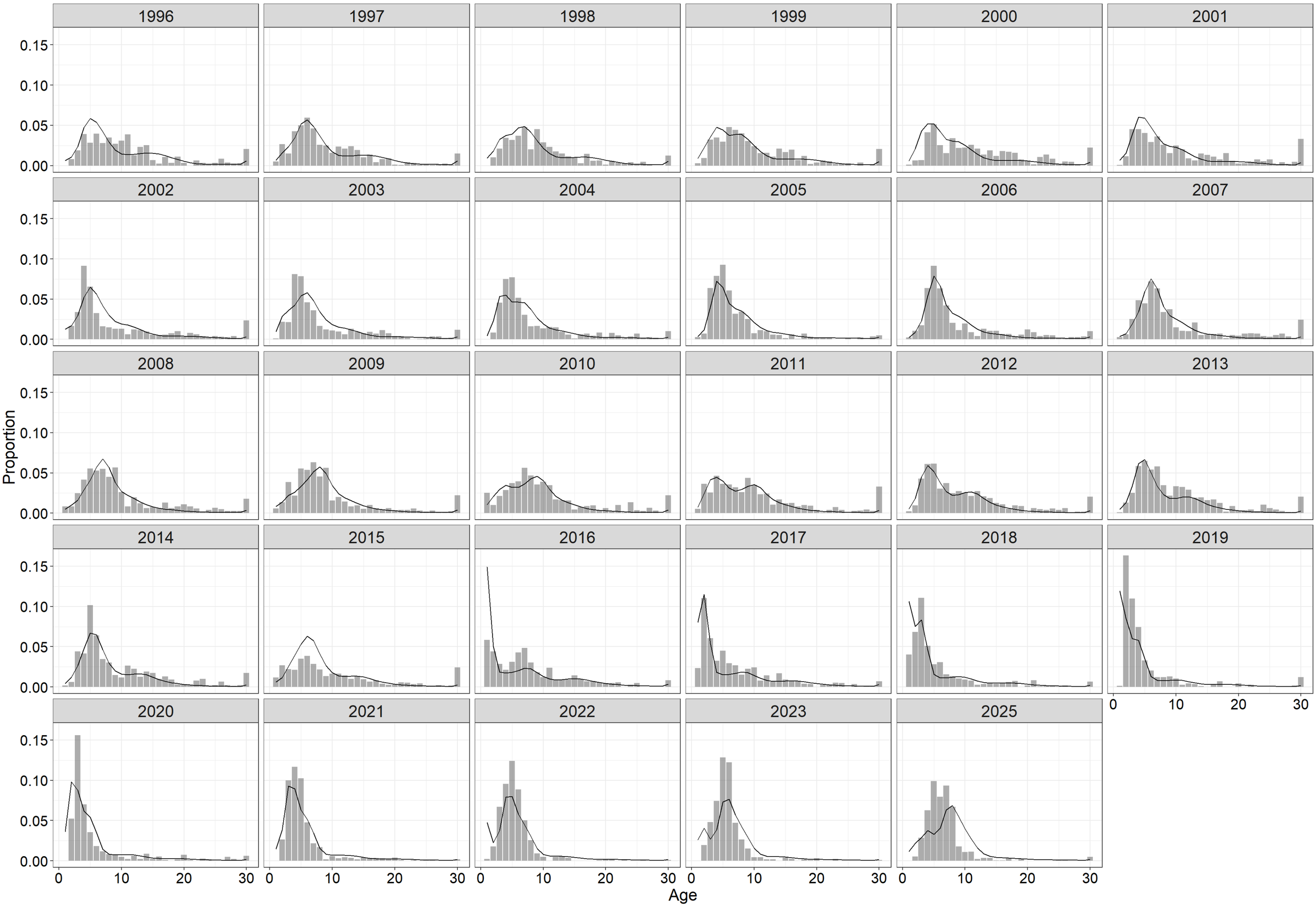
Proportions

Age

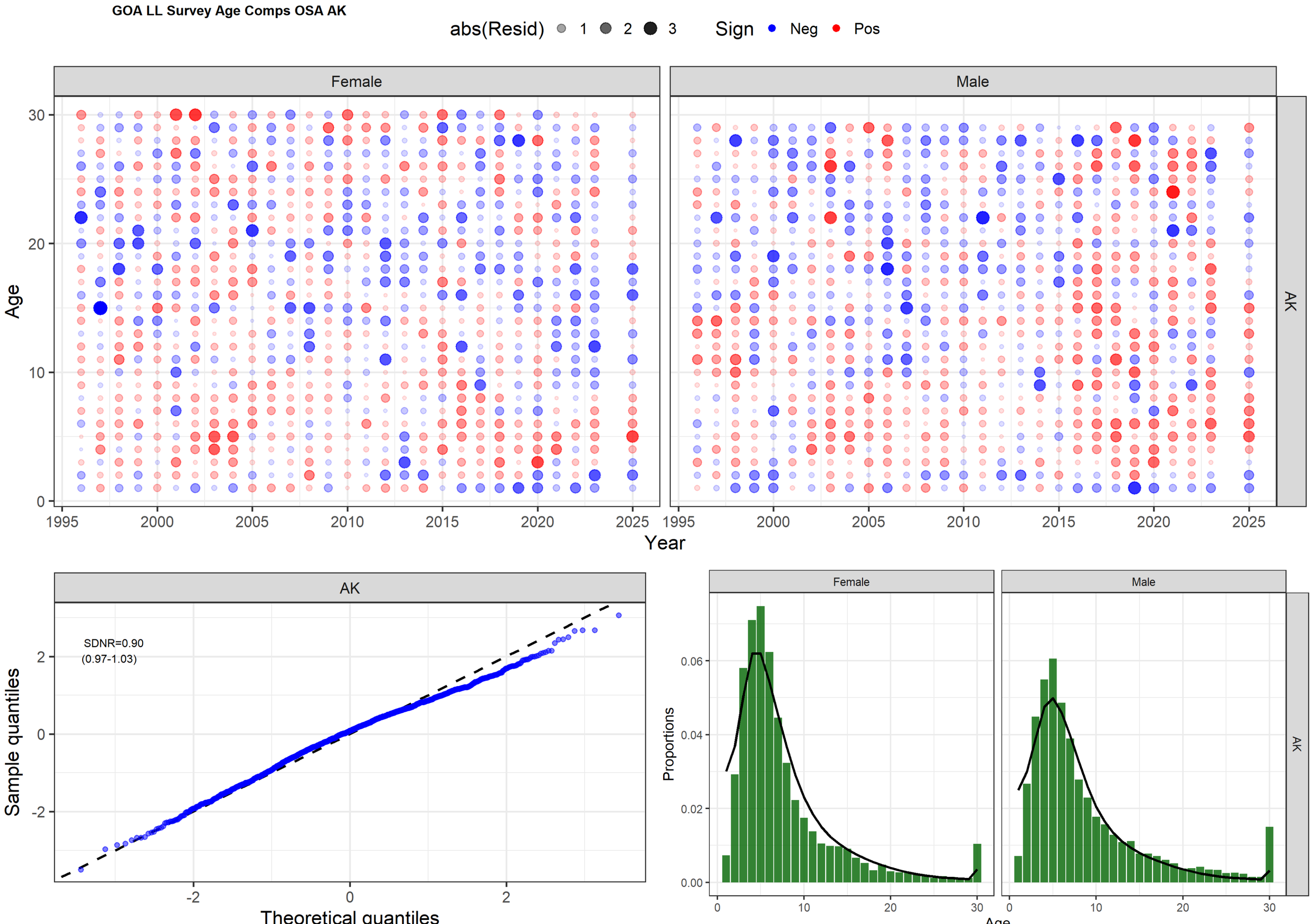
GOA LL Survey Age Comps Female AK

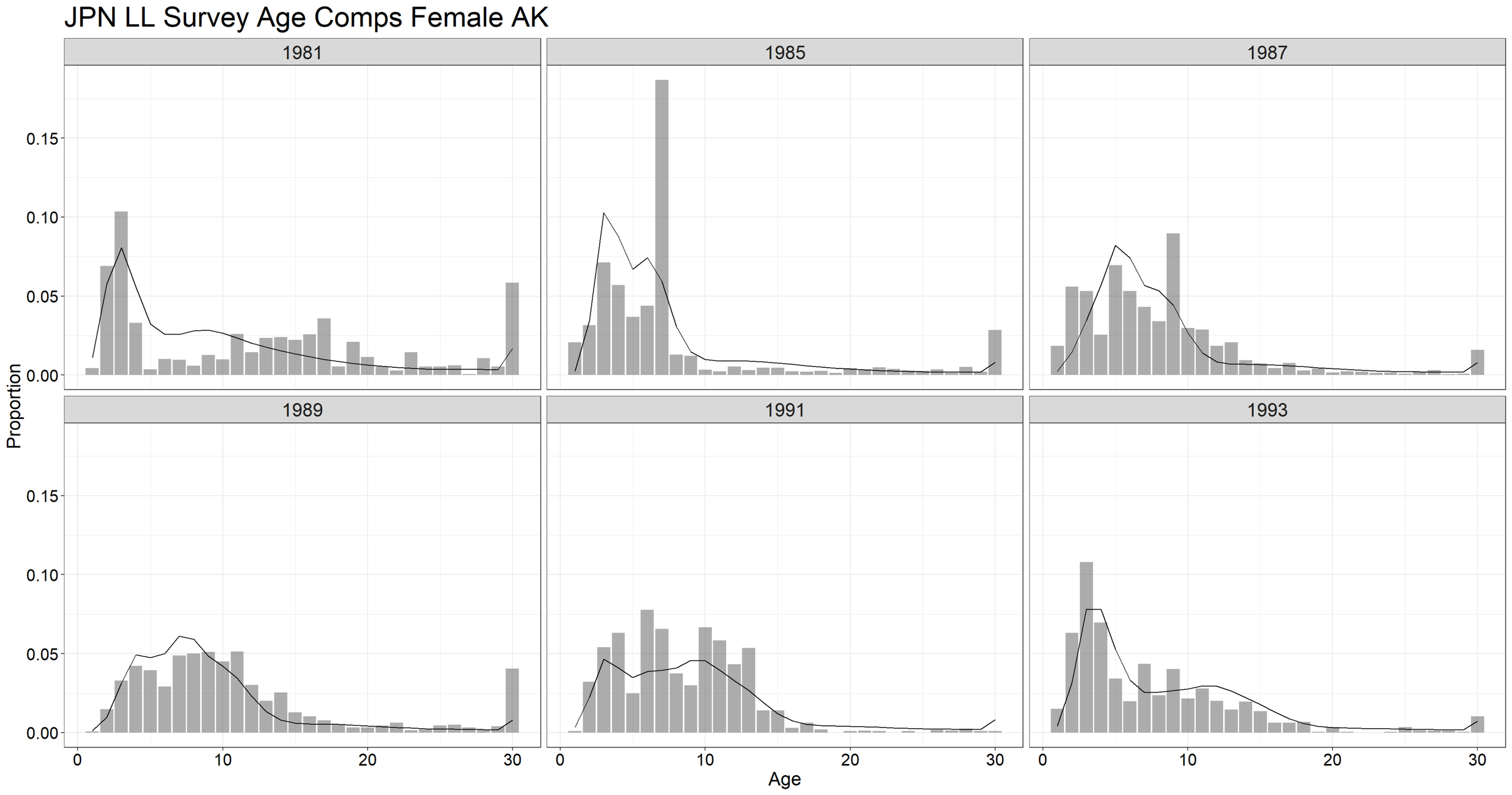


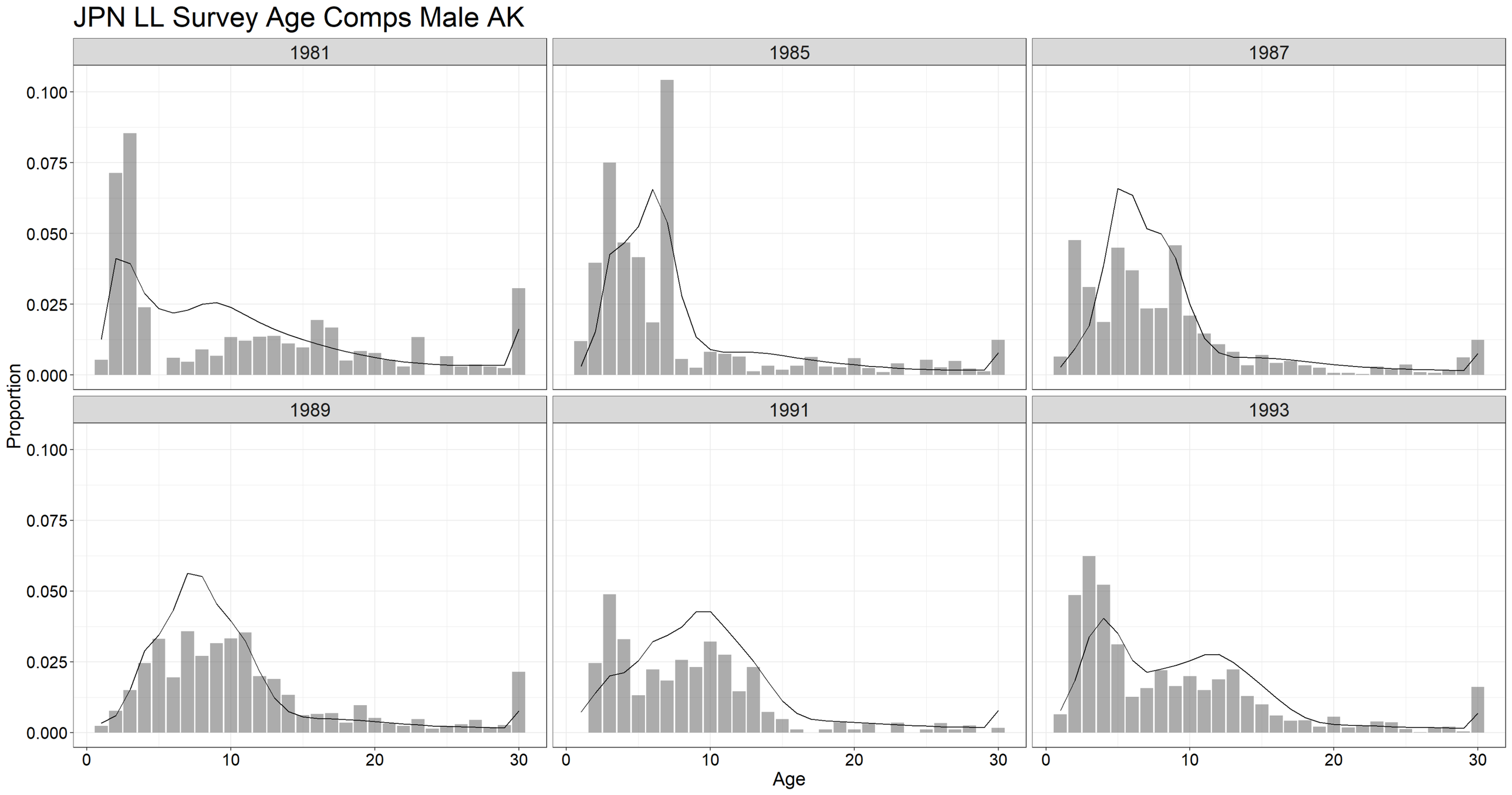
GOA LL Survey Age Comps Male AK

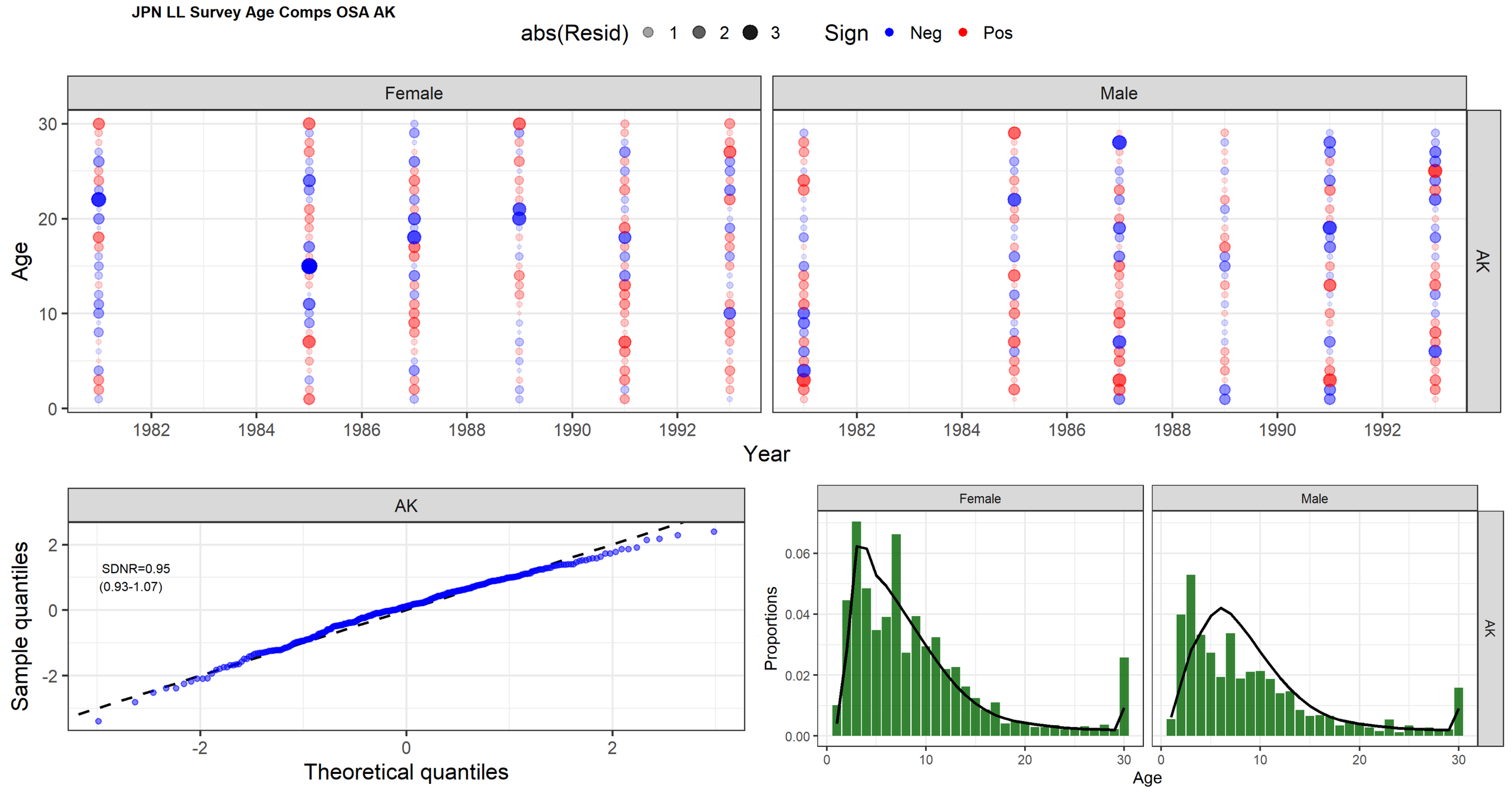


GOA LL Survey Age Comps OSA AK

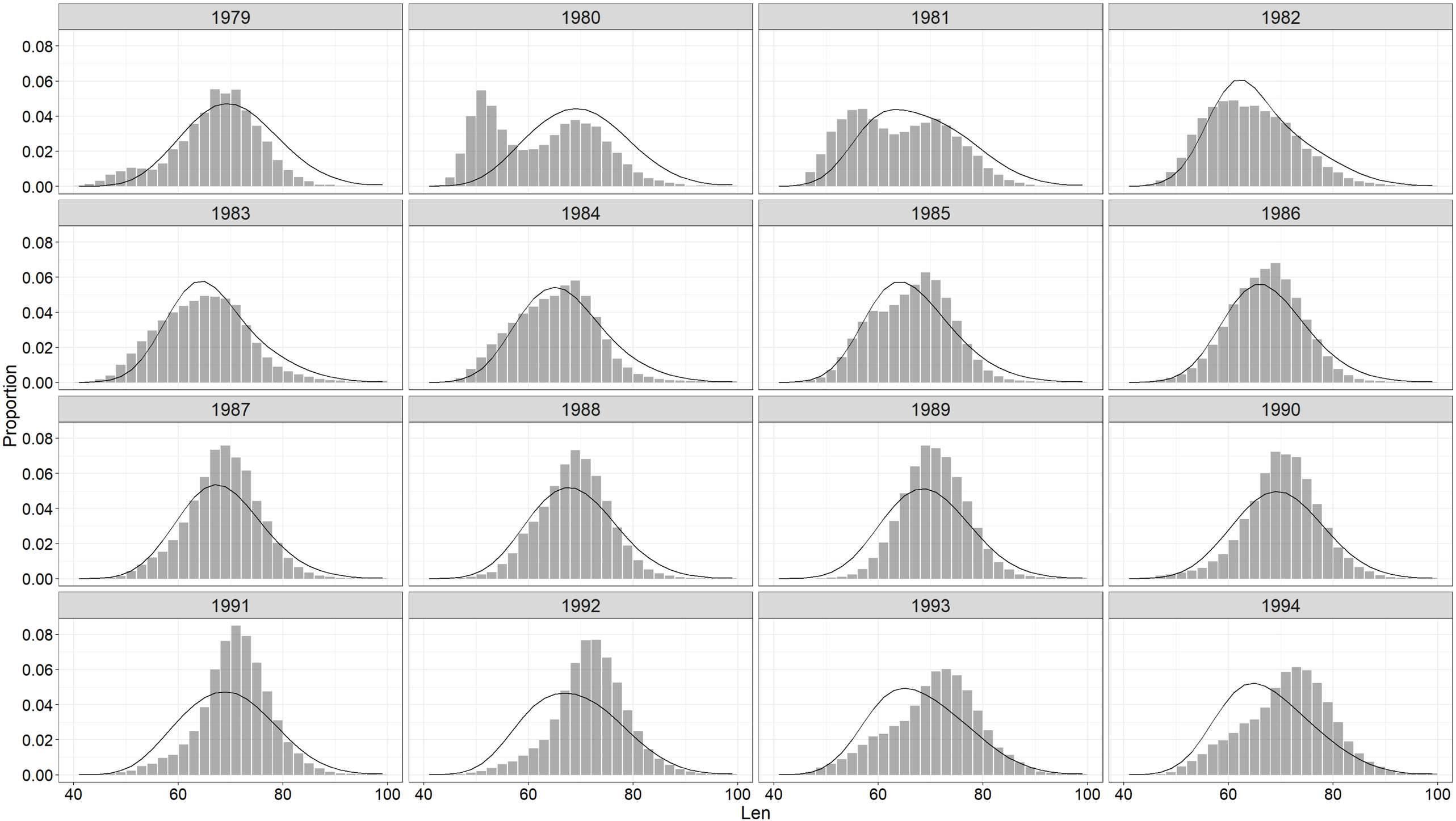



JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
0.100
0.075
0.050
0.025
0.000
Proportion
0
10
20
30
Age


JPN LL Survey Age Comps OSA AK
abs(Resid) 1 2 3
Sign Neg Pos
Female
Male
AK
Age
Year
SDNR=0.95
(0.93-1.07)
Sample quantiles
Theoretical quantiles
Proportions

JPN LL Survey Length Comps Female AK



JPN LL Survey Length Comps Male AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Len

100

80

60

40

1980

1985

1990

Year

AK

SDNR=0.90
(0.95-1.05)

Sample quantiles

Theoretical quantiles

2

0

-2

Female

Male

AK

Proportions

0.06

0.04

0.02

0.00

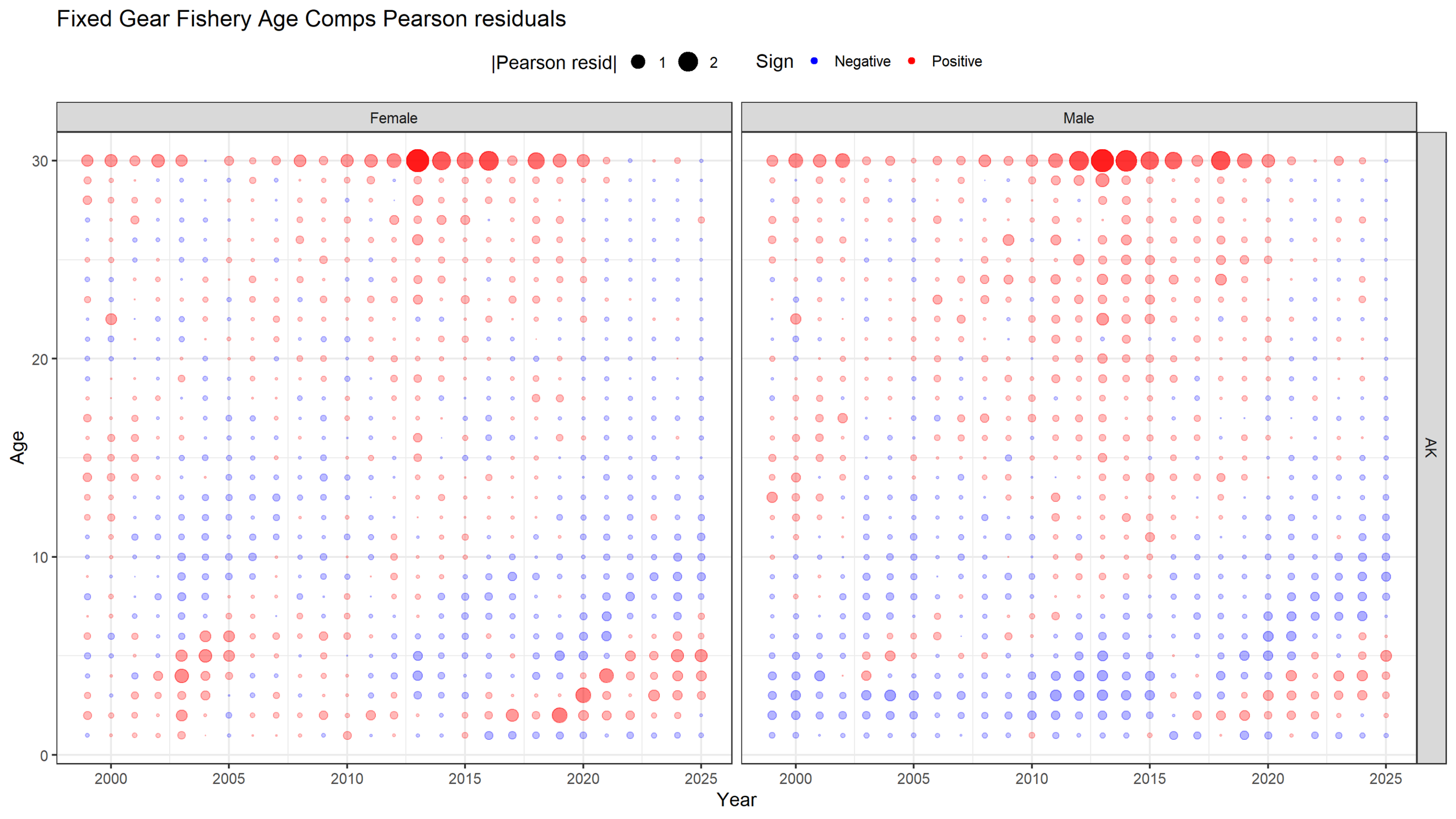
40

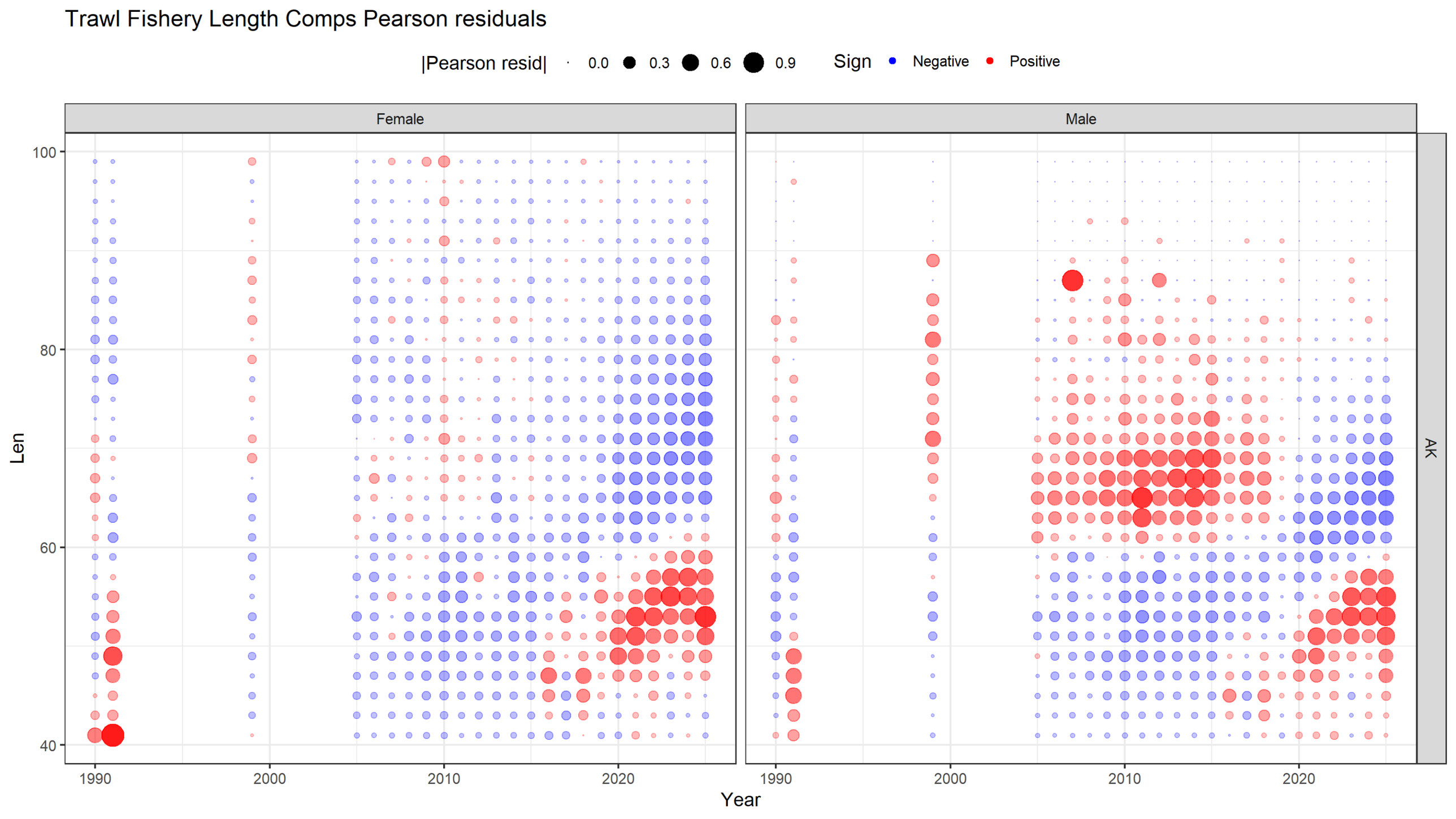
60

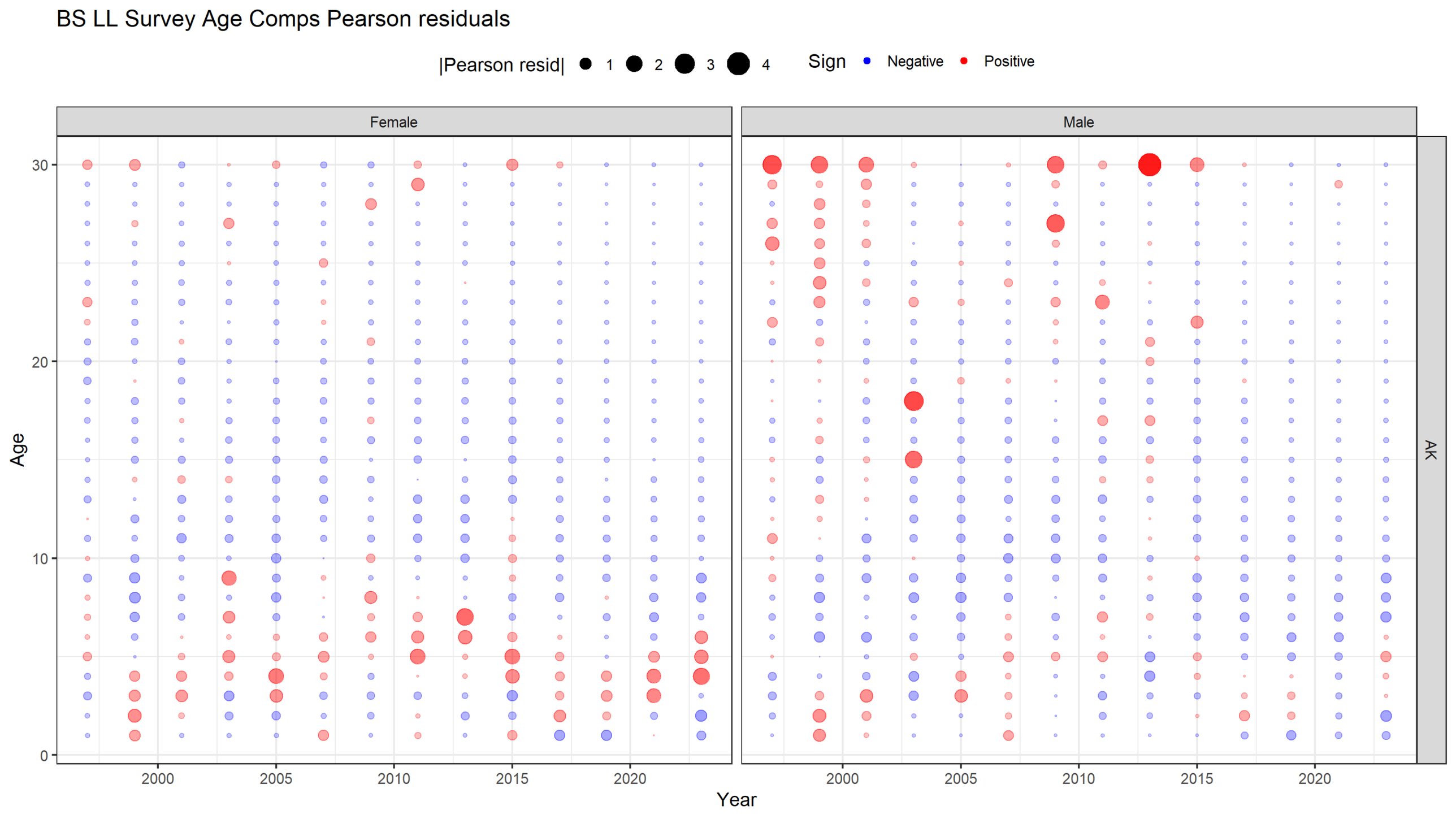
80

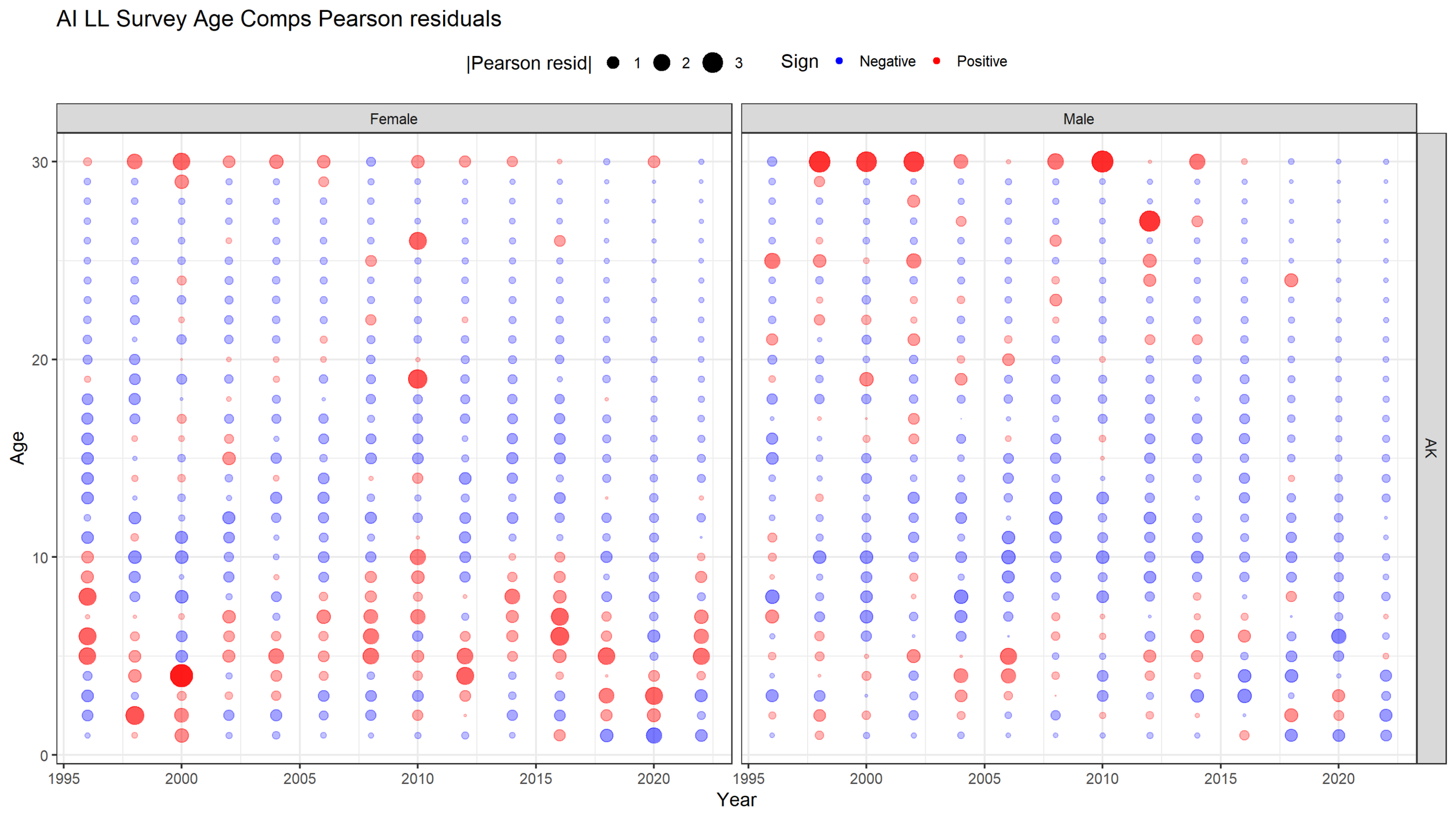
100

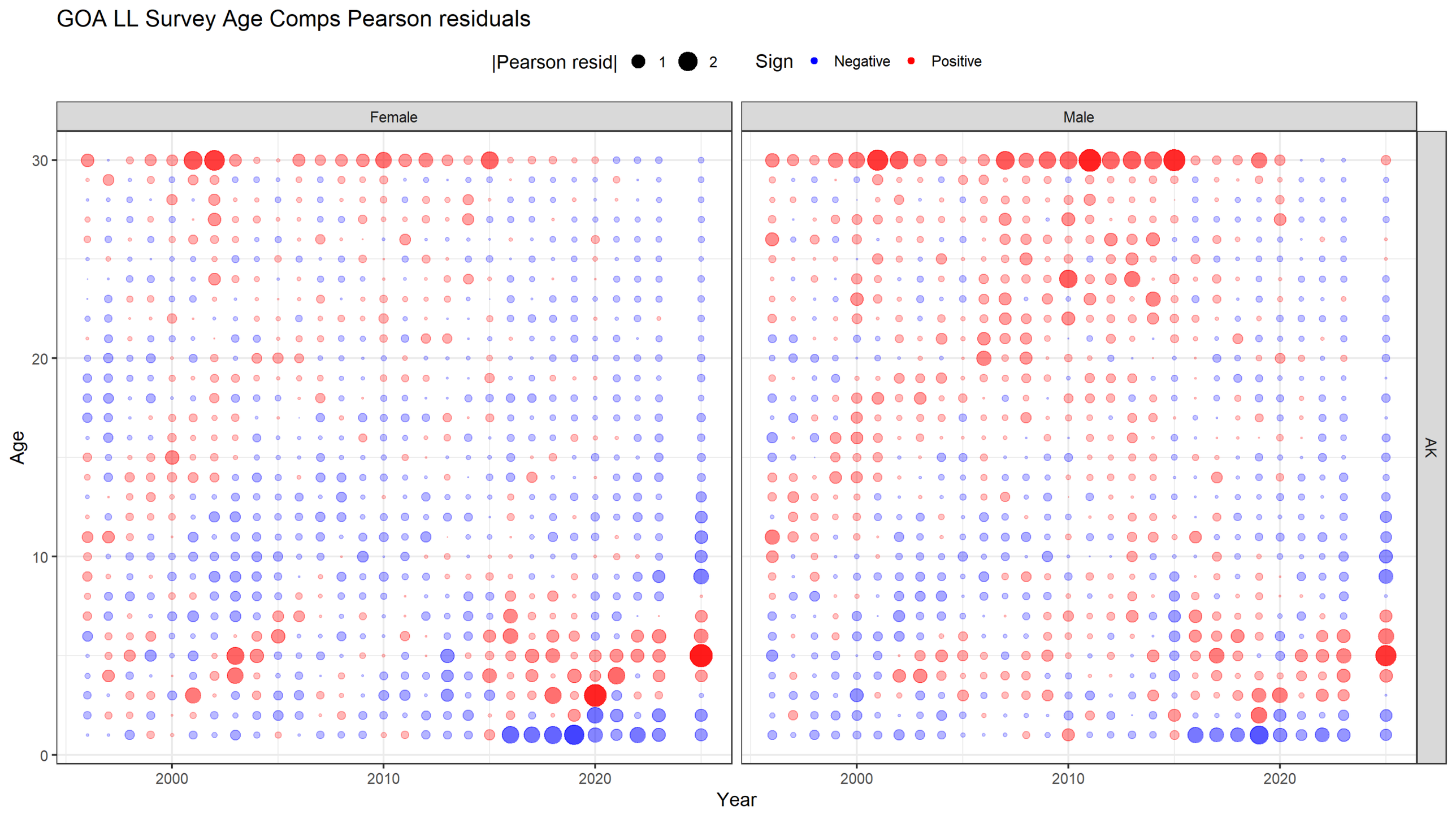
Len

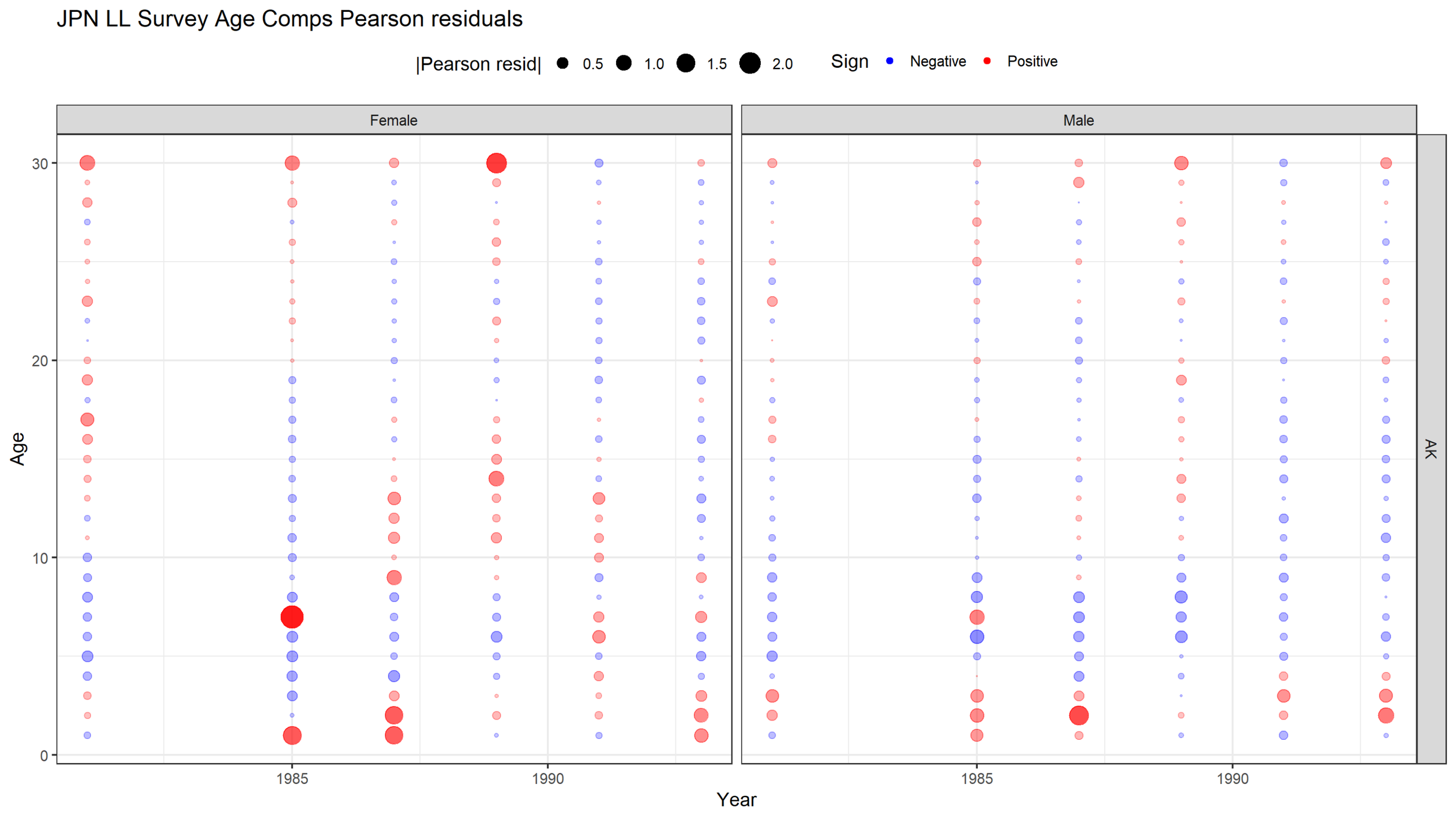

Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid| 1 2
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

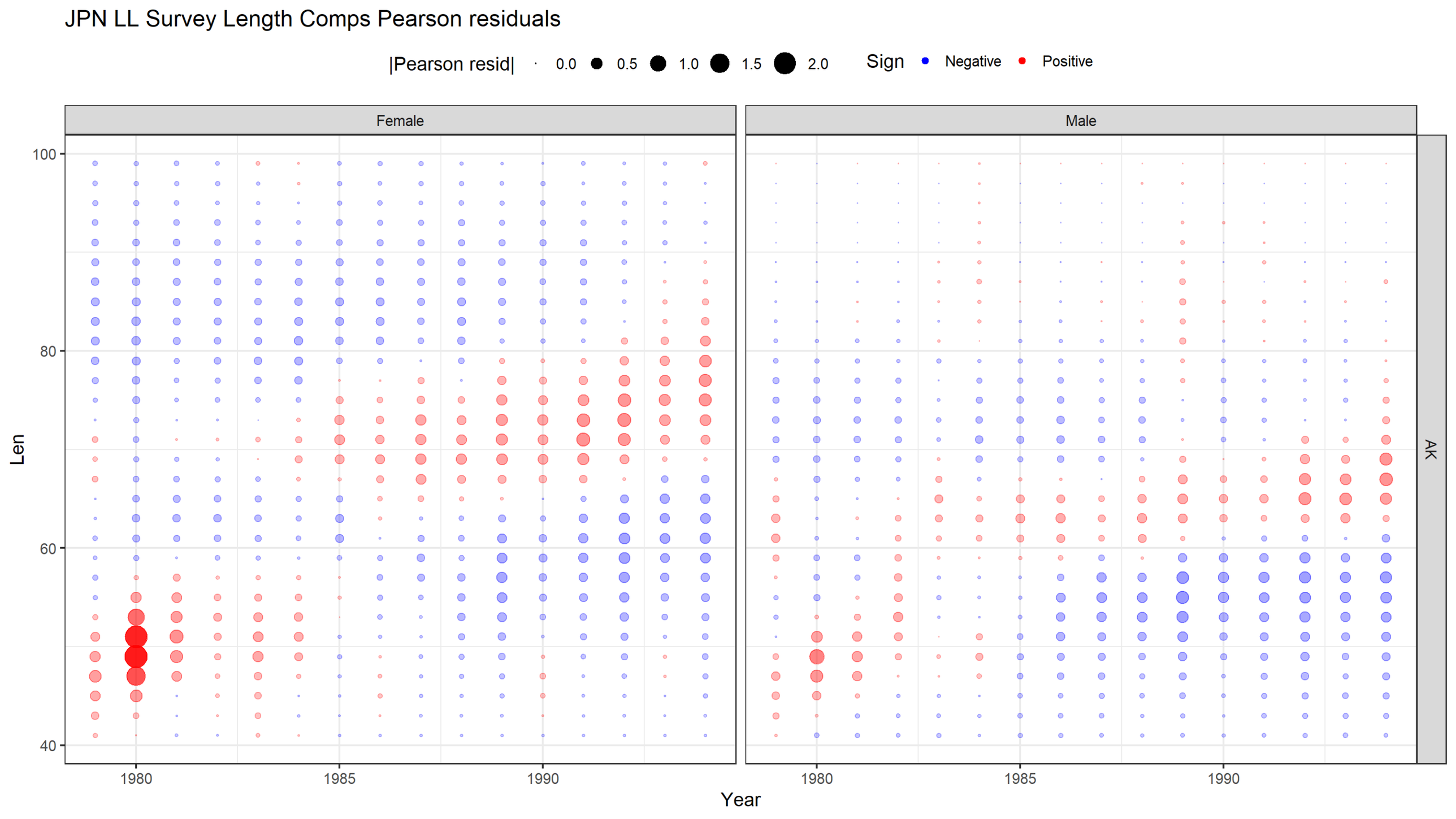

Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.0
0.3
0.6
0.9
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year


BS LL Survey Age Comps Pearson residuals
|Pearson resid|
1
2
3
4
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020

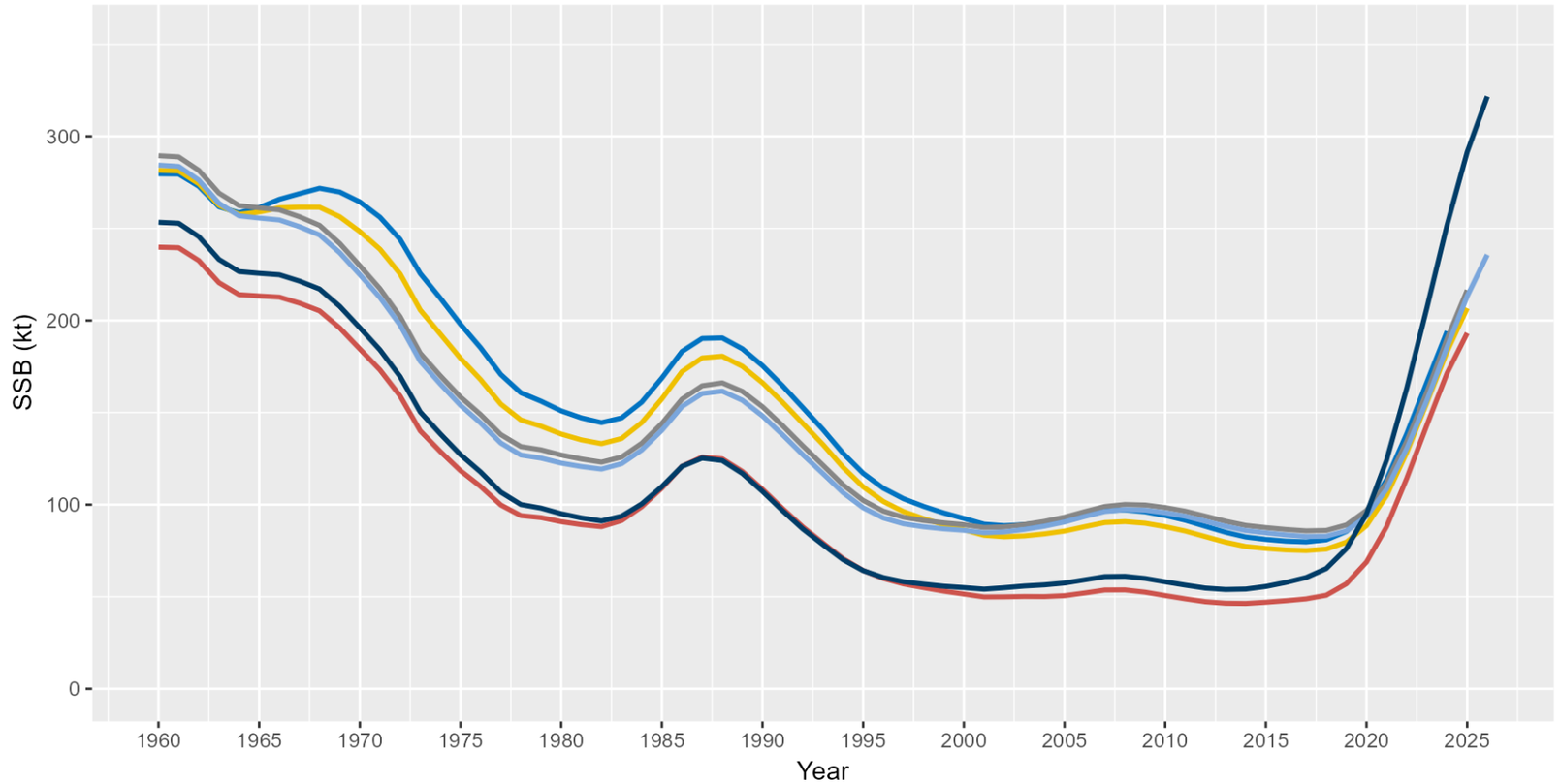

AI LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020


GOA LL Survey Age Comps Pearson residuals
|Pearson resid|
1
2
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

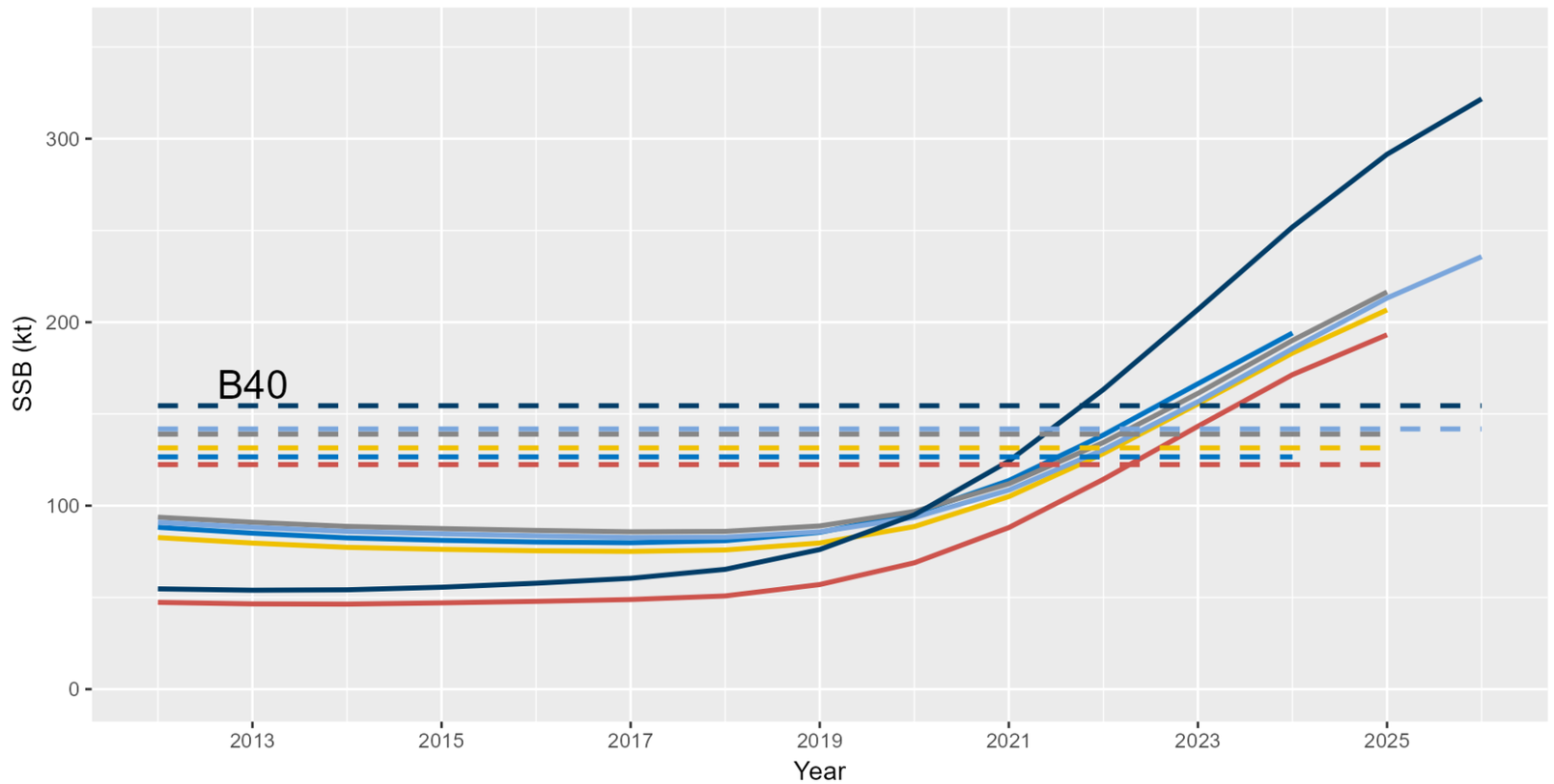

JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year

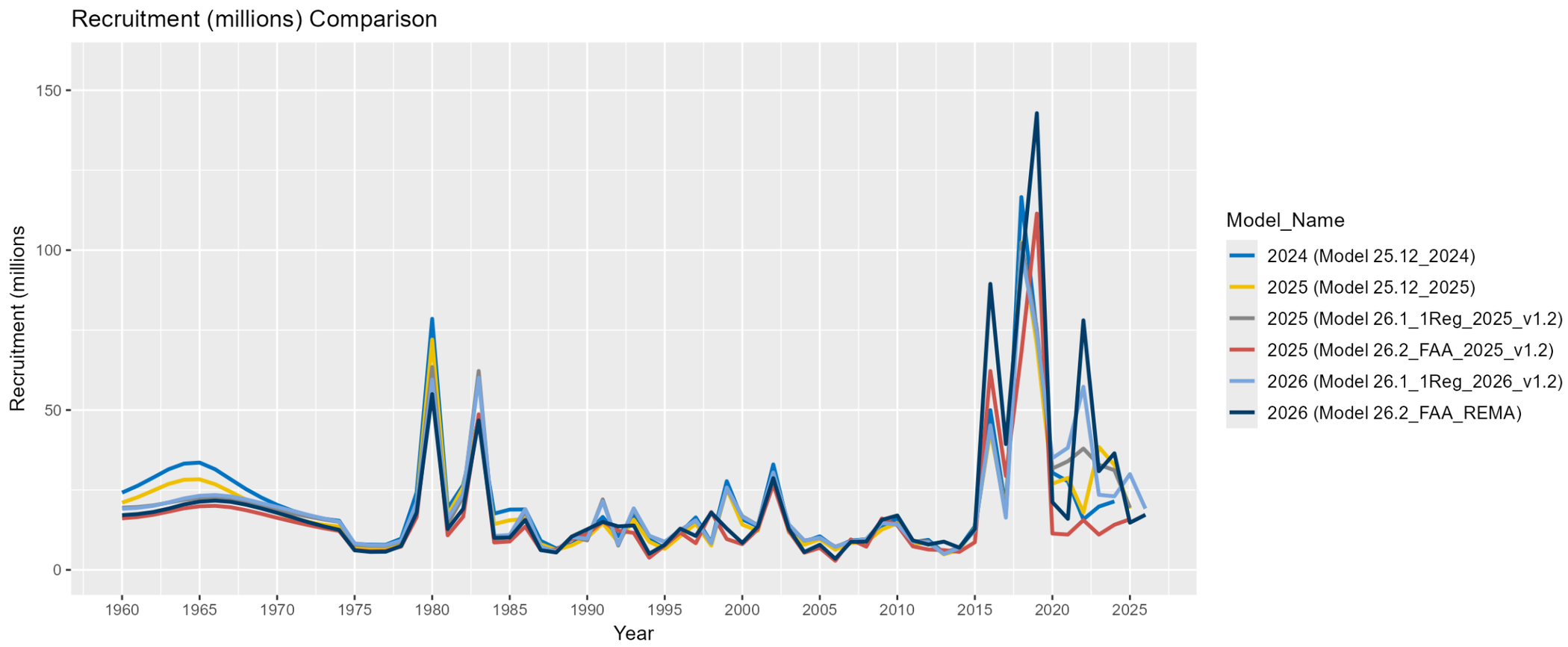

JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

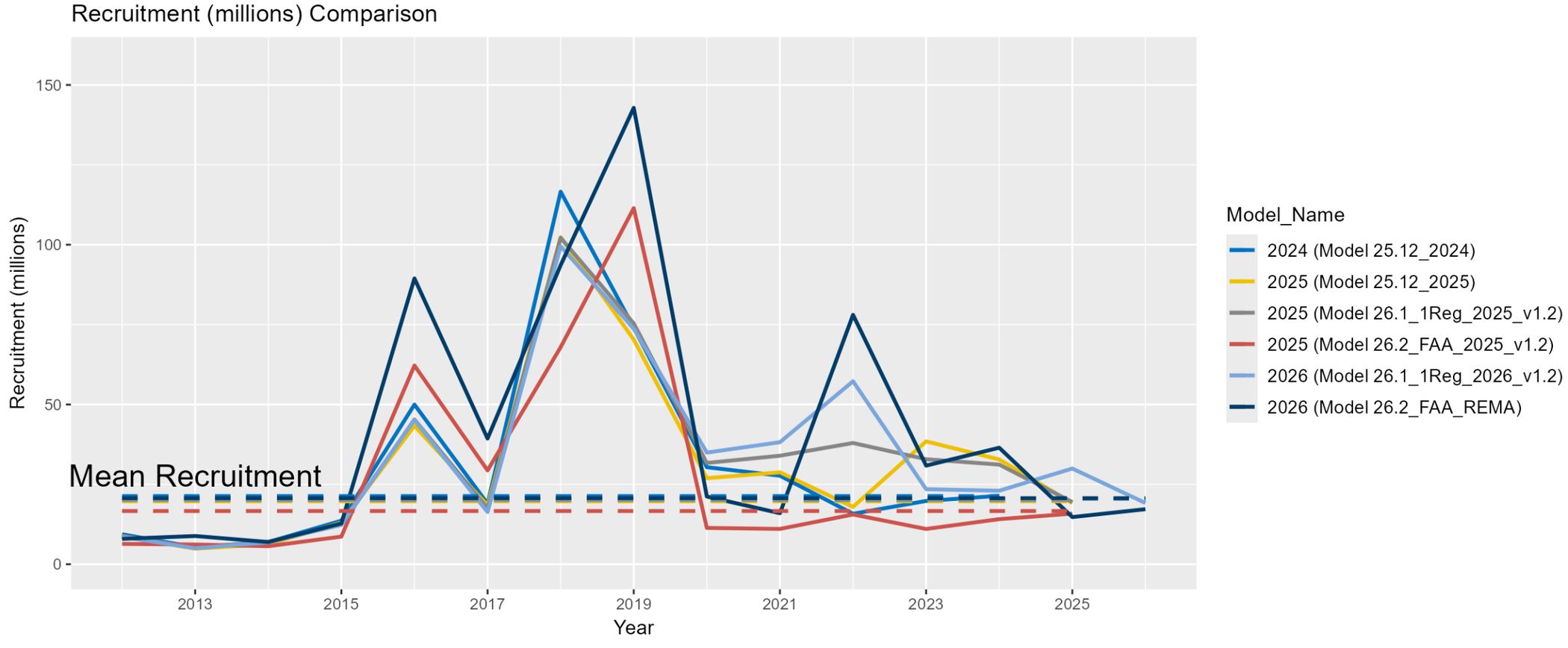
SSB (kt) Comparison

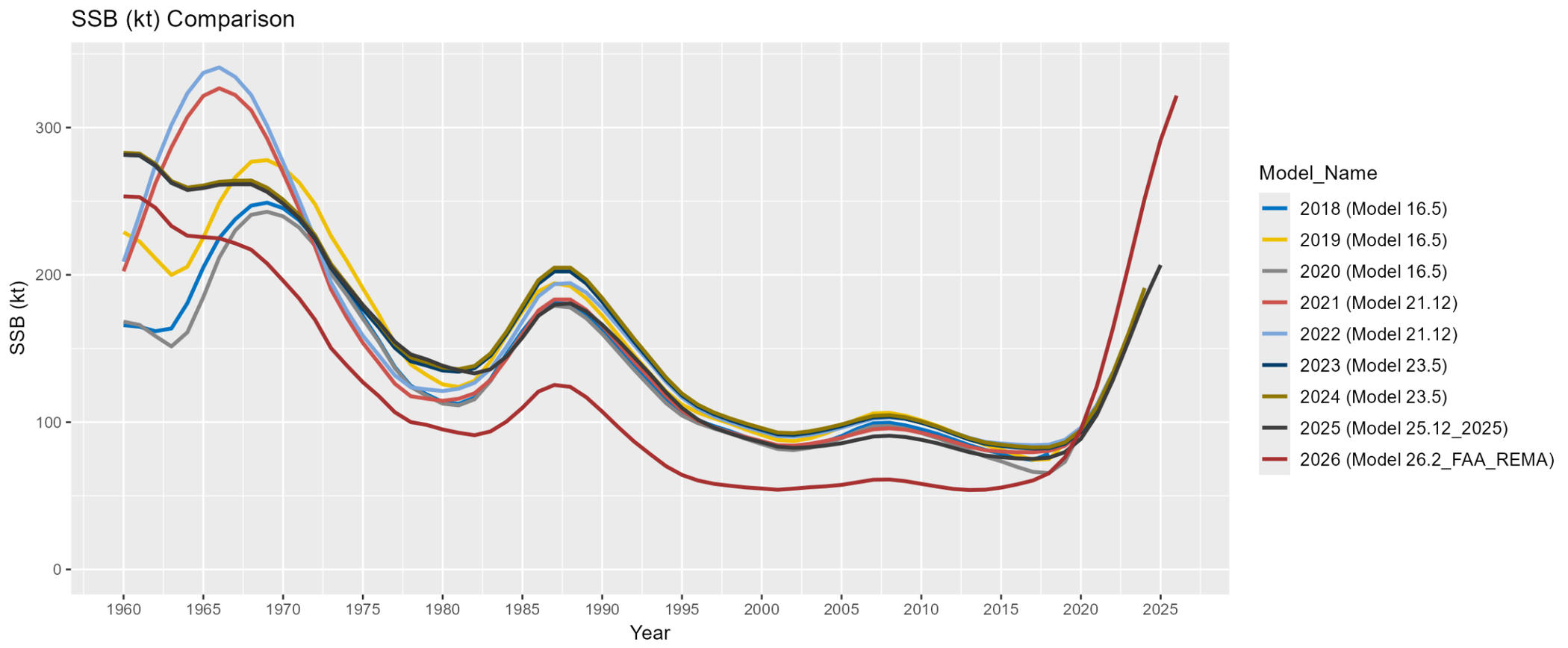


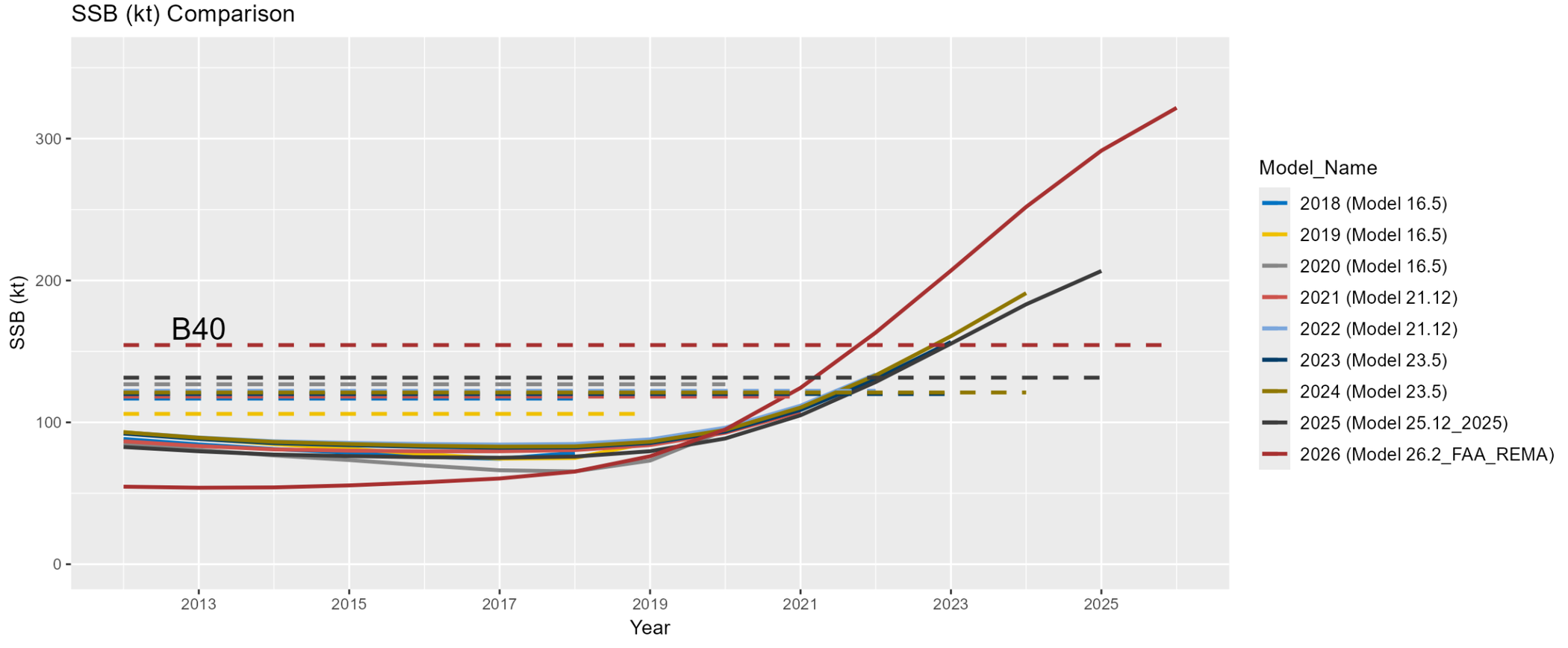
SSB (kt) Comparison

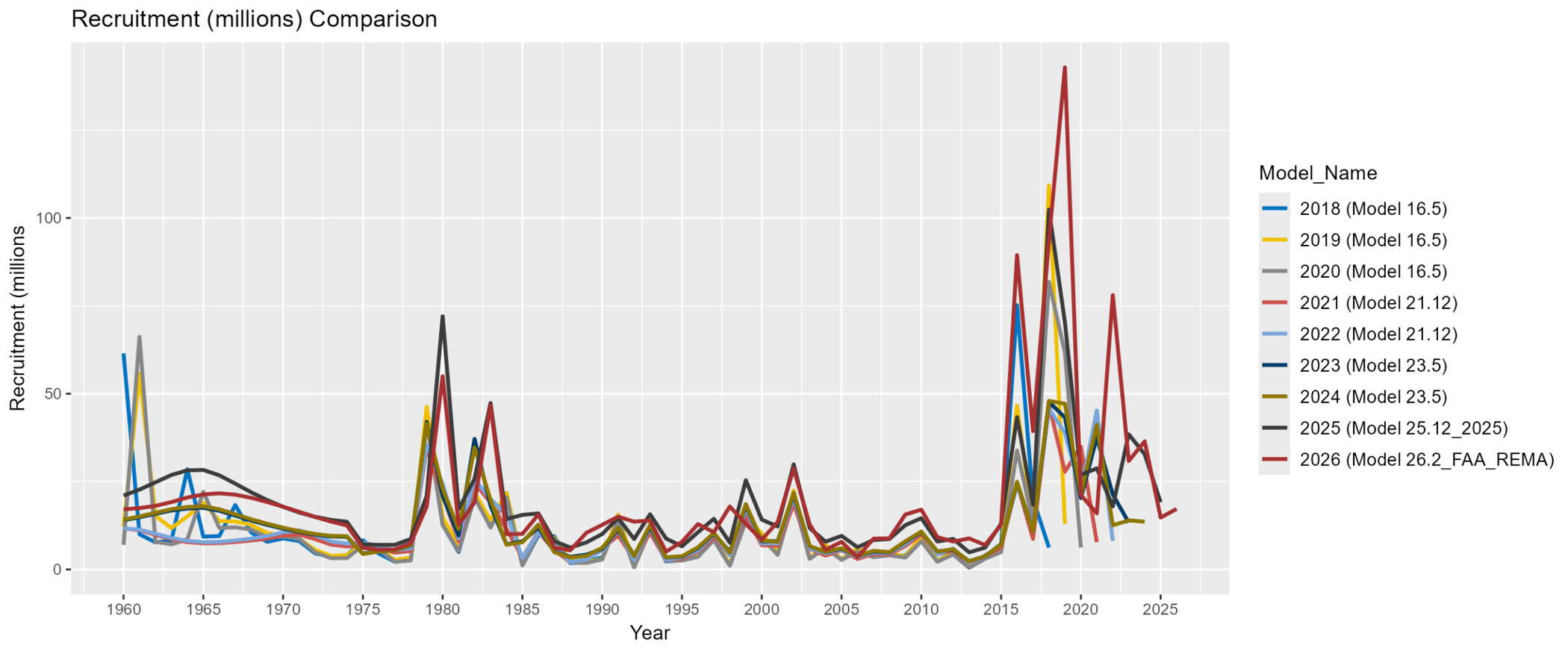


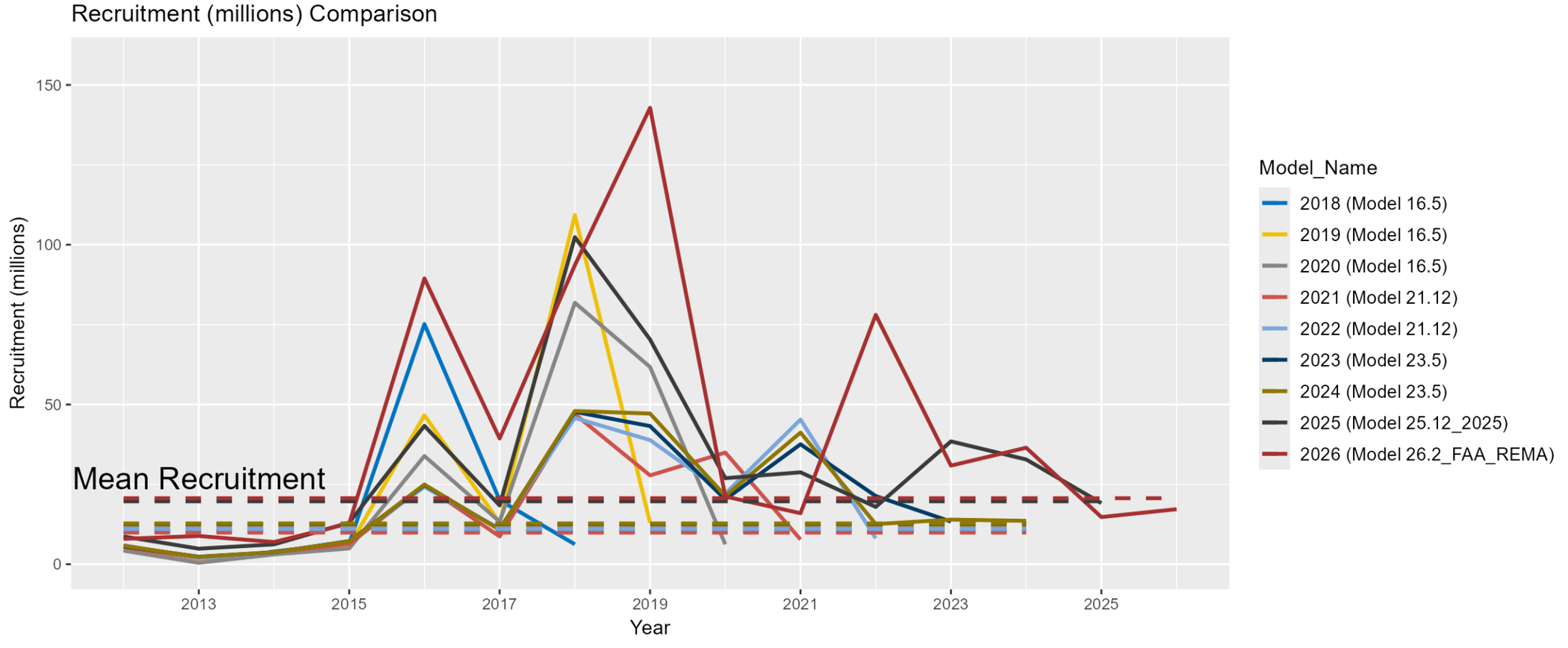

Recruitment (millions) Comparison
Recruitment (millions)
150
100
50
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_REMA)


Recruitment (millions) Comparison
Recruitment (millions)
150
100
50
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_REMA)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_REMA)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
B40
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_REMA)


Recruitment (millions) Comparison
Recruitment (millions
100
50
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_REMA)


Recruitment (millions) Comparison
Recruitment (millions)
150
100
50
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_REMA)

Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

AK

Mohns Rho: -0.3867

Mohns Rho: -0.0273

0.25
0.00
-0.25
-0.50
-0.75

Relative Difference from Terminal Year

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

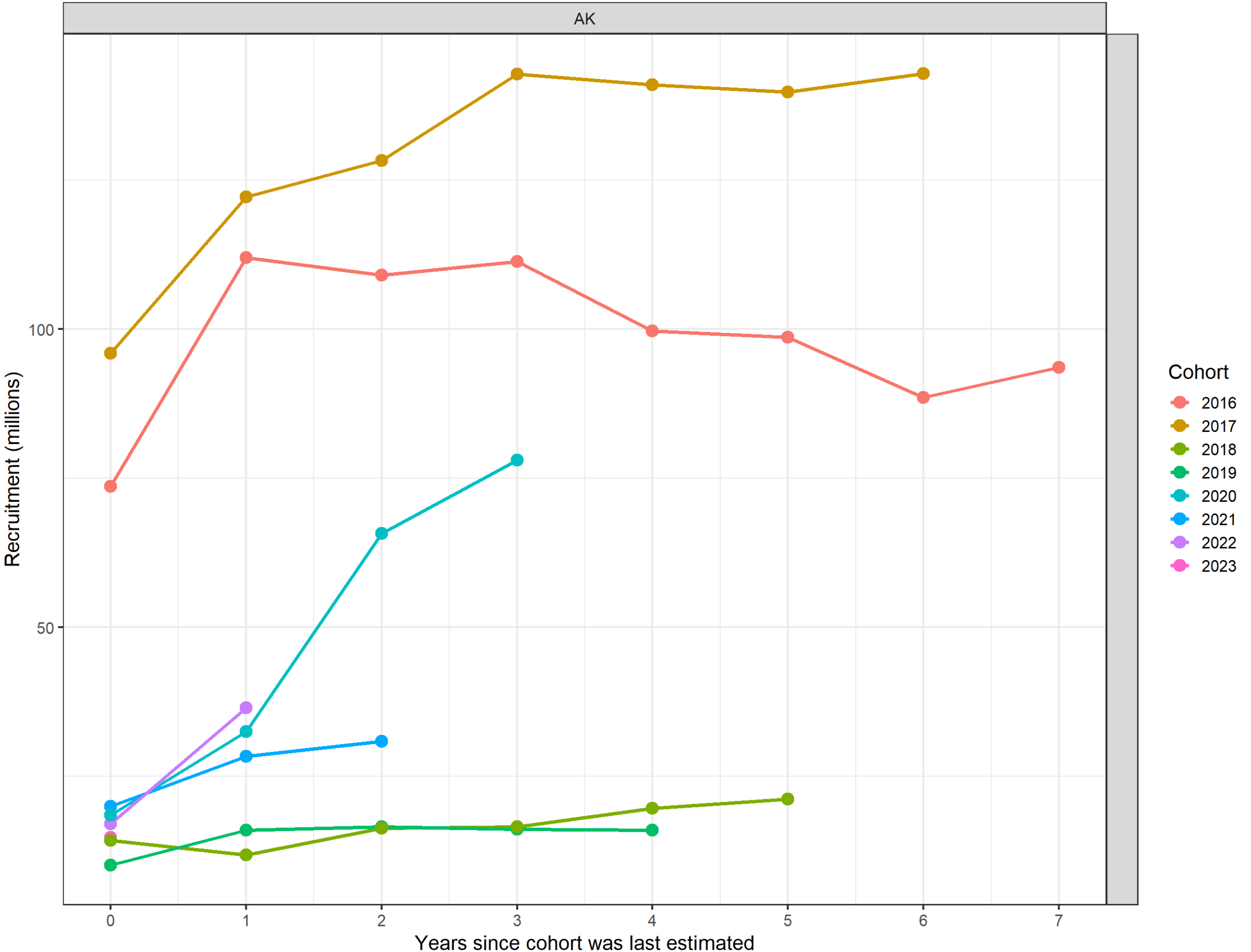
Value

100
50
0

300
200
100
0

1960
1980
2000
2020

Year


AK
Recruitment (millions)
100
50
0
1
2
3
4
5
6
7
Years since cohort was last estimated
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

Mohns Rho: -0.39

Mohns Rho: 1e-04

AK

Relative Difference from Terminal Year

0.3
0.0
-0.3
-0.6
-0.9

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

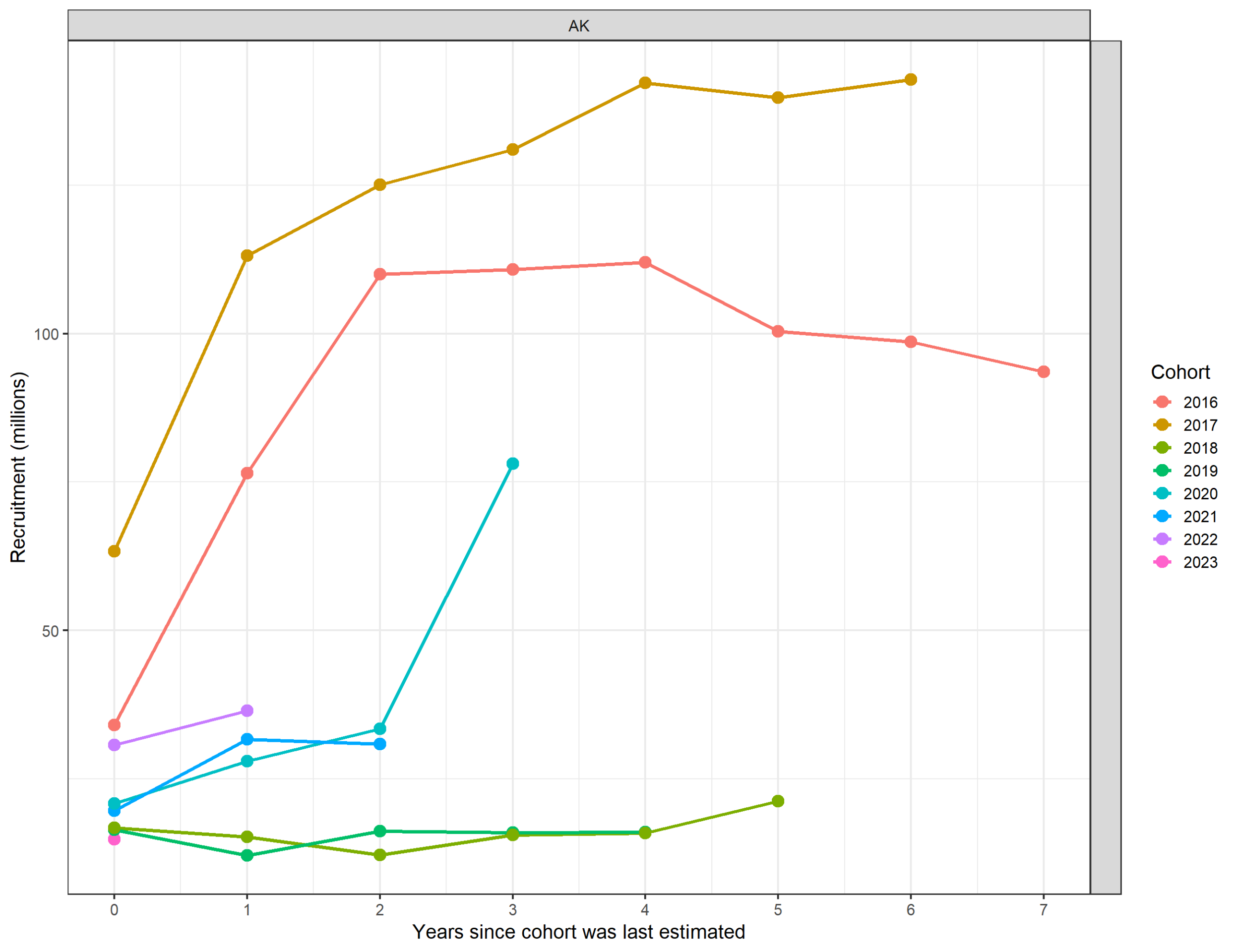
Value

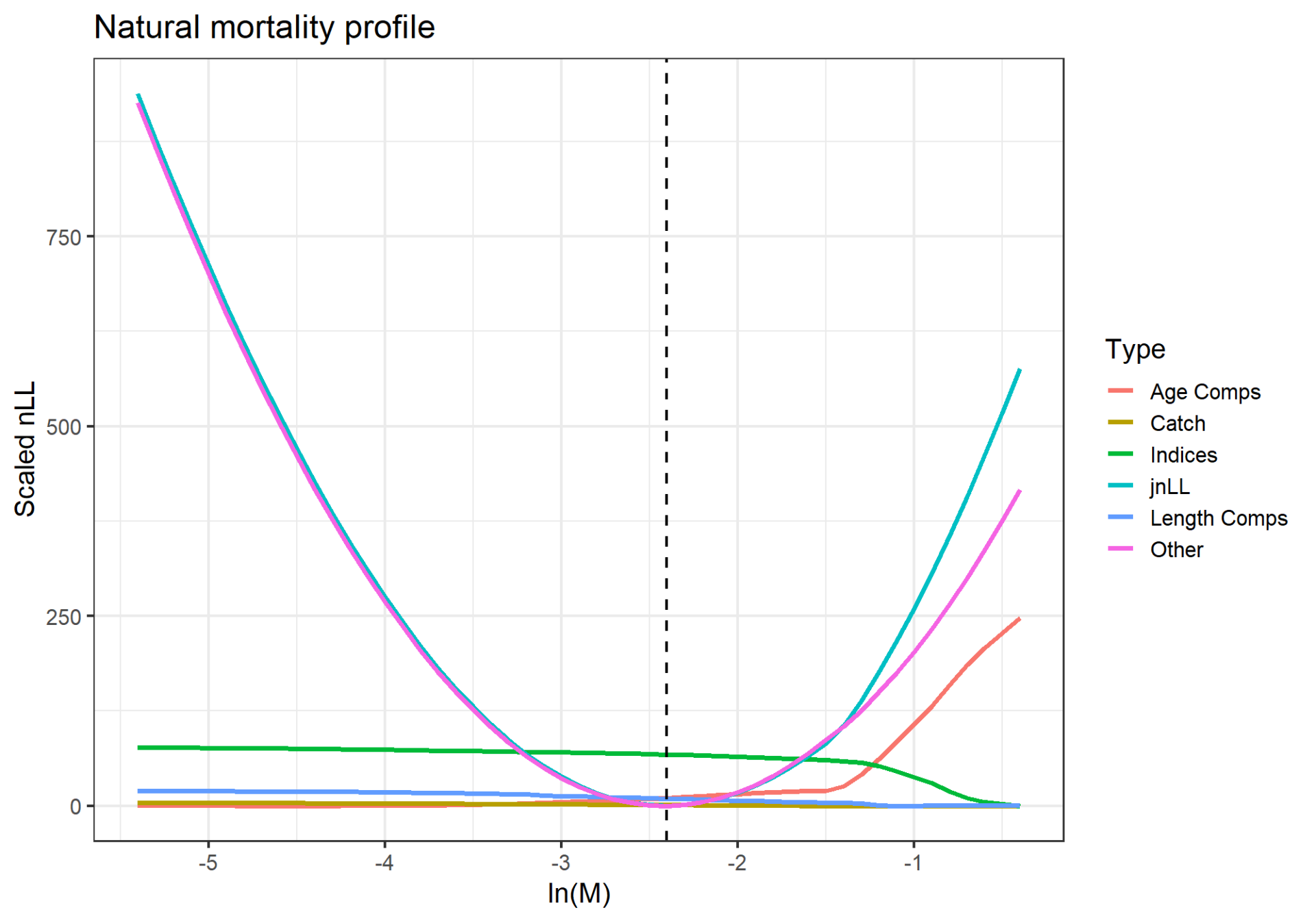
100
50
0

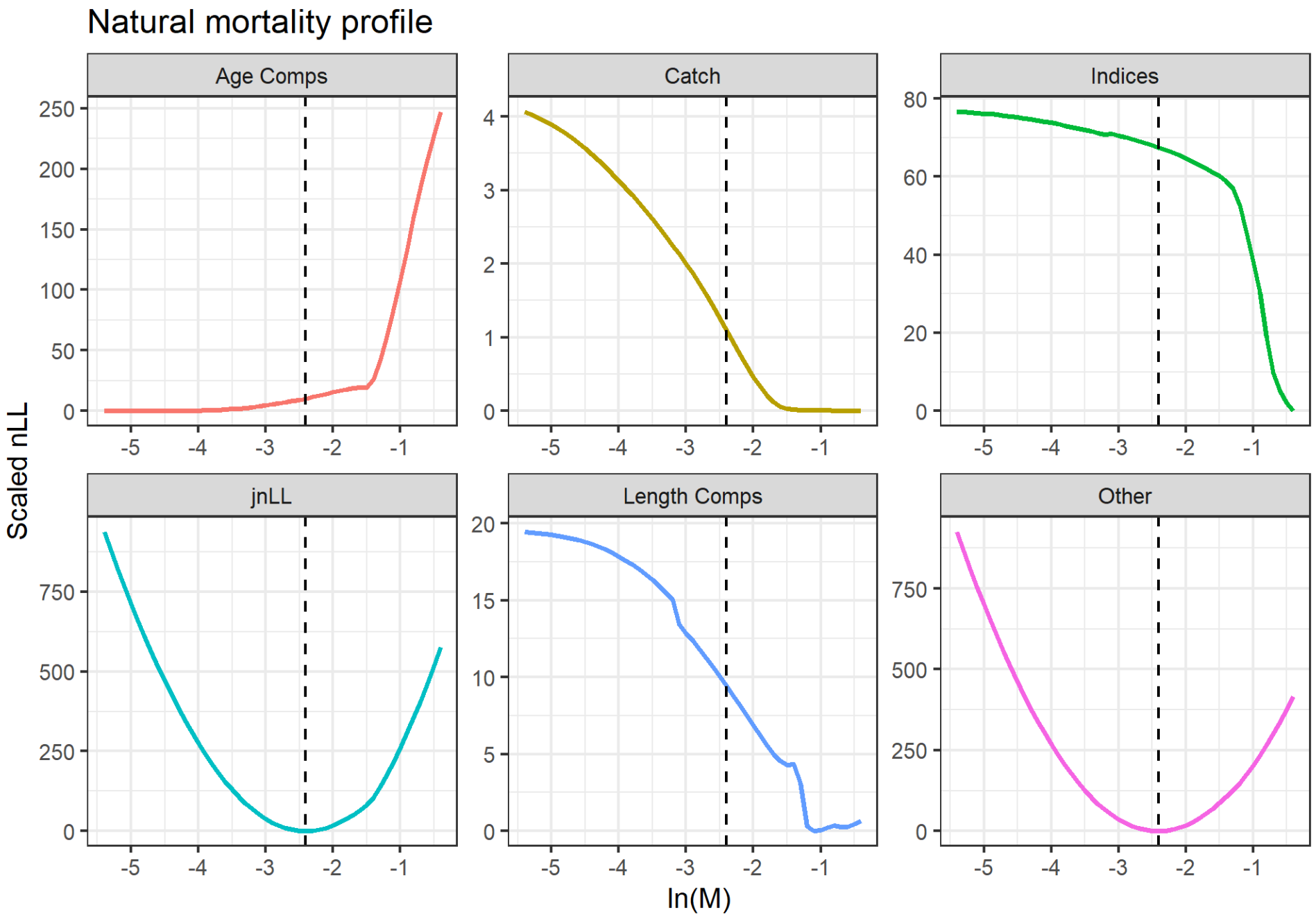
300
200
100
0

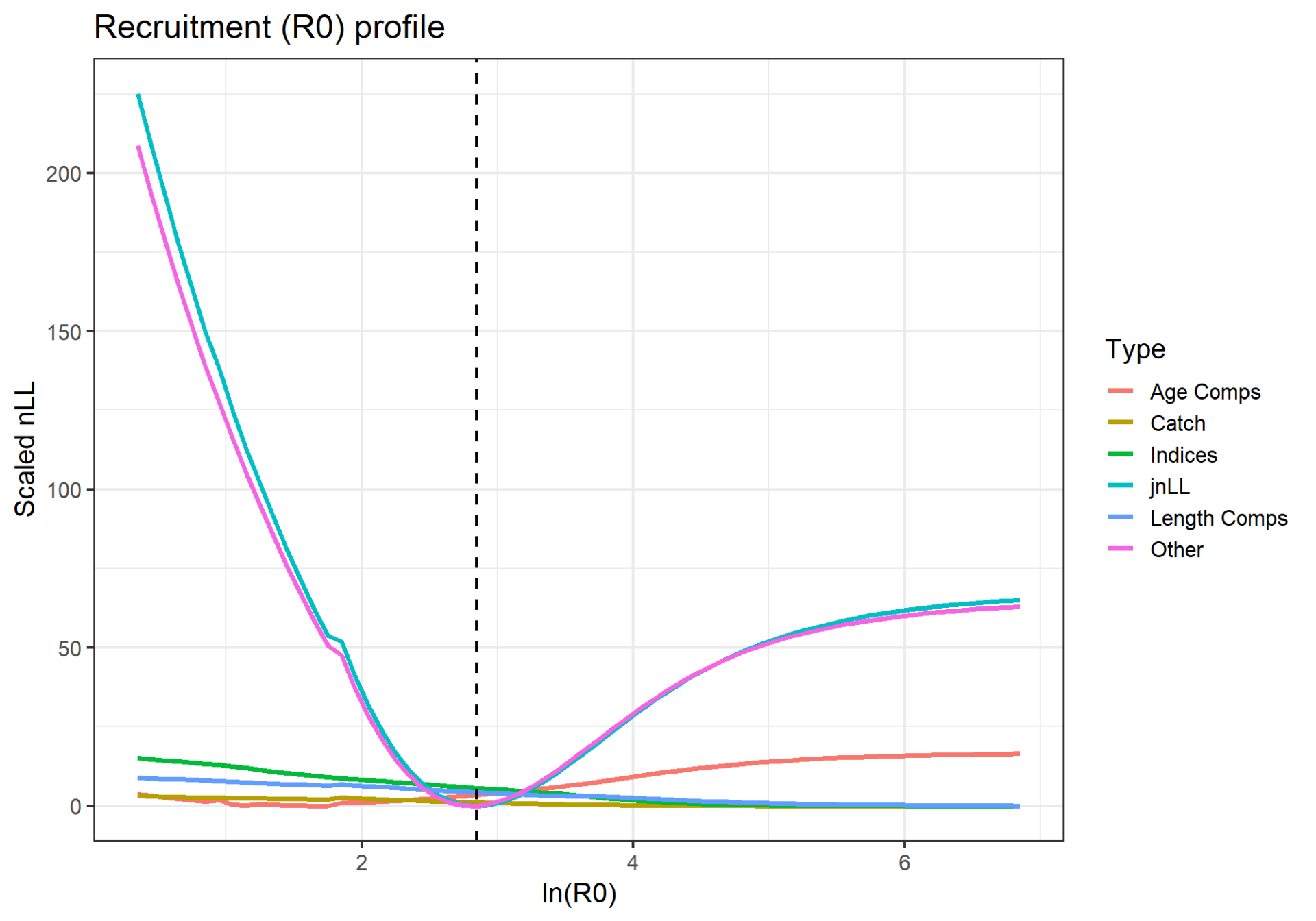
1960
1980
2000
2020

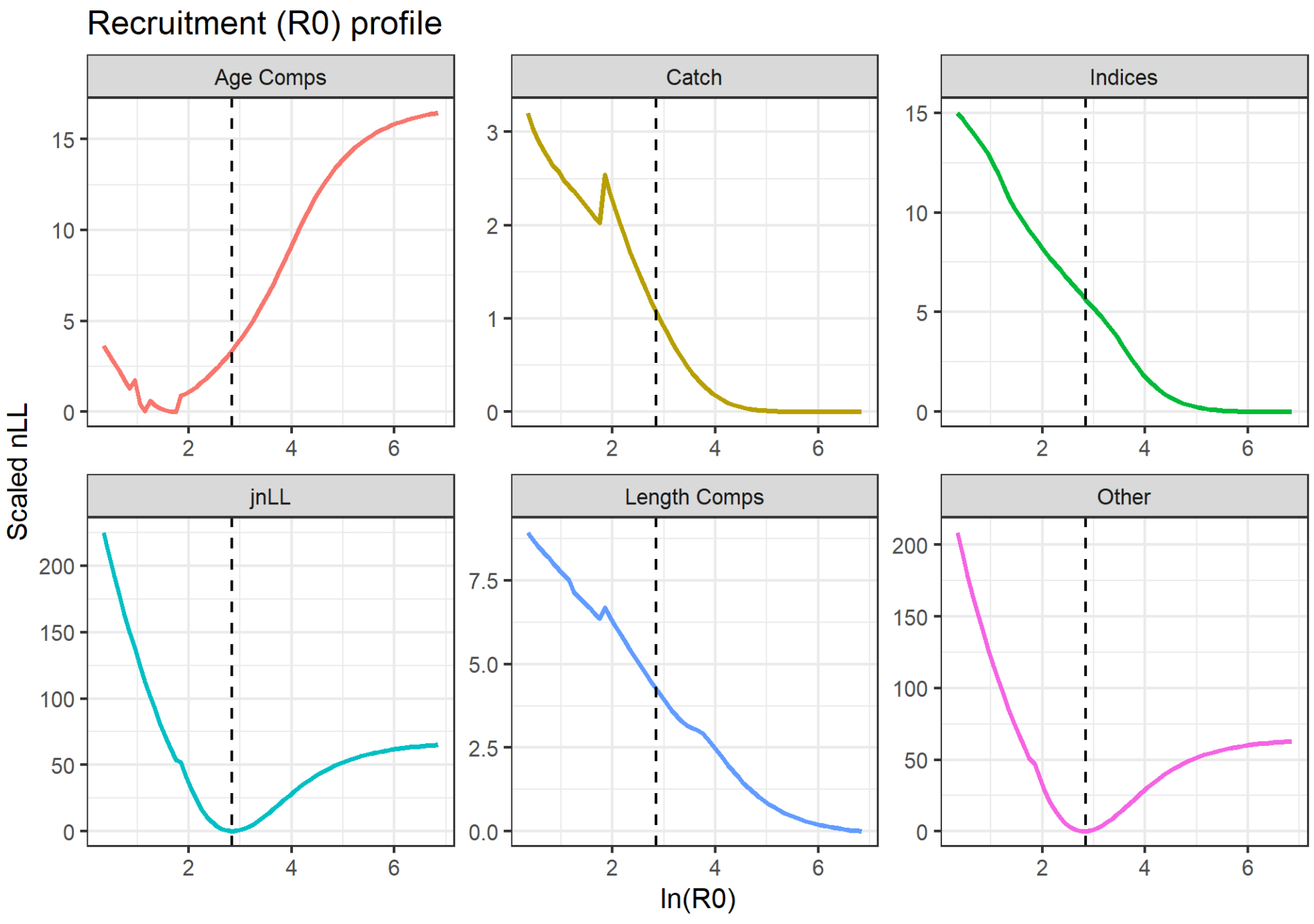
Year

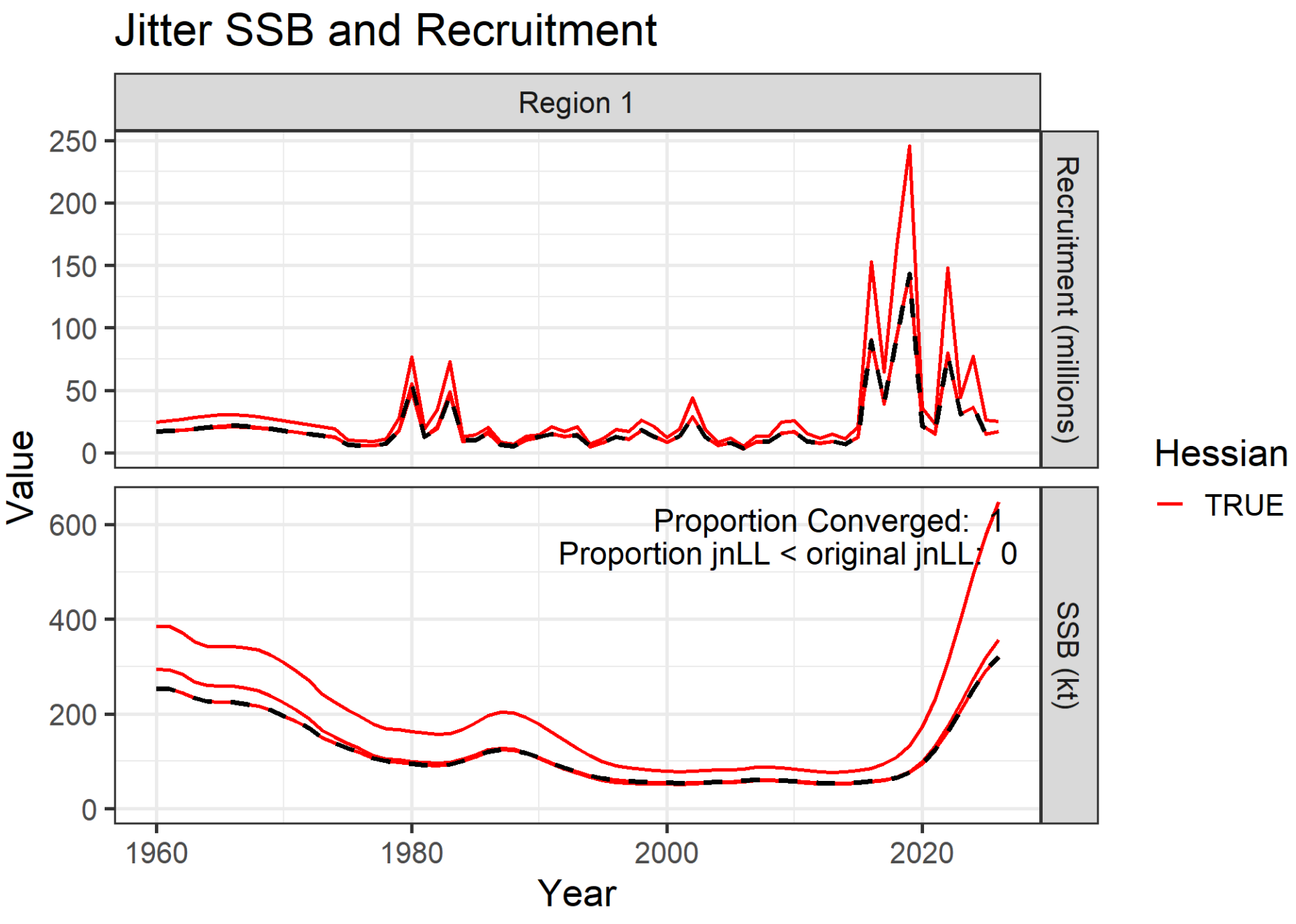

AK
100
50
Recruitment (millions)
0
1
2
3
4
5
6
7
Years since cohort was last estimated
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

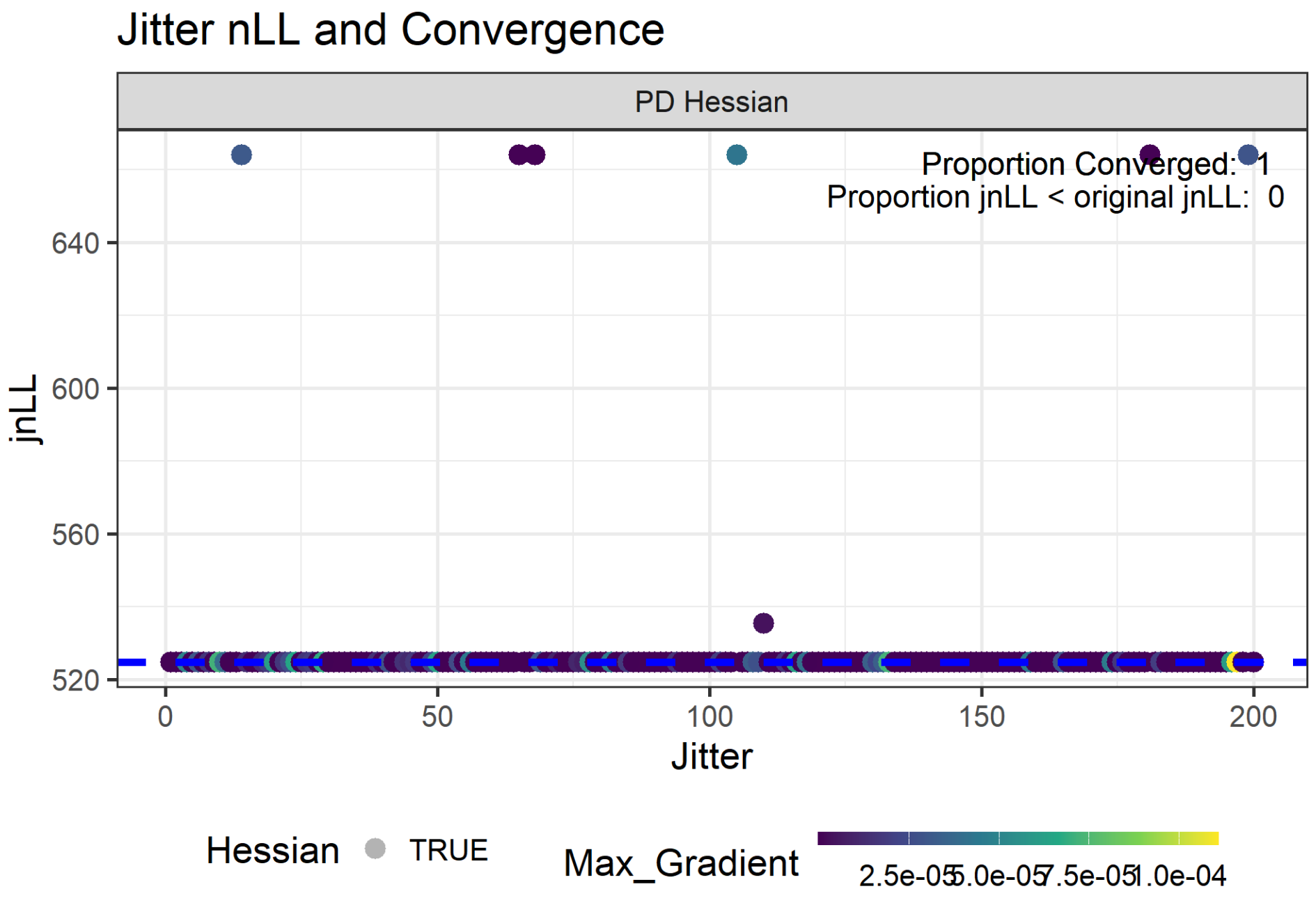

Natural mortality profile
Scaled nLL
750
500
250
0
-5
-4
-3
-2
-1
ln(M)
Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

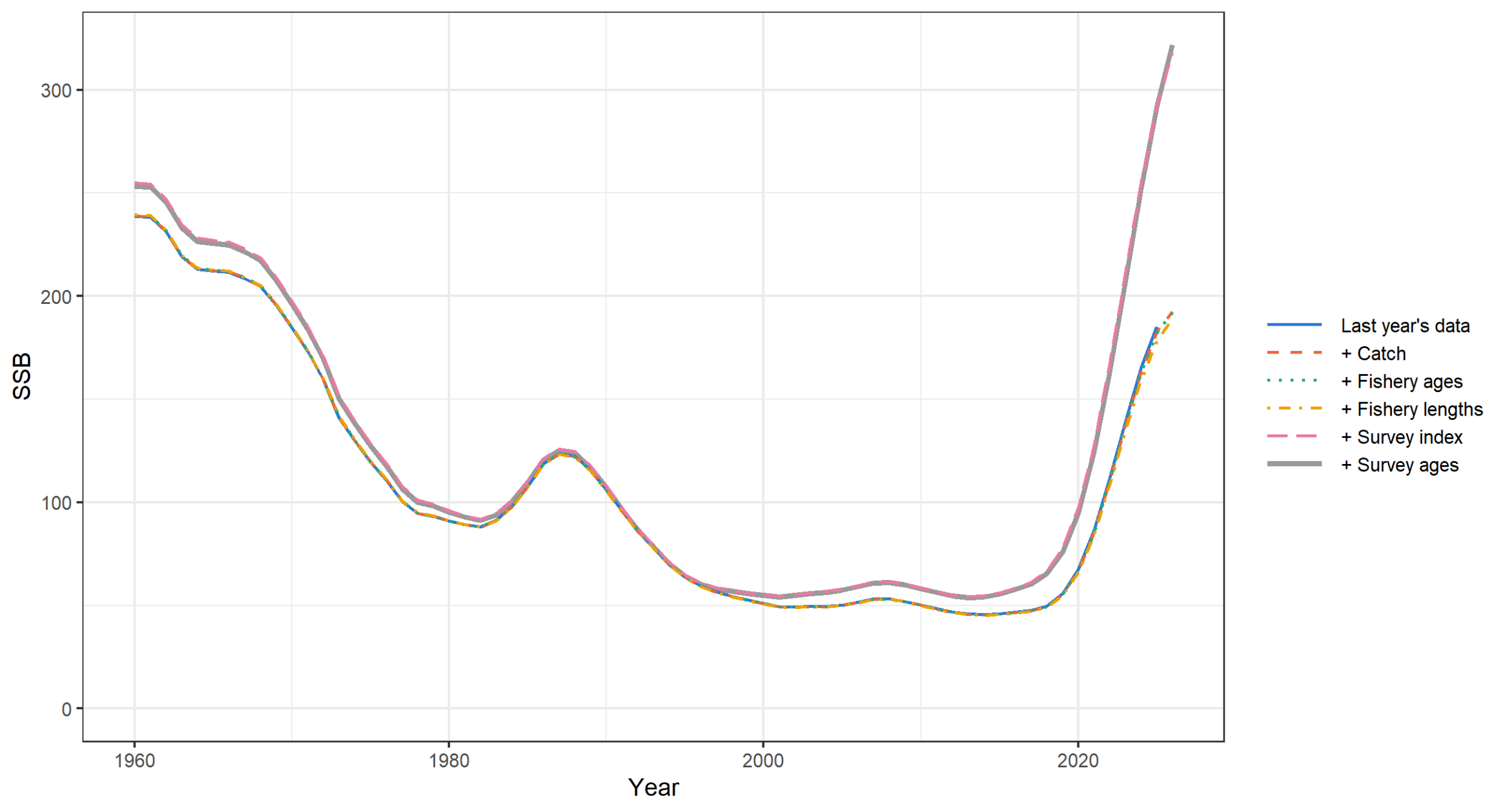

Natural mortality profile
Age Comps
Catch
Indices
jnLL
Length Comps
Other
Scaled nLL
ln(M)

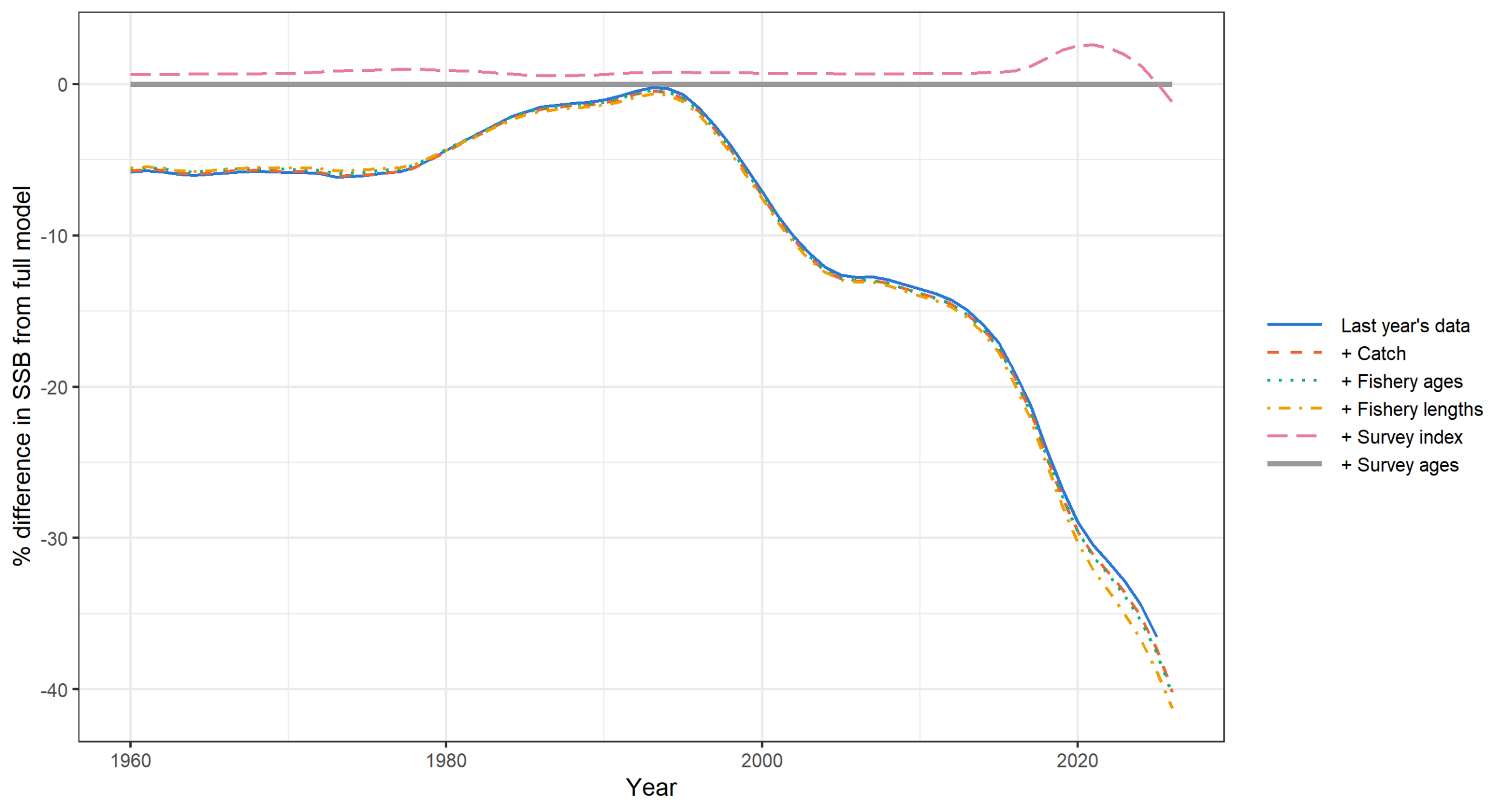

Recruitment (R0) profile
Scaled nLL
200
150
100
50
0
2
4
6
ln(R0)
Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

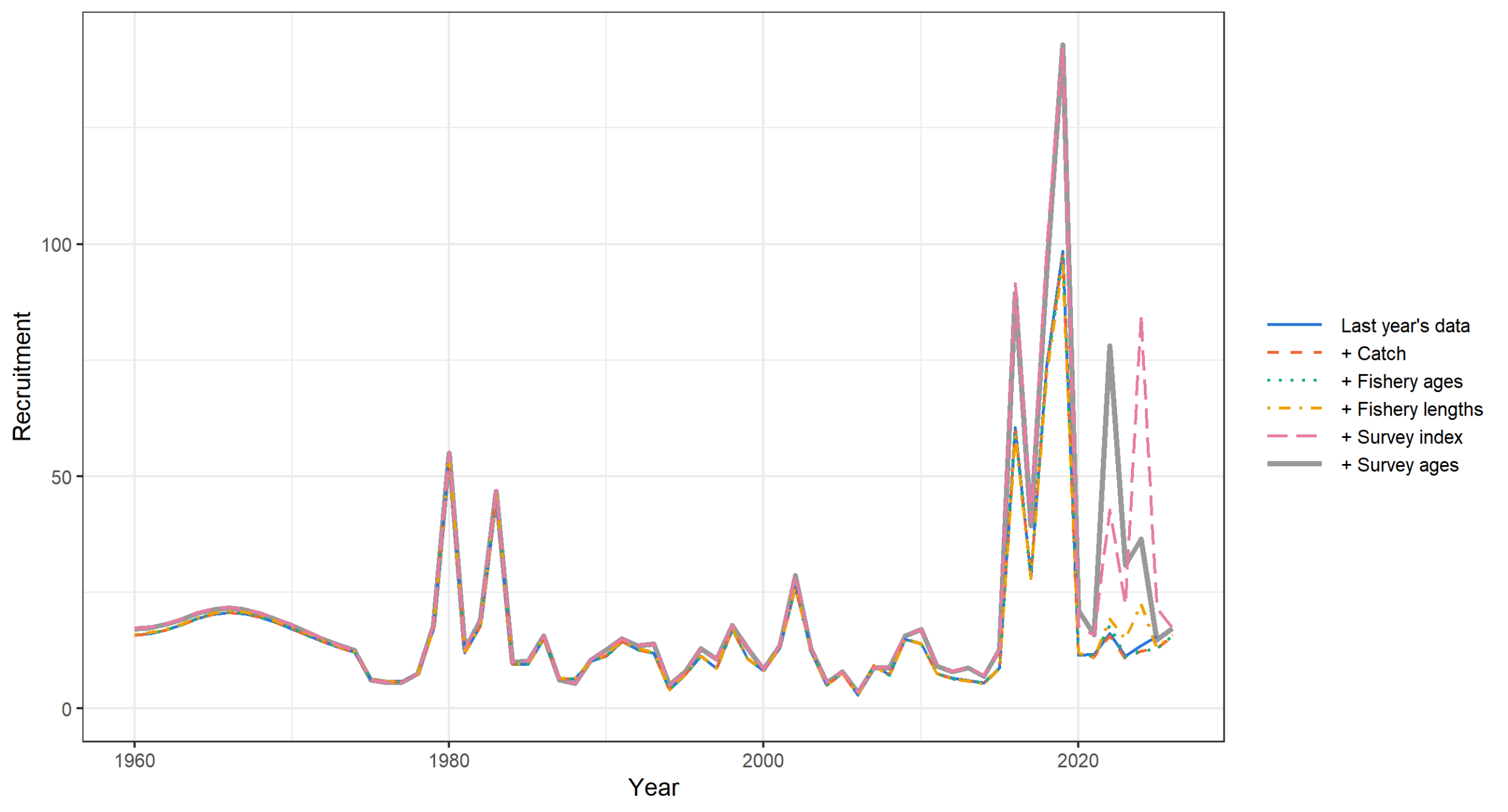

Recruitment (R0) profile
Age Comps
Catch
Indices
jnLL
Length Comps
Other
Scaled nLL
ln(R0)

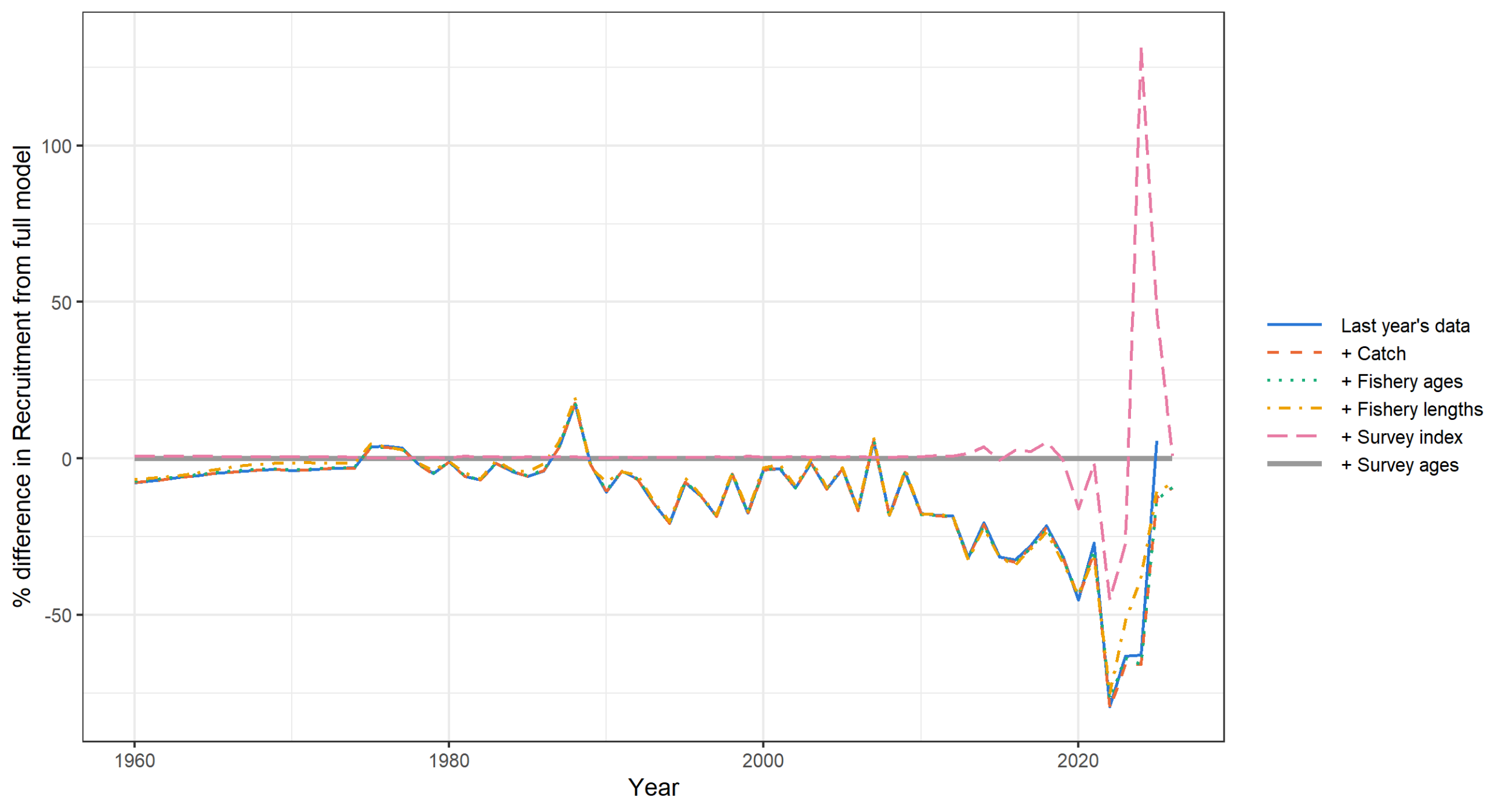

Jitter SSB and Recruitment
Region 1
Recruitment (millions)
SSB (kt)
Value
Year
Proportion Converged: 1
Proportion jnLL < original jnLL: 0
Hessian
TRUE
250
200
150
100
50
0
600
400
200
0
1960
1980
2000
2020

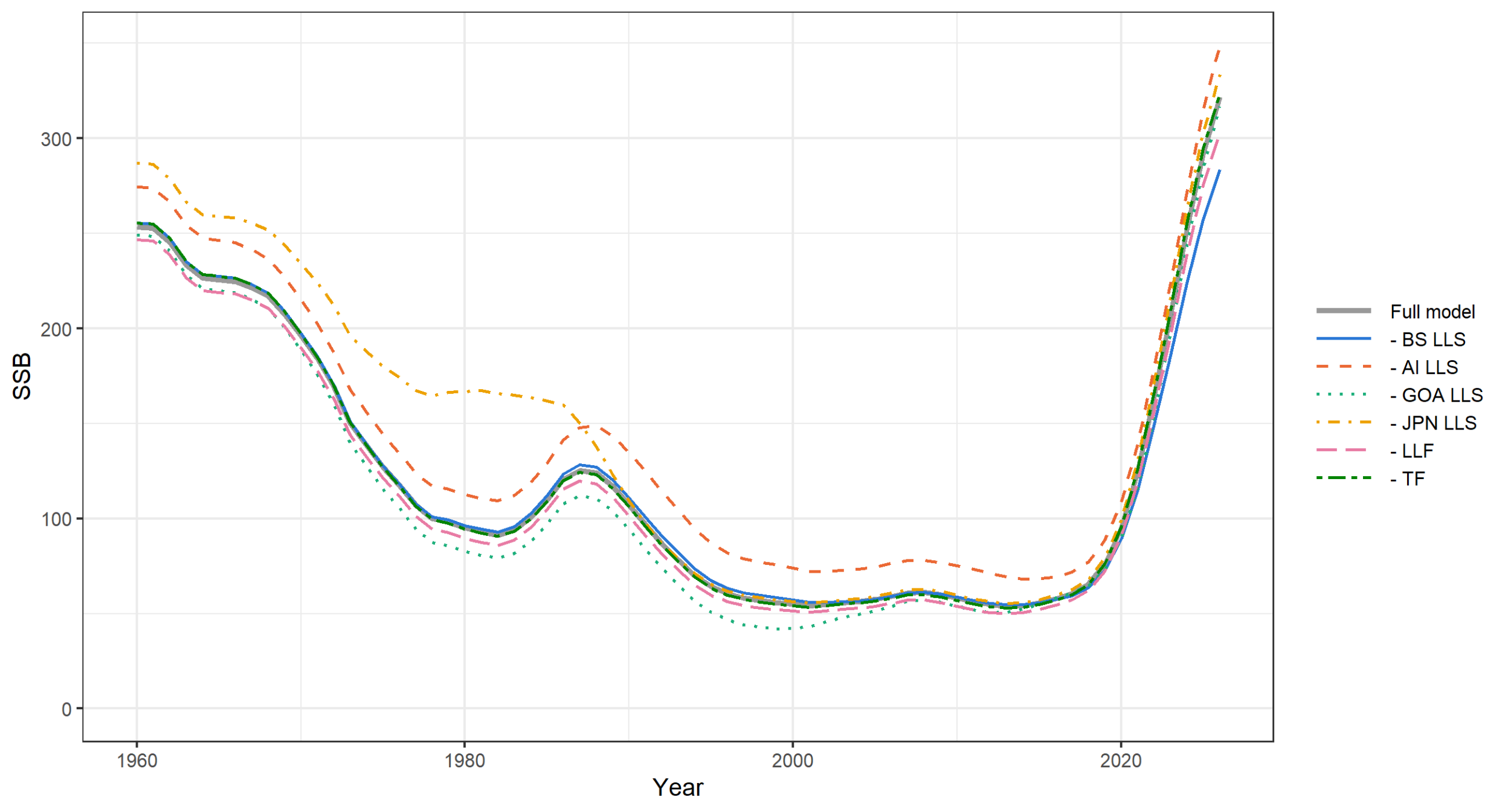

Jitter nLL and Convergence
PD Hessian
Proportion Converged: 1
Proportion jnLL < original jnLL: 0
640
600
560
520
jnLL
0
50
100
150
200
Jitter
Hessian
TRUE
Max_Gradient
2.5e-05
5.0e-05
7.5e-05
1.0e-04

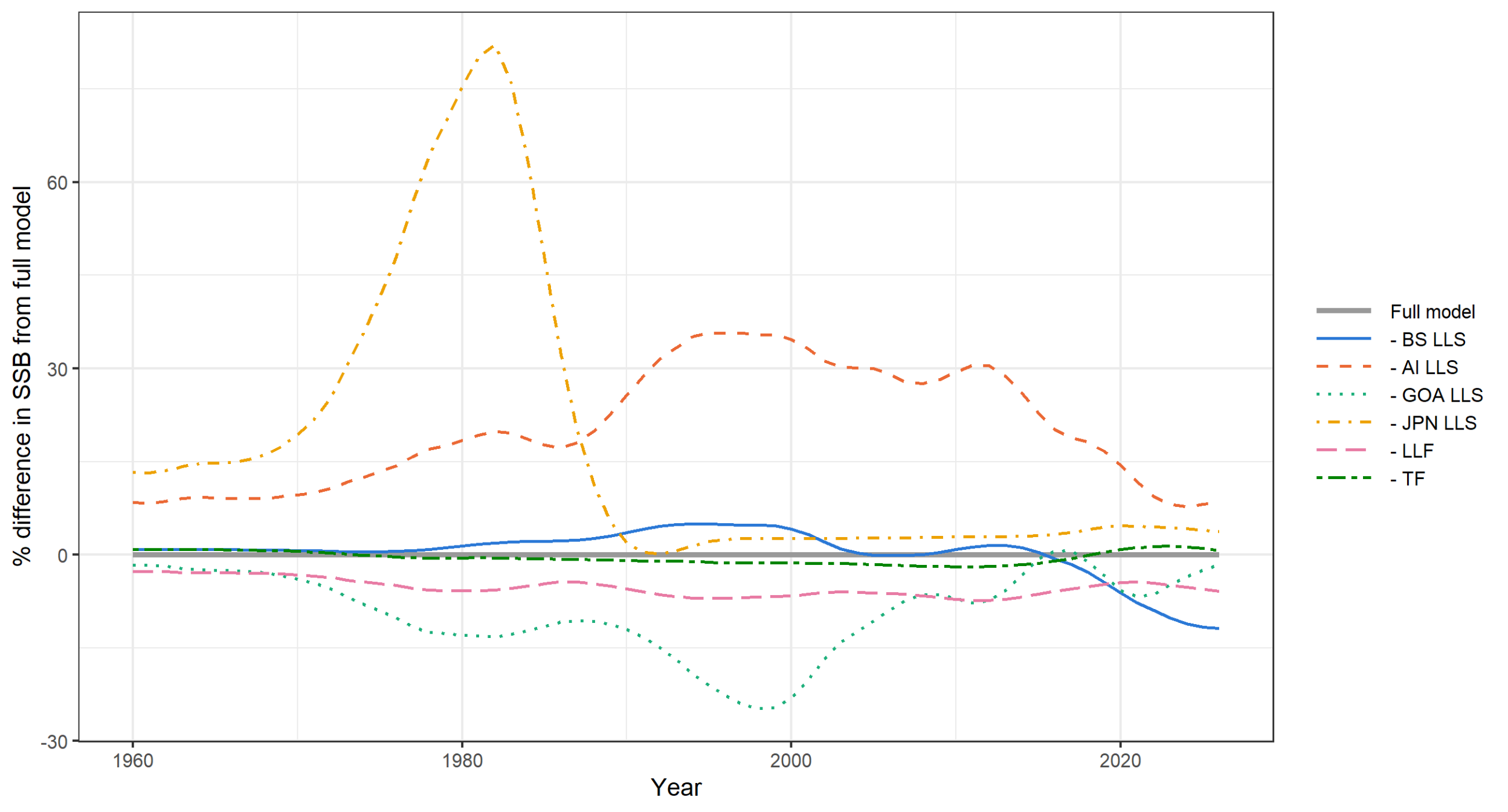

SSB
300
200
100
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

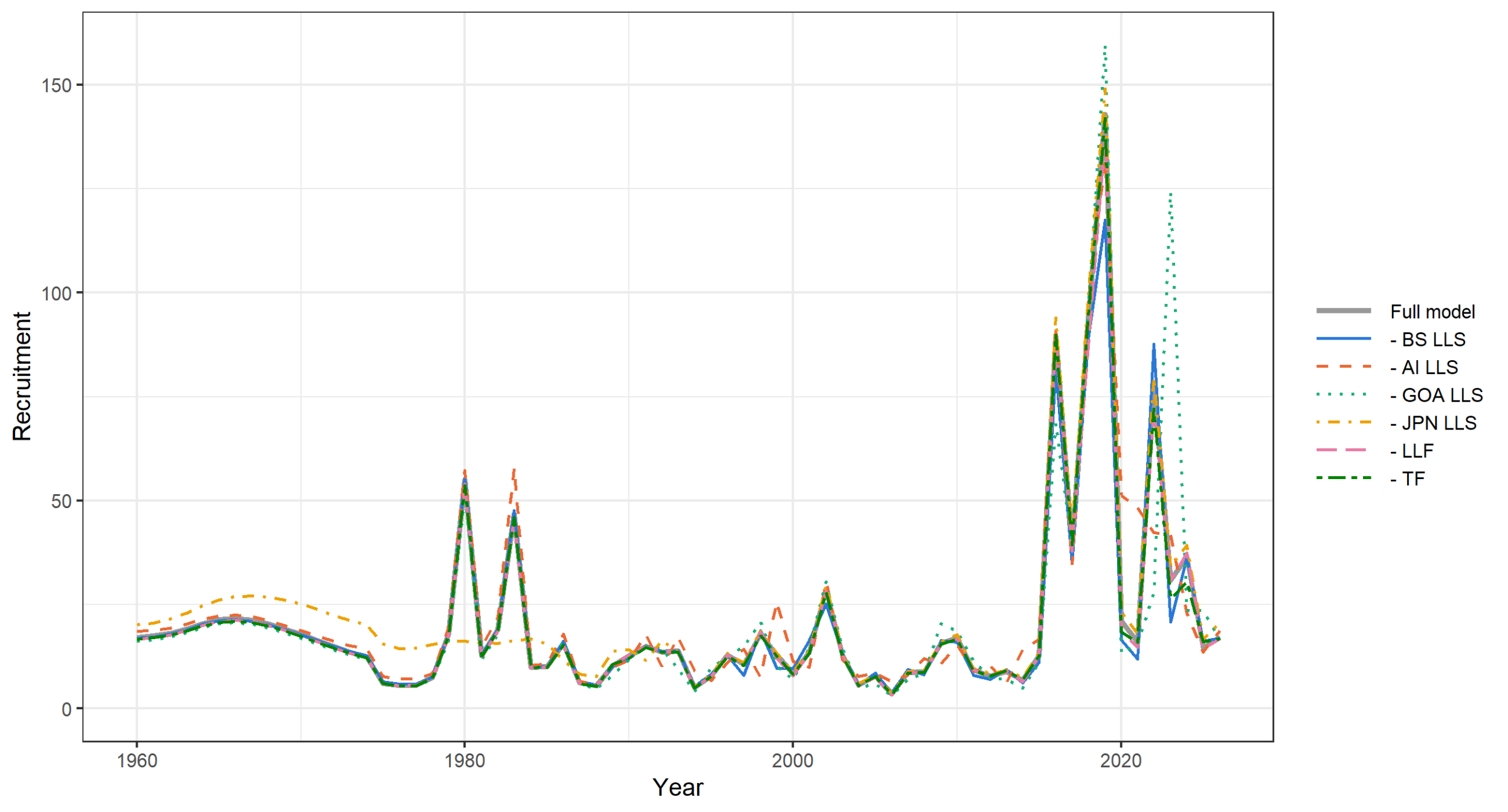

0
-10
-20
-30
-40
% difference in SSB from full model
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

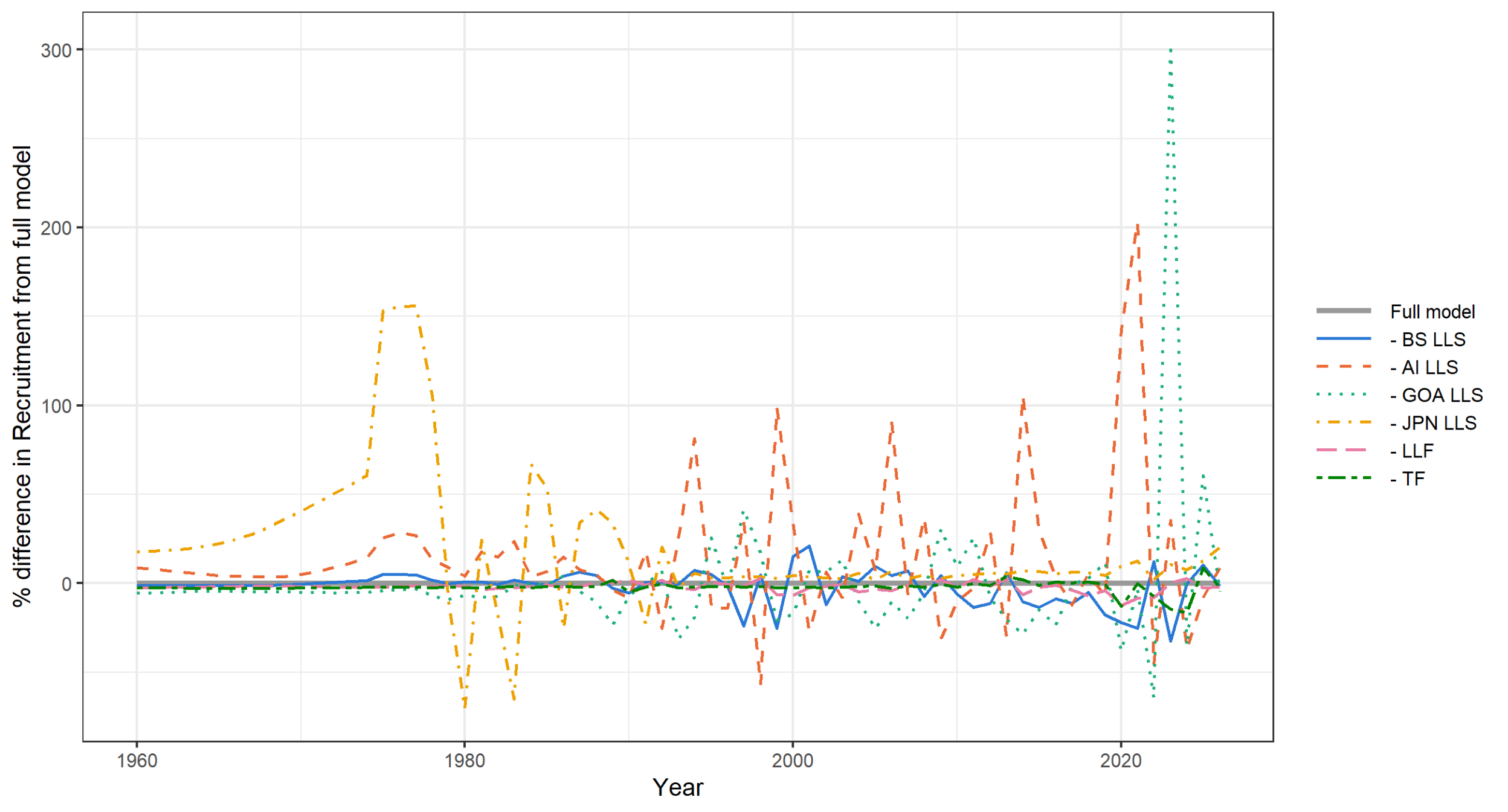

Recruitment
100
50
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


% difference in Recruitment from full model
100
50
0
-50
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


SSB
300
200
100
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


% difference in SSB from full model
60
30
0
-30
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF

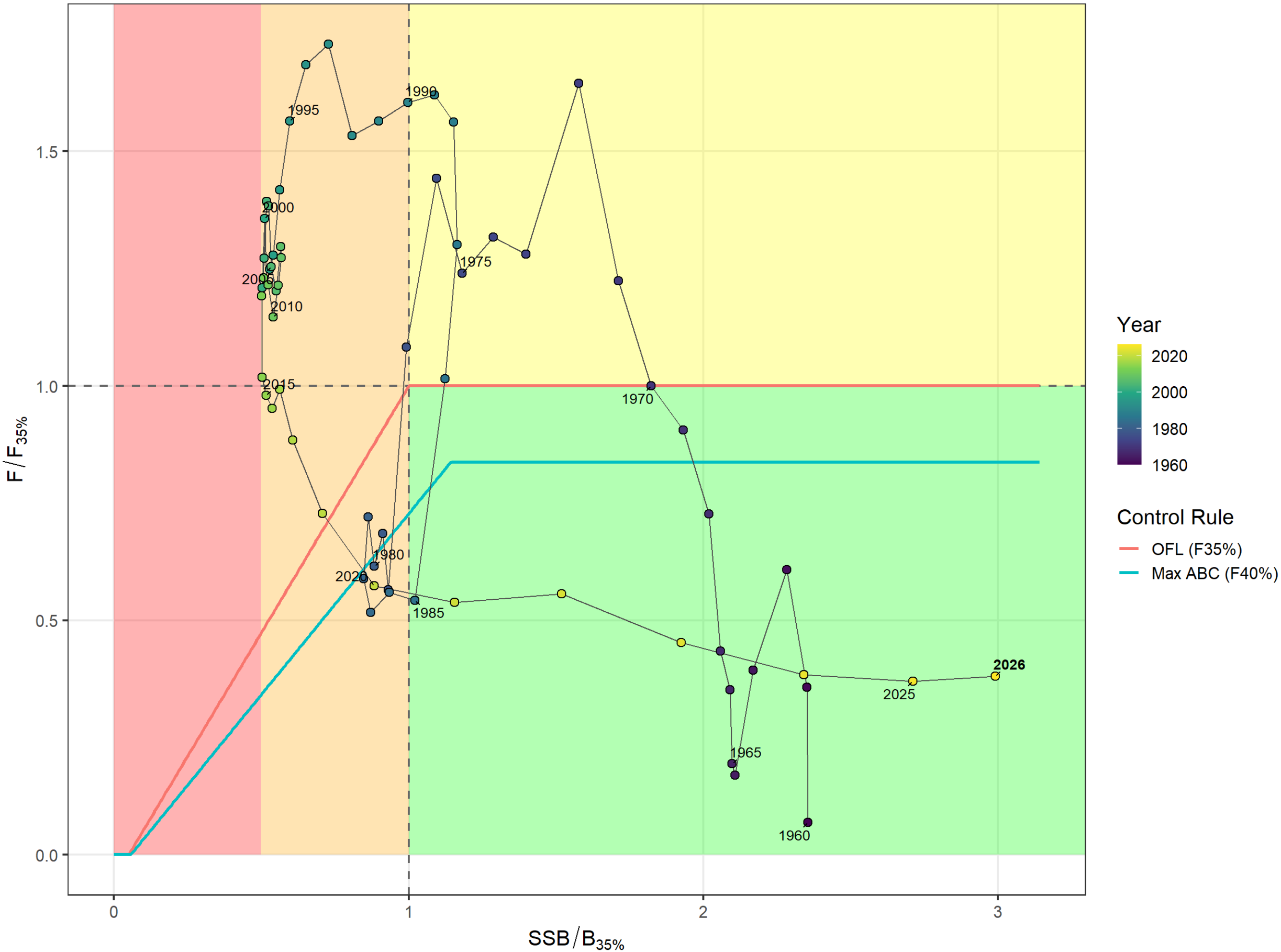

Recruitment
150
100
50
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


300
200
100
0
% difference in Recruitment from full model
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF

Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1.00	NA	NA
AI LL Survey Index	0.23	1.00	NA	NA
GOA LL Survey Index	0.08	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery CPUE	NA	NA	NA	NA
BS LL Survey Ages	NA	0.54	34.07	18.37
AI LL Survey Ages	NA	2.19	35.57	77.85
GOA LL Survey Ages	NA	0.45	67.17	30.55
JPN LL Survey Ages	NA	0.21	83.33	17.70
JPN LL Survey Lengths	NA	0.30	54.62	16.43
Fixed Gear Fishery Ages	NA	0.04	142.85	5.82
Trawl Fishery Lengths	NA	0.08	47.62	3.87

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.2_FAA_REMA	1	321.6868	496.4345	0.03722759	55.13889	154.4758	0.08185	2.08244	0.64799	0.45481

Phase Plane Assuming B40 from ~2018



S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

200

400

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

200

400

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0
300
600
900

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0
300
600
900

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0
50
100

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0

50

100

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

0.15

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
20
40
60

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0

20

40

1960

1980

2000

2020

2040

Year

Fleet

LLF

TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
20
40
60

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0

20

40

2010

2020

2030

2040

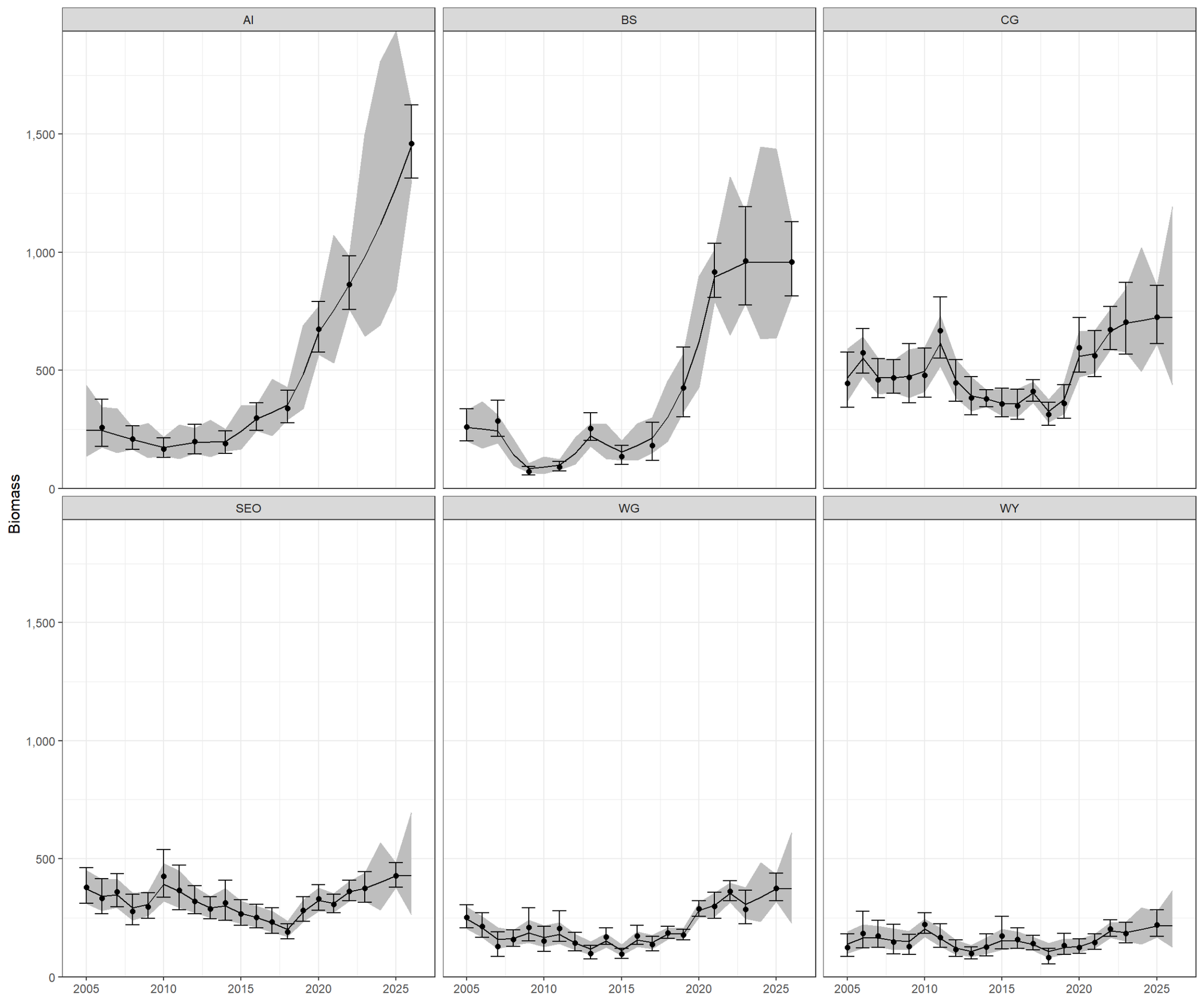
Year

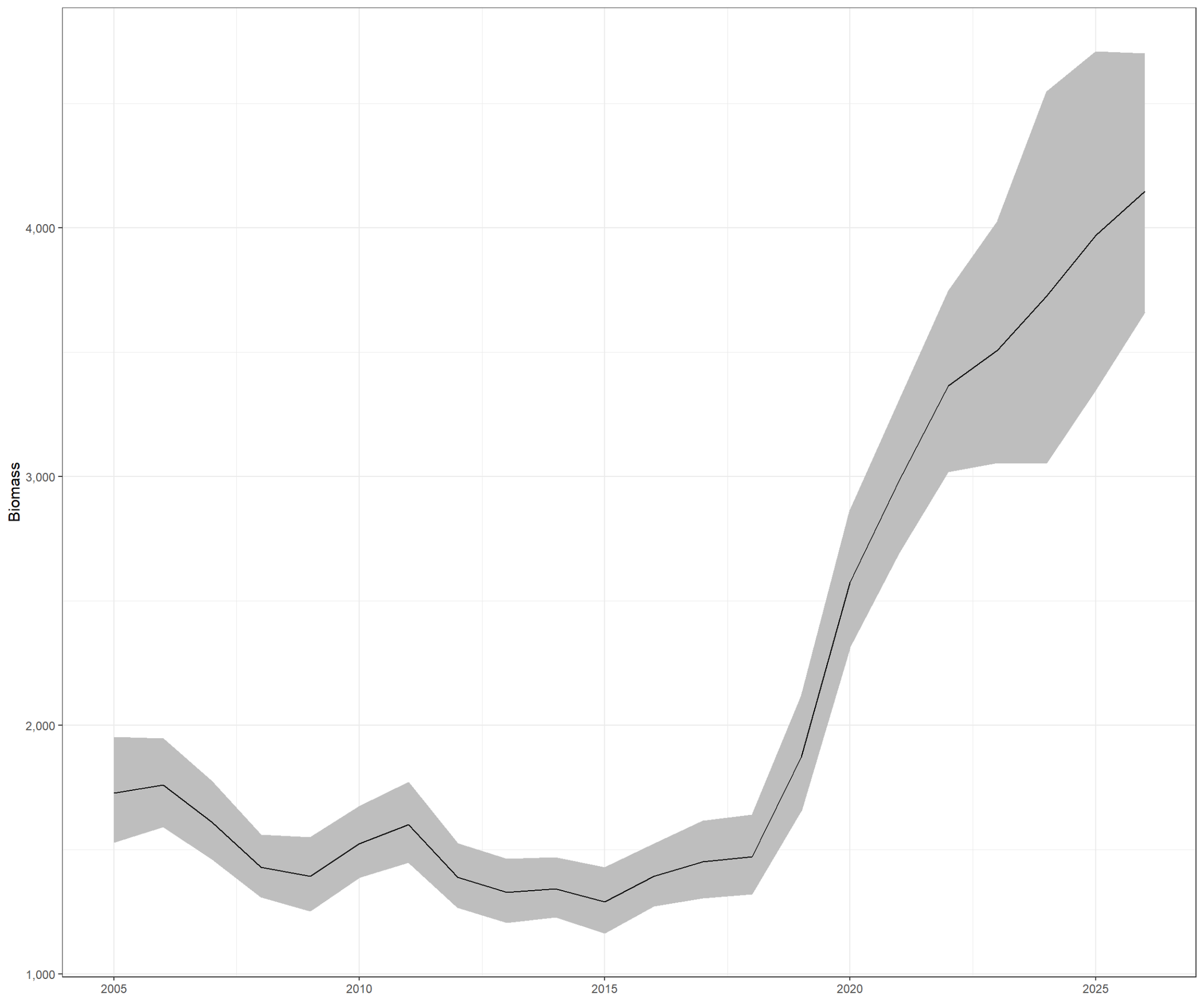
Fleet

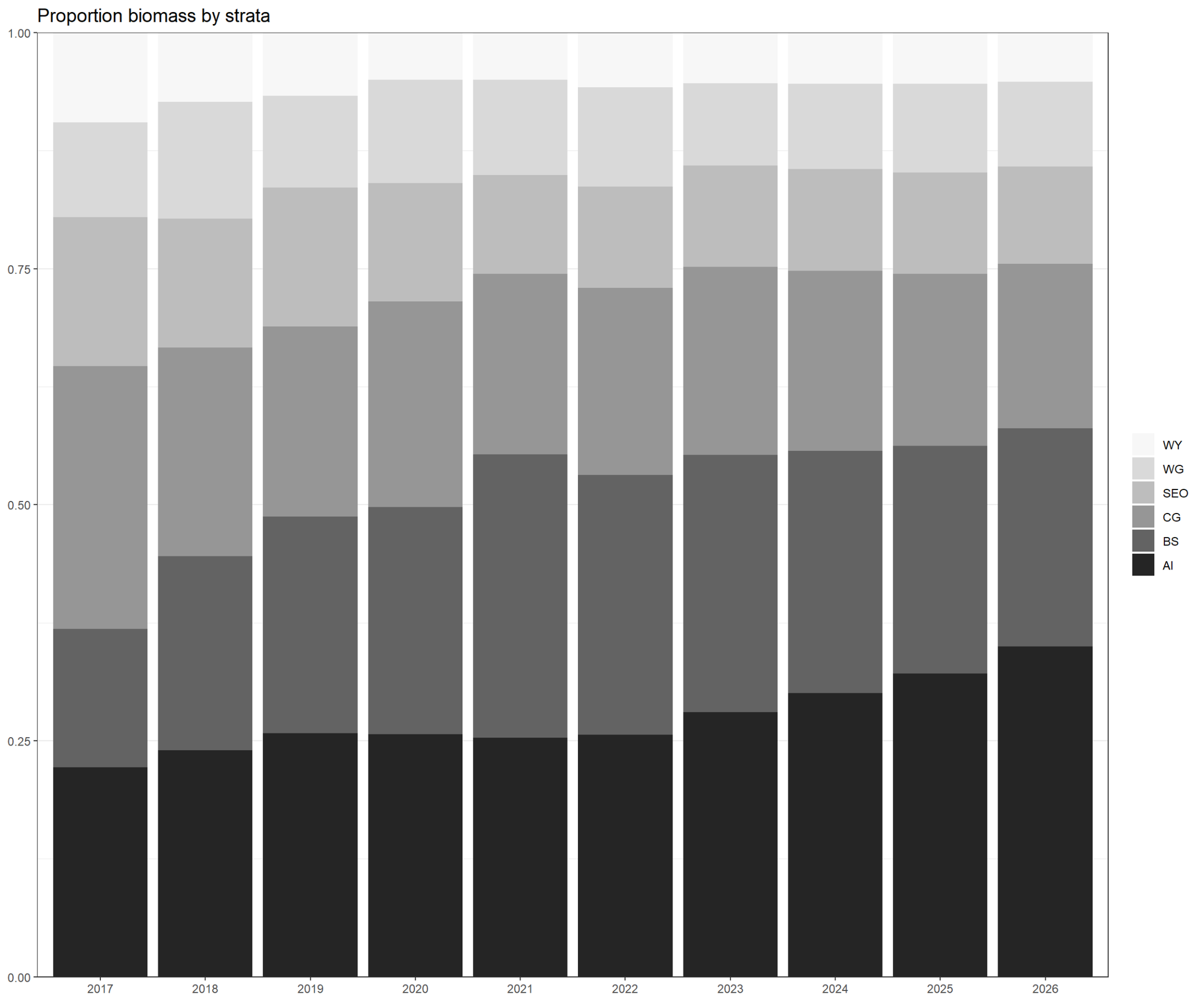
LLF

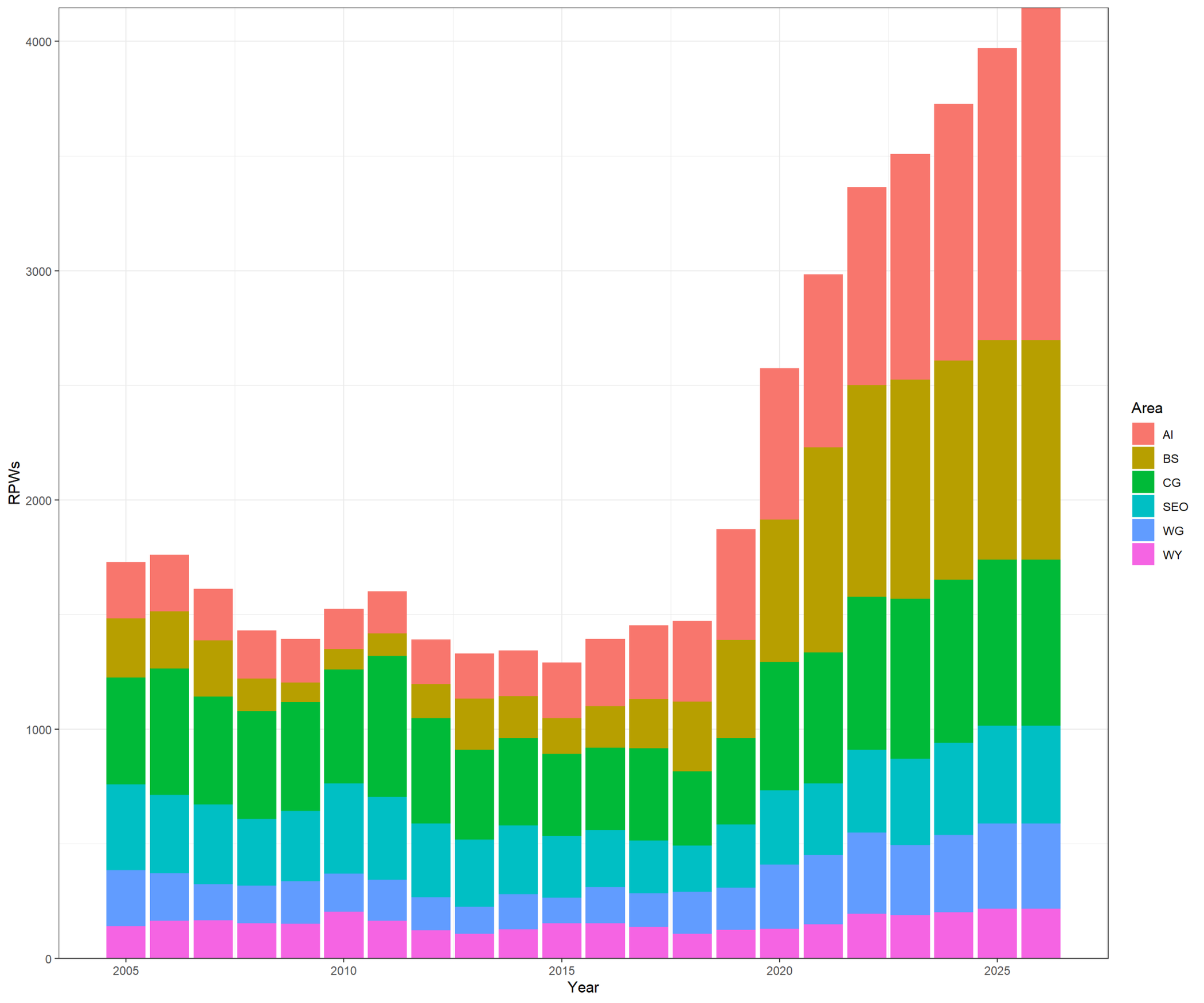
TF

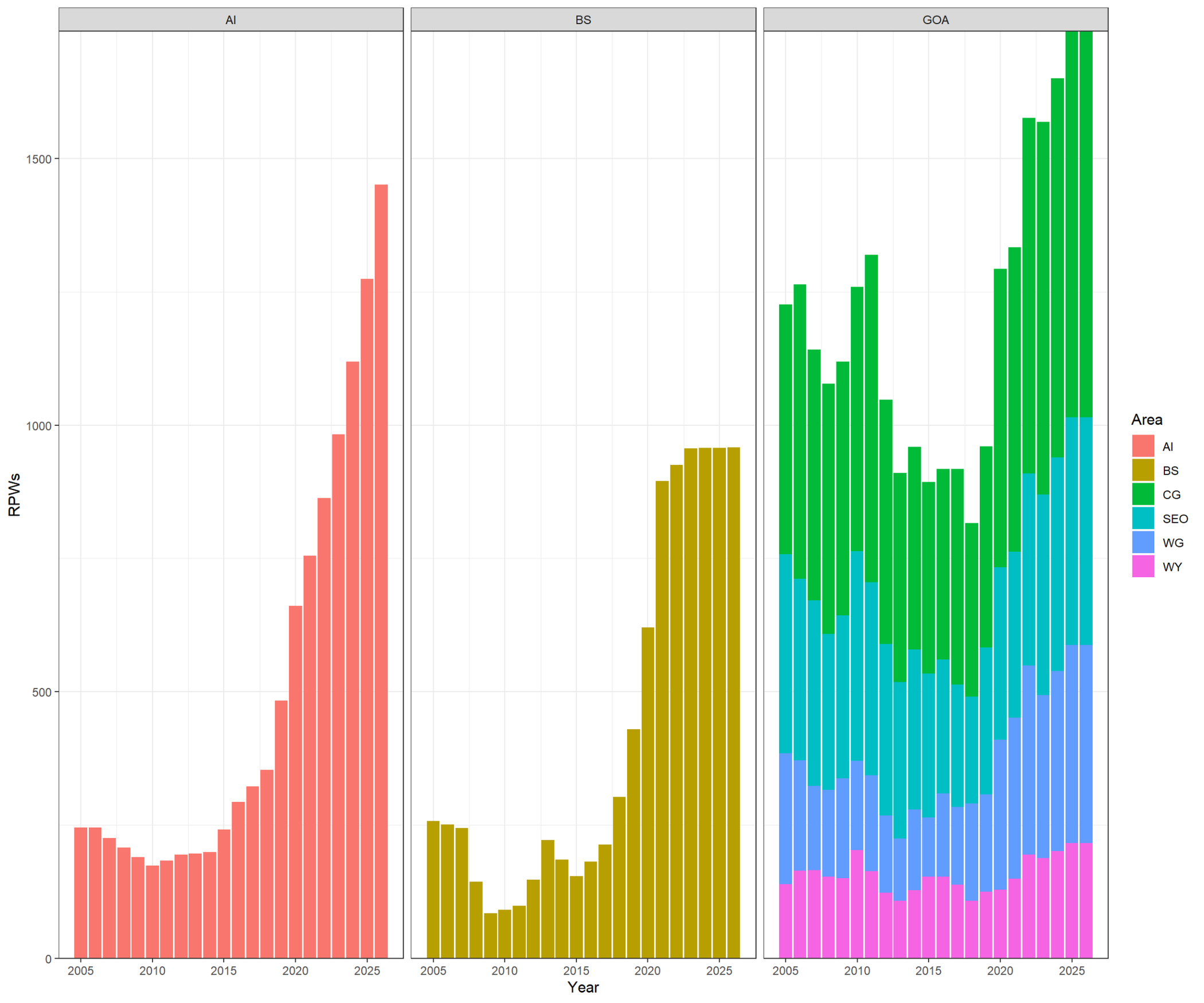
Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.09	0.09
M Male (estimated)	0.114	0.114	0.091	0.091
Average Recruitment (millions)	25.9	25.9	23	23
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	824437	807418
Projected Female Spawning Biomass (t)	219,714	241,217	342387	352952
B100% (t)	302,672	302,672	386189	386189
B40% (t)	121,069	121,069	154476	154476
B35% (t)	105,935	105,935	135166	135166
FOFL	0.102	0.102	0.098	0.098
max FABC	0.087	0.087	0.082	0.082
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	65361	64395
OFLw (t)	58,532	57,797	65230	64264
max ABC (t)	50,283	49,651	55136	54325
ABC (t)	50,283	49,651	55136	54325
ABCw (t)	50,111	49,482	55005	54194
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

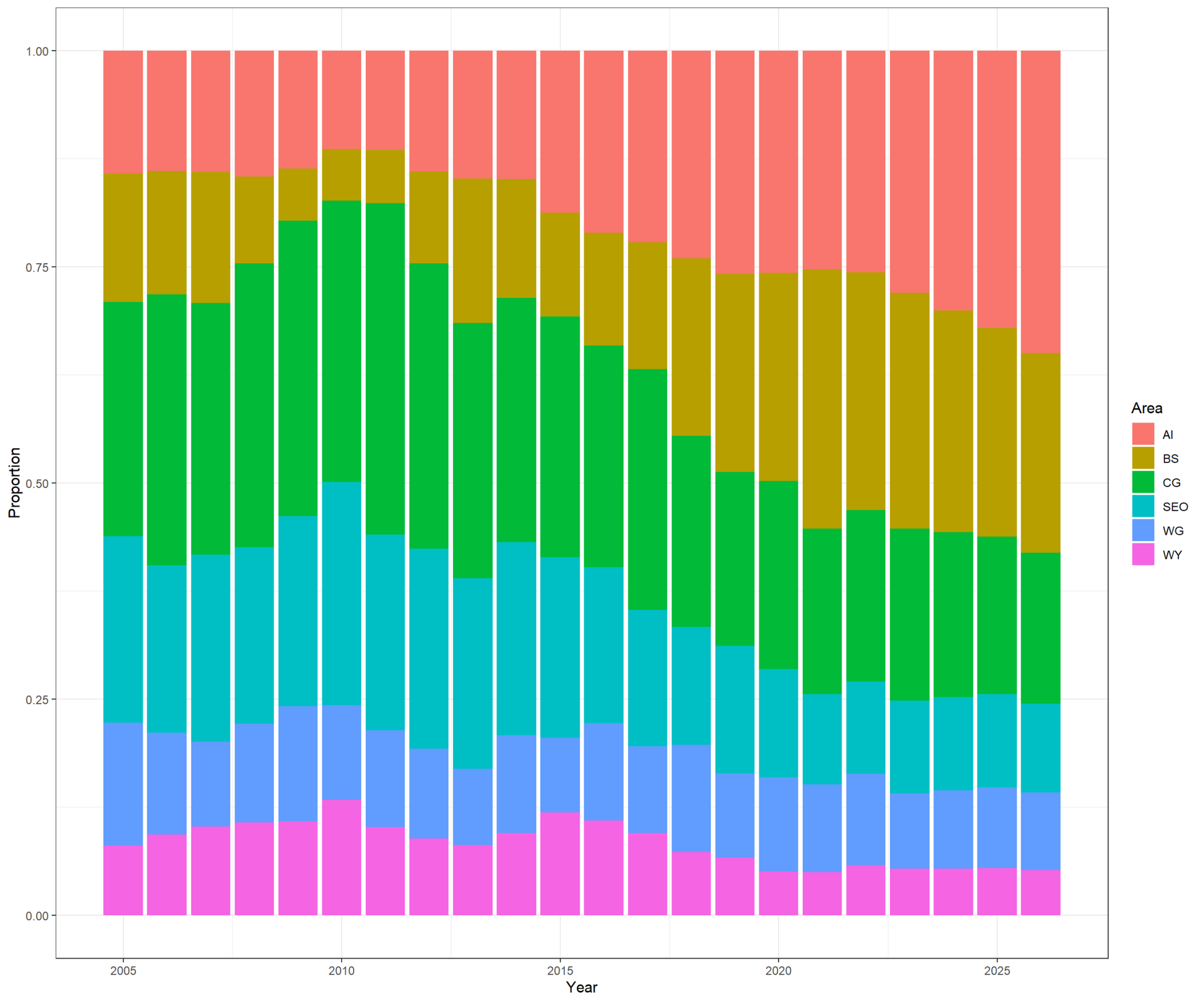

AI
BS
CG
SEO
WG
WY
Biomass
0
500
1,000
1,500
2005
2010
2015
2020
2025

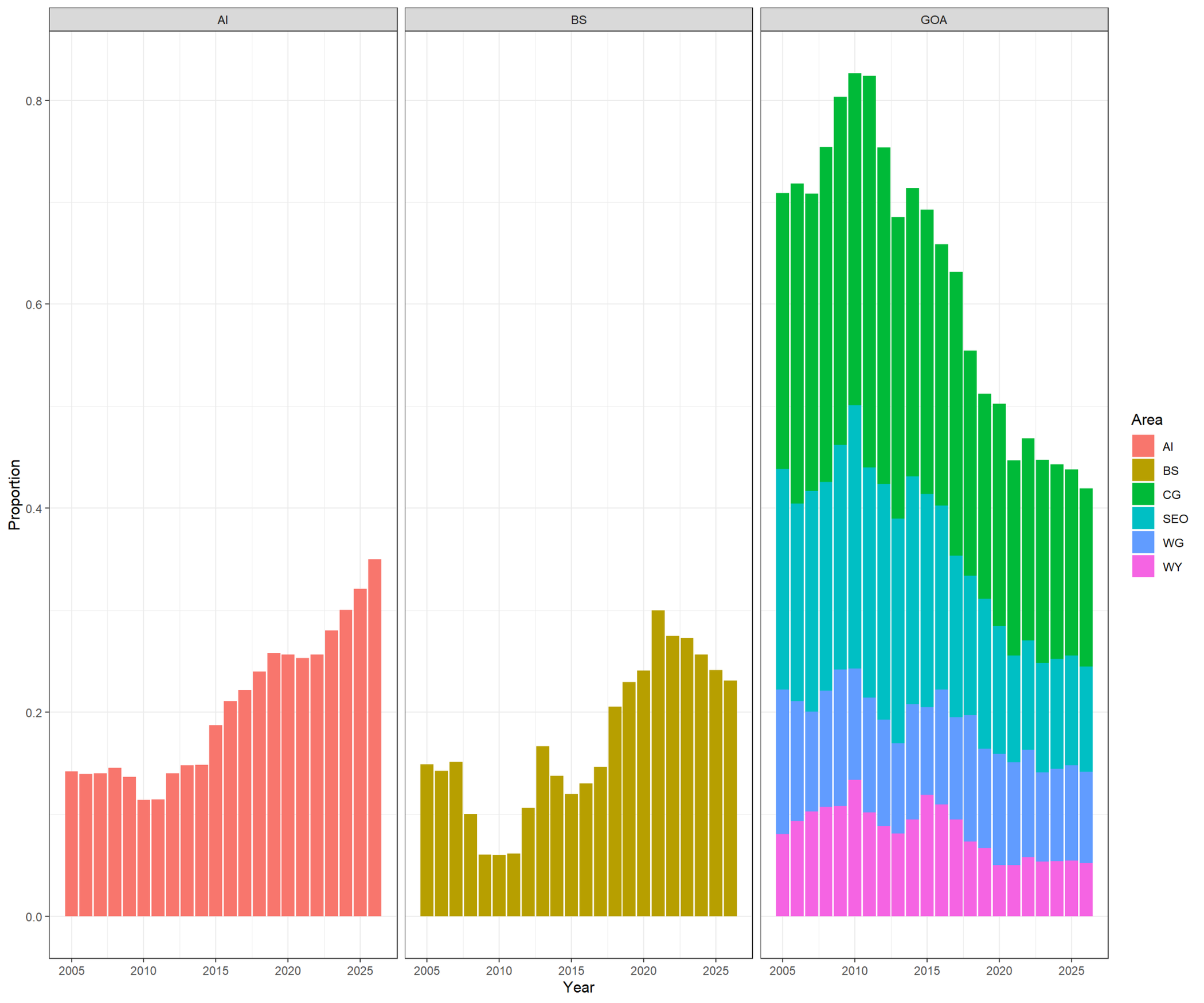

Biomass
4,000
3,000
2,000
1,000
2005
2010
2015
2020
2025

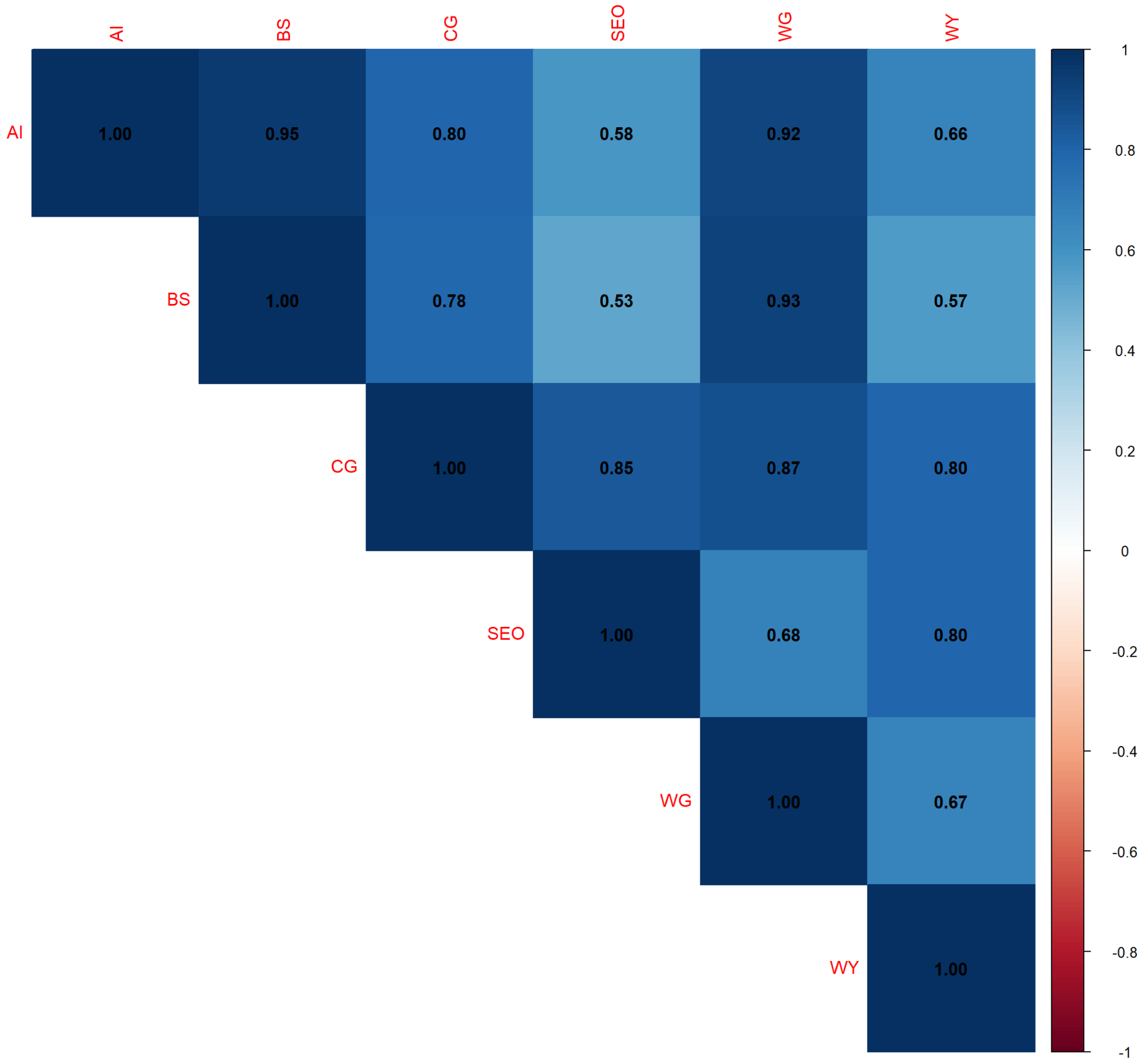

Proportion biomass by strata
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
WY
WG
SEO
CG
BS
AI

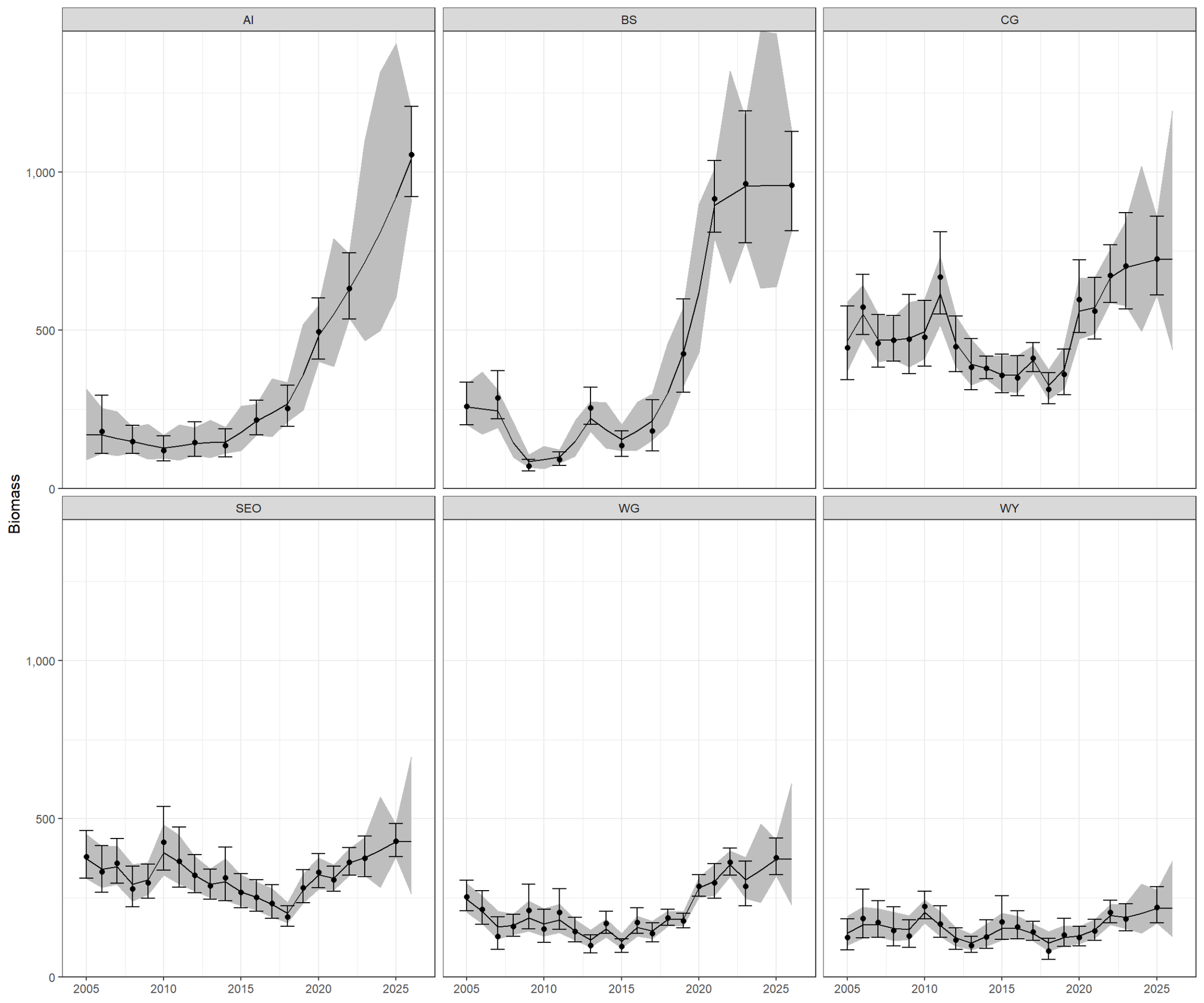

4000
3000
2000
1000
0
RPWs
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY

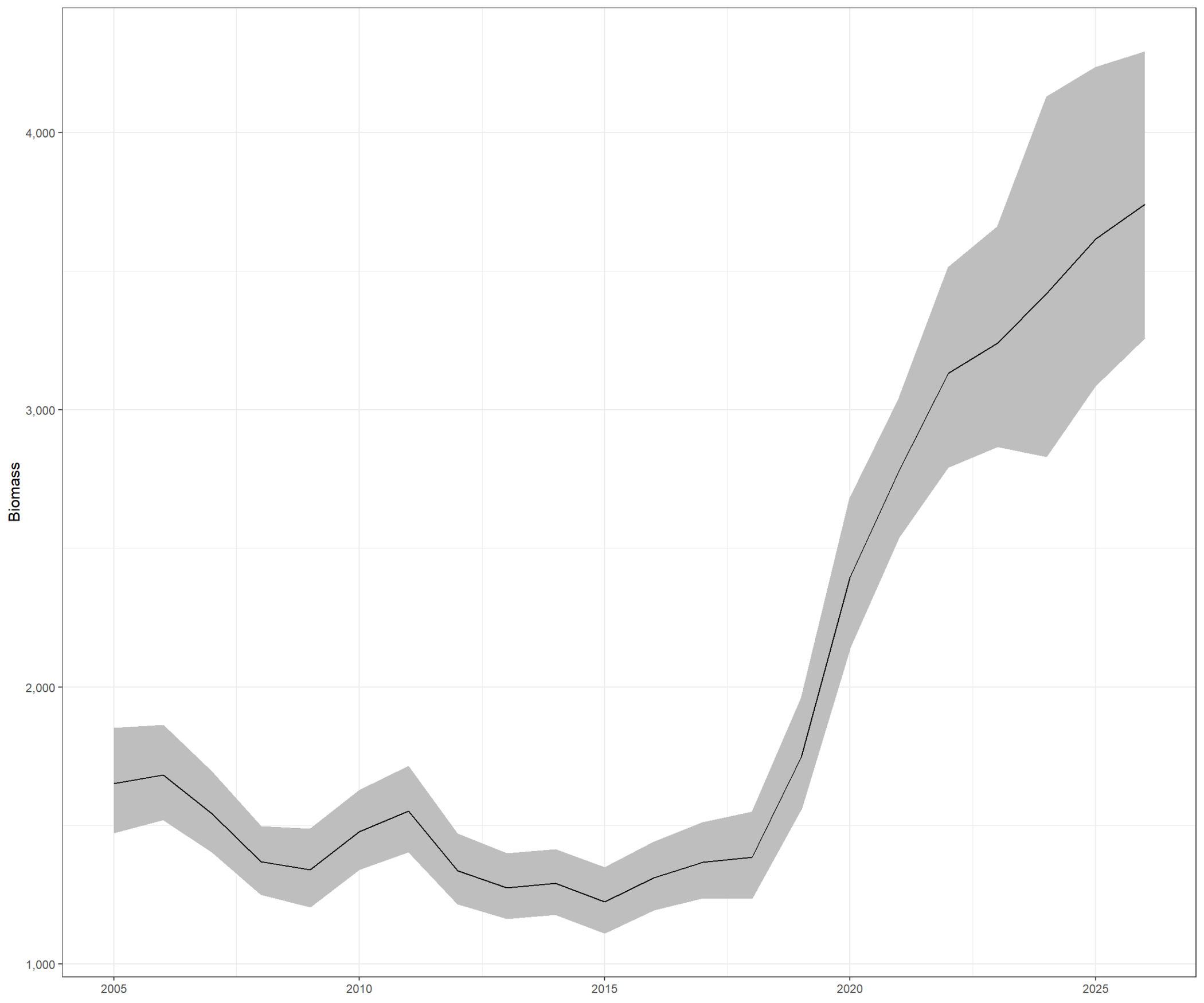

AI
BS
GOA
RPWs
0
500
1000
1500
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY

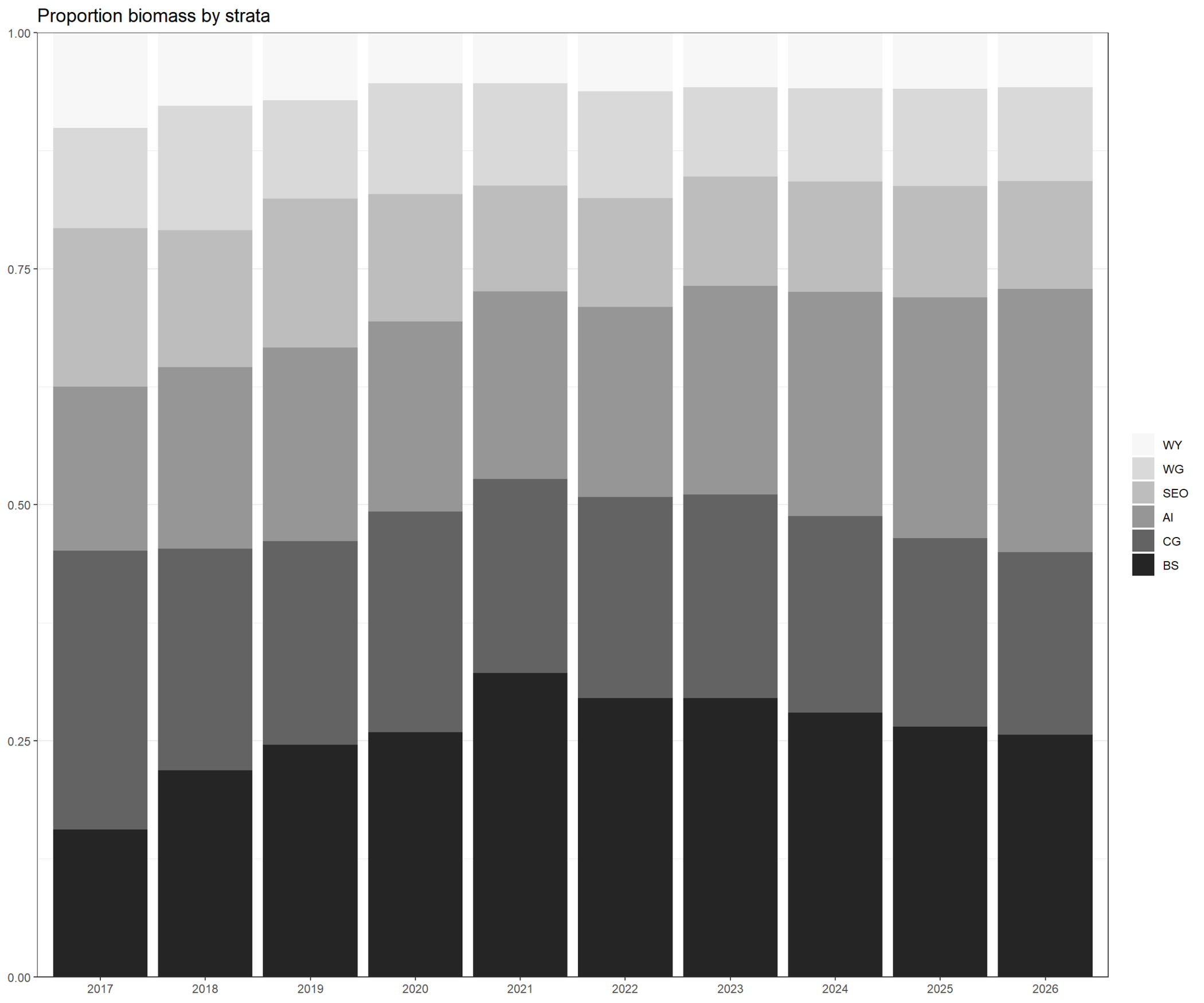

1.00
0.75
0.50
0.25
0.00
Proportion
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY

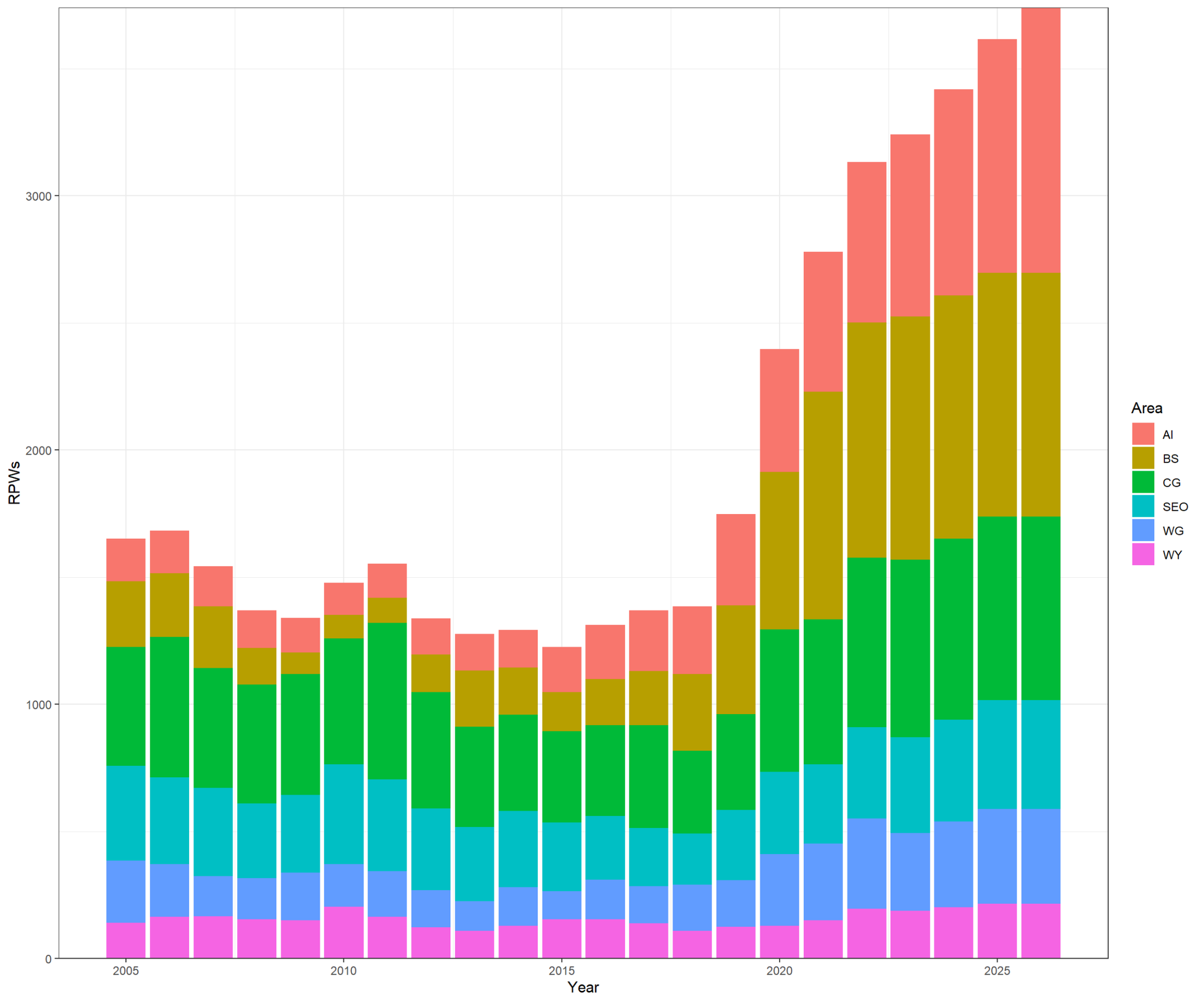

AI
BS
GOA
Proportion
0.0
0.2
0.4
0.6
0.8
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY

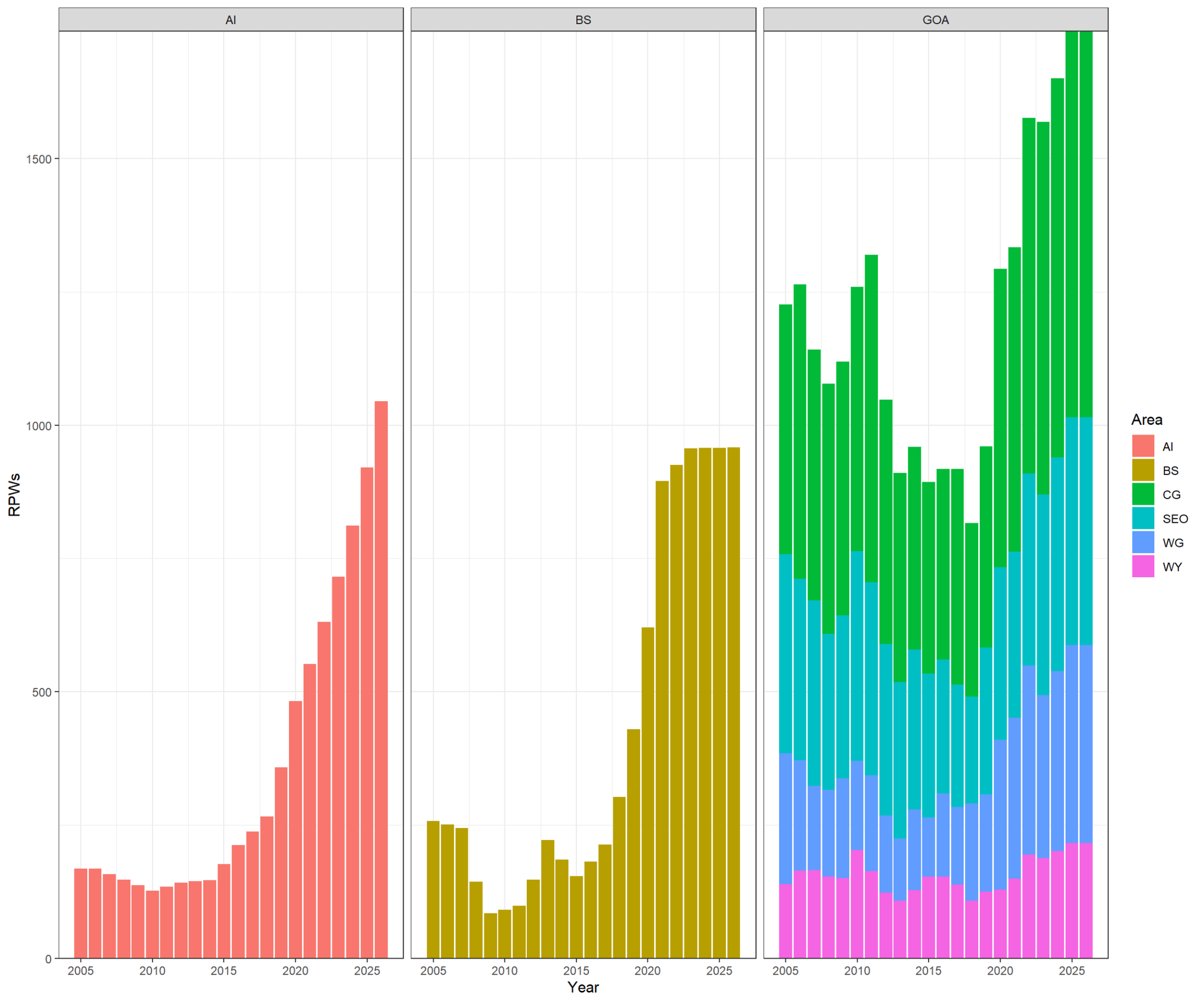

AI
BS
CG
SEO
WG
WY
1.00
0.95
0.80
0.58
0.92
0.66
1.00
0.78
0.53
0.93
0.57
1.00
0.85
0.87
0.80
1.00
0.68
0.80
1.00
0.67
1.00
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

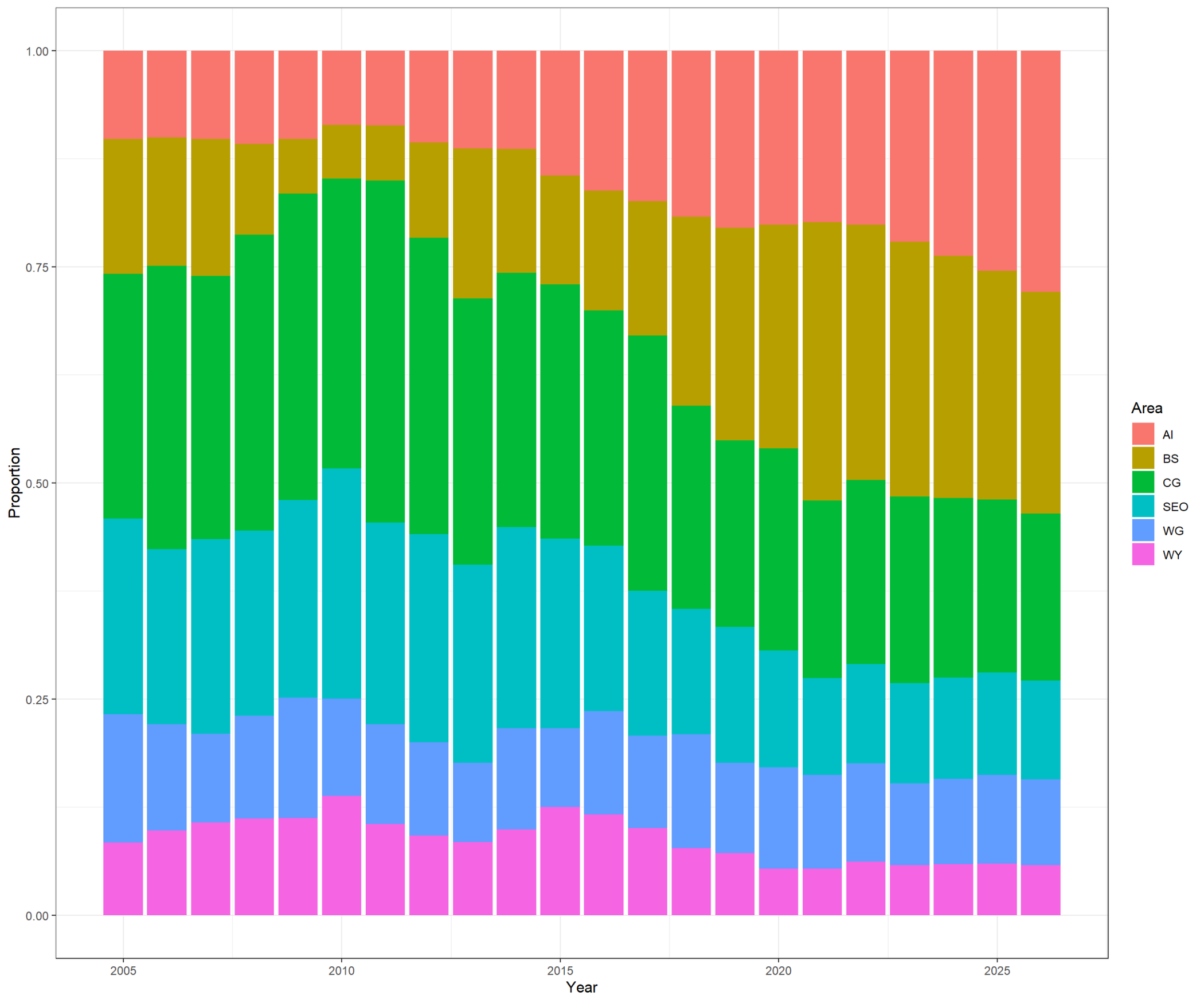

AI
BS
CG
SEO
WG
WY
Biomass
0
500
1,000
2005
2010
2015
2020
2025

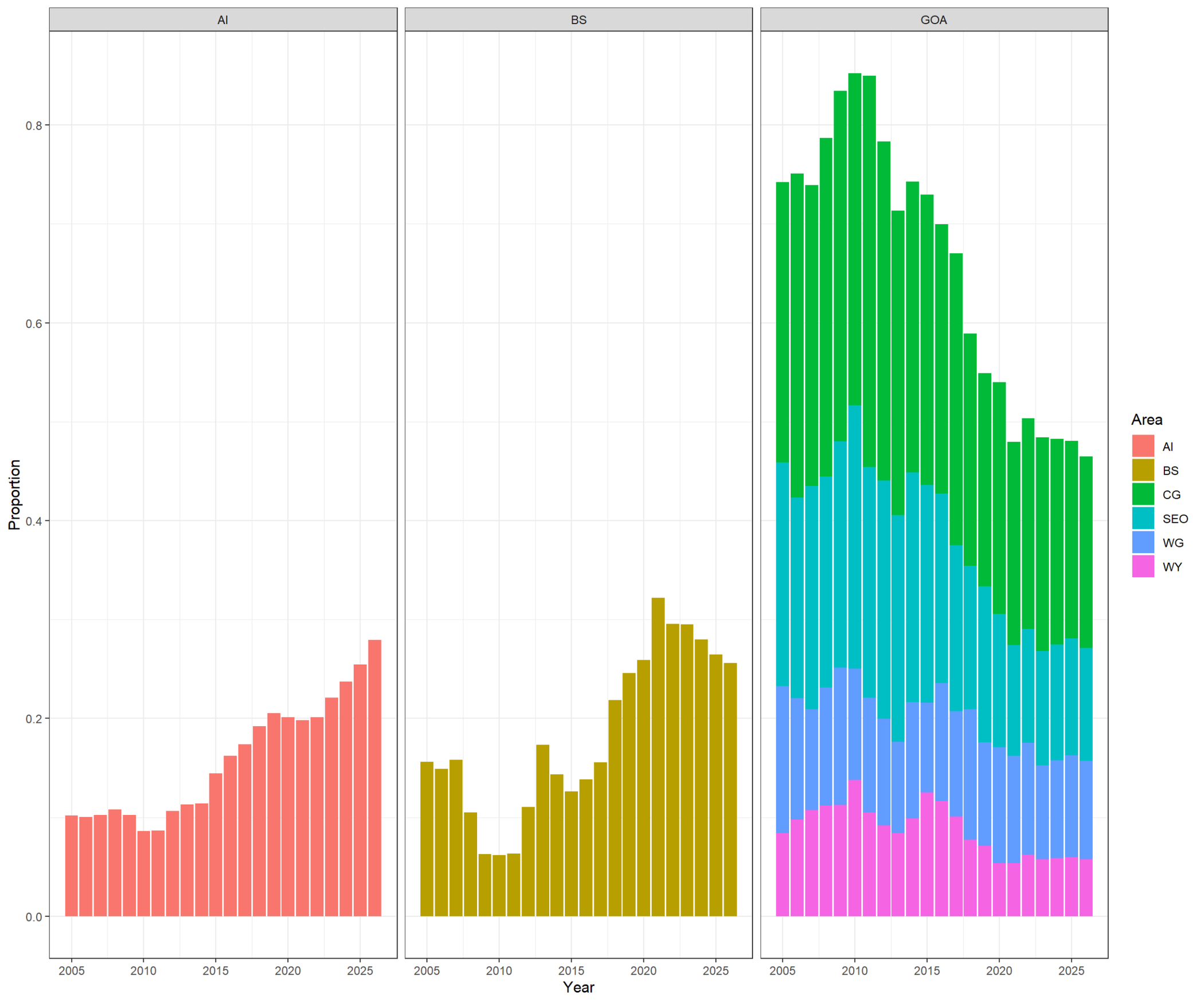

Biomass
4,000
3,000
2,000
1,000
2005
2010
2015
2020
2025

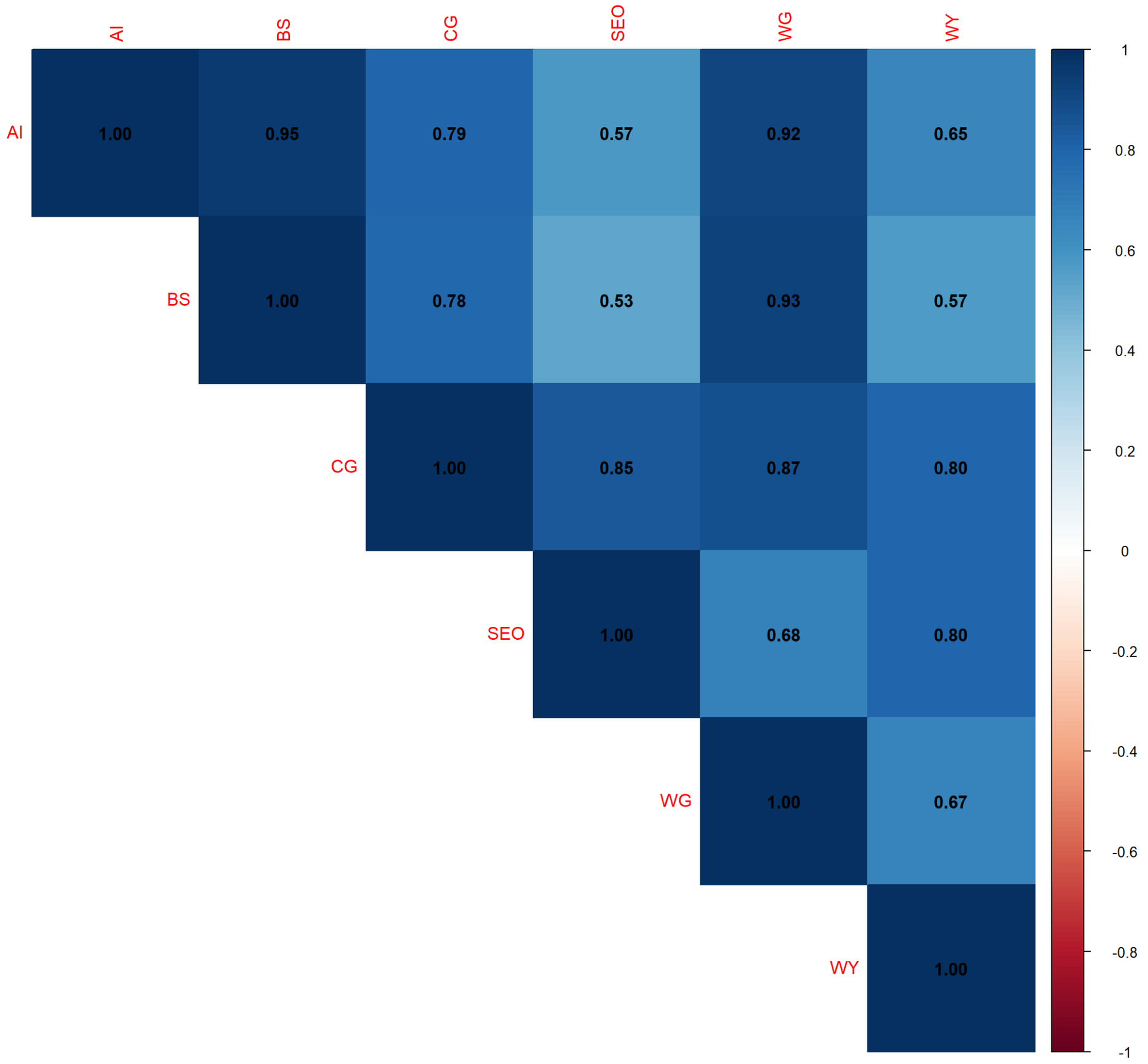

Proportion biomass by strata
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
WY
WG
SEO
AI
CG
BS


RPWs
3000
2000
1000
0
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY


AI
BS
GOA
RPWs
0
500
1000
1500
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY


1.00
0.75
0.50
0.25
0.00
Proportion
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY


AI
BS
GOA
Proportion
0.0
0.2
0.4
0.6
0.8
2005
2010
2015
2020
2025
Year
Area
AI
BS
CG
SEO
WG
WY


AI
BS
CG
SEO
WG
WY
1.00
0.95
0.79
0.57
0.92
0.65
1.00
0.78
0.53
0.93
0.57
1.00
0.85
0.87
0.80
1.00
0.68
0.80
1.00
0.67
1.00
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

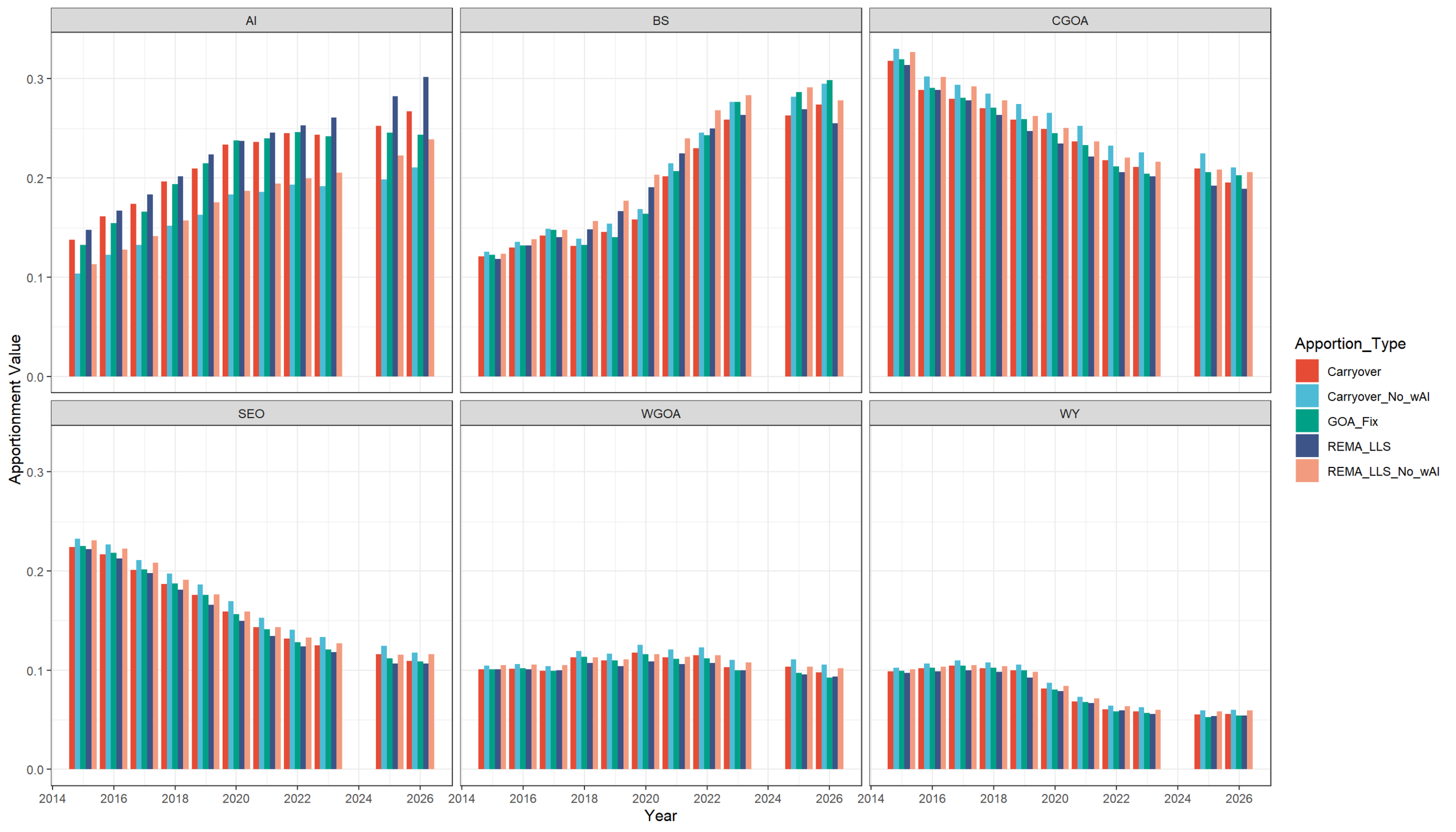
Year	Apportion_Type	BS	AI	WGOA	CGOA	WY	SEO
2026	Carryover_No_wAI	0.30	0.20	0.11	0.21	0.06	0.12
2026	REMA_LLS_No_wAI	0.28	0.23	0.10	0.21	0.06	0.12
2026	Carryover	0.27	0.26	0.10	0.20	0.06	0.11
2026	REMA_LLS	0.26	0.30	0.09	0.19	0.05	0.11
2026	GOA_Fix	0.30	0.24	0.09	0.20	0.05	0.12

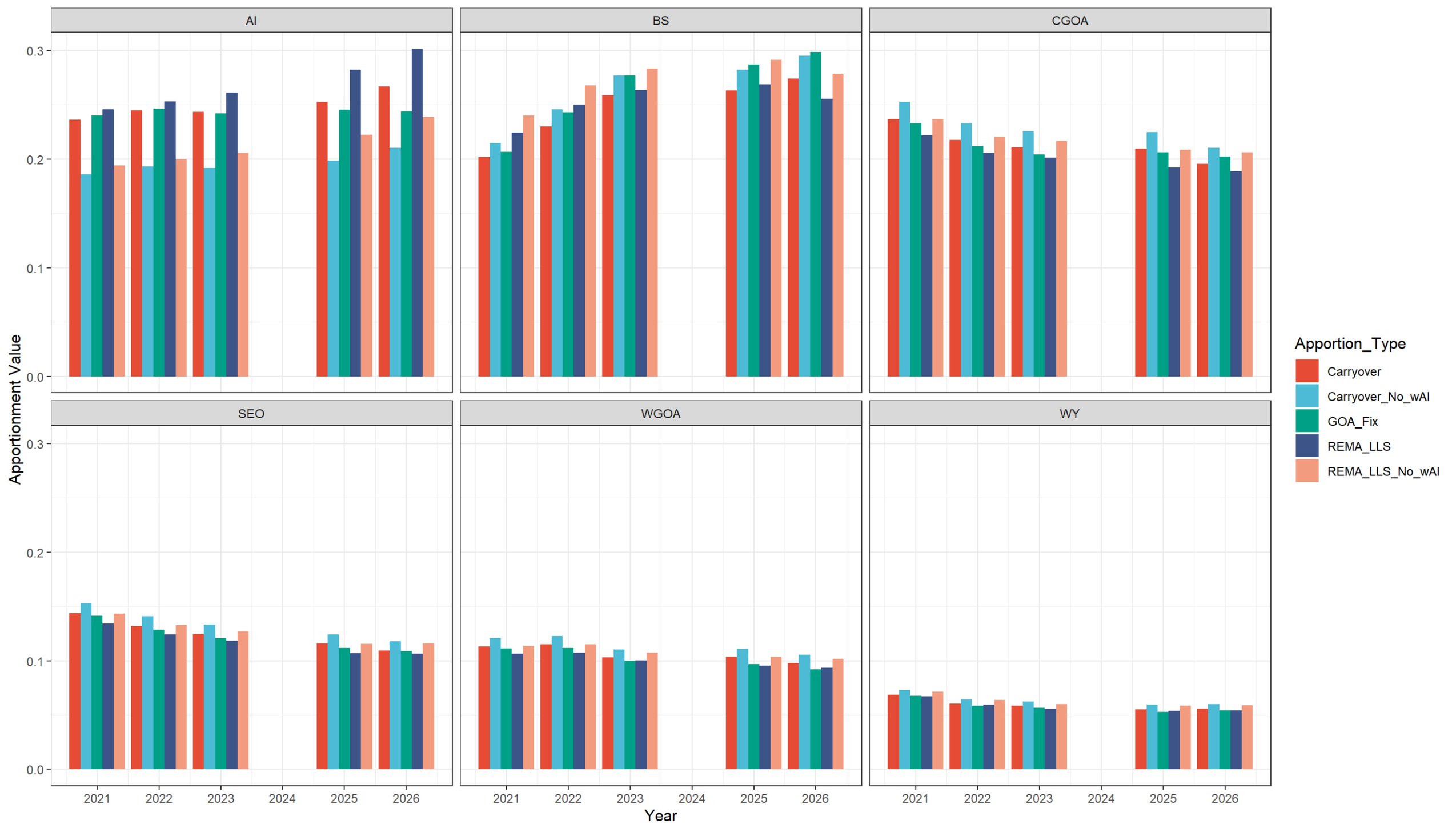
Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	13037	11421	22550	4687	9622	2652	5589	47008
2027	Carryover_No_wAI	16502	11000	27503	6051	11551	3300	6601	55005
2027	REMA_LLS_No_wAI	15401	12651	26952	5501	11551	3300	6601	55005
2027	Carryover	14851	14301	25852	5501	11001	3300	6051	55005
2027	REMA_LLS	14301	16502	24202	4950	10451	2750	6051	55005
2027	GOA_Fix	16502	13201	25302	4950	11001	2750	6601	55005
2028	Carryover_No_wAI	16258	10839	27097	5961	11381	3252	6503	54194
2028	REMA_LLS_No_wAI	15174	12465	26555	5419	11381	3252	6503	54194
2028	Carryover	14632	14090	25471	5419	10839	3252	5962	54194
2028	REMA_LLS	14090	16258	23845	4877	10297	2710	5962	54194
2028	GOA_Fix	16258	13007	24929	4877	10839	2710	6503	54194

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	0	0	0	0	0	0	0	0
2027	Carryover_No_wAI	27	-4	22	29	20	24	18	17
2027	REMA_LLS_No_wAI	18	11	20	17	20	24	18	17
2027	Carryover	14	25	15	17	14	24	8	17
2027	REMA_LLS	10	44	7	6	9	4	8	17
2027	GOA_Fix	27	16	12	6	14	4	18	17
2028	Carryover_No_wAI	25	-5	20	27	18	23	16	15
2028	REMA_LLS_No_wAI	16	9	18	16	18	23	16	15
2028	Carryover	12	23	13	16	13	23	7	15
2028	REMA_LLS	8	42	6	4	7	2	7	15
2028	GOA_Fix	25	14	11	4	13	2	16	15

Year	Region	OFLw	ABCw	TAC	Catch	Year2	Region	OFLw	ABCw	Year3	Region	OFLw	ABCw
2026	BS	NA	13037	8996	3339	2027	BS	NA	16502	2028	BS	NA	16258
2026	AI	NA	11421	7440	632	2027	AI	NA	11000	2028	AI	NA	10839
2026	GOA	NA	22550	22550	18191	2027	GOA	NA	27503	2028	GOA	NA	27097
2026	WGOA	NA	4687	4687	3458	2027	WGOA	NA	6051	2028	WGOA	NA	5961
2026	CGOA	NA	9622	9622	7597	2027	CGOA	NA	11551	2028	CGOA	NA	11381
2026	WY	NA	2652	2652	2271	2027	WY	NA	3300	2028	WY	NA	3252
2026	SEO	NA	5589	5589	4865	2027	SEO	NA	6601	2028	SEO	NA	6503
2026	Total	57797	47008	38986	22161	2027	Total	65230	55005	2028	Total	64264	54194

Region	Year	Biomass (4+)	OFLw	ABCw	TAC	Catch
GOA	2025	305000	NA	24038	22836	18387
GOA	2026	299000	NA	23737	22550	18191
GOA	2027	361861	NA	27503	NA	NA
GOA	2028	350639	NA	27097	NA	NA
BS	2025	175000	NA	13898	8496	3909
BS	2026	172000	NA	13723	8996	3339
BS	2027	198147	NA	16502	NA	NA
BS	2028	192002	NA	16258	NA	NA
AI	2025	153000	NA	12175	7940	948
AI	2026	151000	NA	12022	7440	632
AI	2027	218143	NA	11000	NA	NA
AI	2028	211378	NA	10839	NA	NA


AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2014
2016
2018
2020
2022
2024
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI


AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2021
2022
2023
2024
2025
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI